A Data Science Analysis of Erdős Problem 30

The maximum size of a Sidon set in an interval — a computational and analytic investigation

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with Anthropic Claude Fable 5.1 and Claude Opus 5.5

Version 6 · 21 October 2026

extreme values of Singer constructions · symmetric certificates reach $2\sqrt{2}/3$ · the pair-level and three-point relaxations stop at $2\sqrt{2}/3$ · an almost-all lower bound · an evidence ledger

Contents

Executive summary

Erdős Problem 30 asks whether h(N), the largest size of a Sidon set contained in {1,…,N}, satisfies h(N) = N^(1/2) + O_ε(N^ε) for every ε > 0. The problem has been open since Erdős and Turán's 1941 paper and carries a $1000 prize. The lower half of the conjecture reduces to prime gaps and is believed; the difficulty lies in the upper bound, where eighty-five years of work has moved the constant in front of N^(1/4) and never the exponent.

This document collects the results of an investigation carried out in stages. Versions 3 to 5 located the floor of the Hou–Zhao smoothing framework, proved that it is $2\sqrt{2}/3$ exactly, carried the question to a semidefinite relaxation that keeps what the smoothing argument discards, surveyed the algebraic constructions and proved the first gap theorems of Stage 1. Version 6 completes the map of the upper-bound wall and adds three analyses from the data side: an extreme-value study of the strongest algebraic family, an almost-all lower bound, and an explicit account of what data can and cannot decide. The principal findings:

- **The upper-bound wall, completed (Parts IV–VI, Appendices B–D).** Three independent barriers now stop at exactly $2\sqrt{2}/3$, each with a written proof (none refereed). Every certificate of the vector-valued smoothing framework has $\gamma \geq 2\sqrt{2}/3$ (Theorem 4.1), and the original symmetric framework reaches that value (Appendix D), so $\gamma^*_{sym} = \gamma^* = 2\sqrt{2}/3$. The pair-level moment relaxation satisfies $\hat{h}_2(N) = \sqrt{N} + (2\sqrt{2}/3)\cdot N^{1/4} + O(1)$, also with distinct-sums and Sherali–Adams constraints added (Theorem 6.6; Conjecture 6.6 of Version 5). The three-point relaxation satisfies the same, $\hat{h}_3(N) = \sqrt{N} + (2\sqrt{2}/3)\cdot N^{1/4} + O(1)$ (Theorem 6.8). Keeping the variance term, keeping the difference slack and adding three-point constraints each gain only O(1).
- **A new verified constant (Part V).** Symmetric certificates of the original Hou–Zhao framework, built from centred uniform kernels whose mixture reproduces the extremal autocorrelation, give $\gamma < 0.942809493$ at m = 1024, checked in exact arithmetic on 6145 covering inequalities. Hence $h(N) \leq N^{1/2} + 0.9428095\cdot N^{1/4} + O(1)$ using only Hou and Zhao's Theorem 2.1, without the two-sided extension; this is 4.5×10^{-7} above the proved limit. Haoyu Chen's unrefereed forum claim of the explicit bound at $2\sqrt{2}/3$ itself remains the best bound claimed.
- **Extreme values of Singer constructions (Part VIII).** Every multiplier class of the characteristic-2 Singer sets was enumerated at $q = 2^{11}–2^{14}$, and random samples of 2×10^6 classes were drawn at $q = 2^{15}–2^{18}$, reaching $N \approx 6.9\times10^{10}$. The gap at the origin follows Gamma(2, 2q), as the Frobenius map predicts; every record arc straddles the origin; the per-class tail is exponential, with a formal one-sided 95% bound $\xi \leq -0.011$ on its shape and a drift of at most N^0.03 in its scale. The best excess grows like $c\cdot\ln N$ with $c \approx 1.6–2.0$. The family with the strongest structural effect, and so the likeliest algebraic source of a power law, shows none.
- **The lower side for typical N (Part IX).** Singer sets and exceptional-set theorems for primes in short intervals give $\sqrt{N} - h(N) \leq N^{1/43+\varepsilon}$ for all N outside a set of density zero, and, under the Lindelöf Hypothesis, $\sqrt{N} - h(N) < N^{\delta}$ for every $\delta > 0$ outside $O(Y^{1-\delta+\varepsilon})$ integers $N \leq Y$. The uniform figure improves from N^0.2625 to N^0.26 on R. Li's preprint. In the almost-all sense, Erdős #30 is a question about the upper bound alone.
- **Arithmetic: one success, three failures (Part VI).** In a cyclic model where every moment relaxation fails, integer counts on a single quotient certify 98 of 105 impossible configurations. On the interval, three analogues — integrality on short differences, on end counts and residues, and

genuine end atoms tested against distinct sums — certify nothing and cost O(1). Arithmetic that could move the constant has to be tied to positions, not to the difference set alone.

- **What the data can decide (Part X).** The statement is Π_3, and N^ε cannot be told from 1 + ε·ln N until ε·ln N ≈ 1, so no dataset decides it. An explicit Bayes ledger moves P(true) from an illustrative 0.60 to 0.74, with a ceiling of 0.746 however strong the construction-side evidence becomes: 94% of the remaining probability of "false" sits on routes no data here reaches. Gain-vs-N regressions triage proposed strengthenings of the upper-bound relaxations before they are built.
- **Stage 1 (Part VII), unchanged from Version 5.** Every Singer set, for every multiplier, has a gap of at least 2q (Theorem 7.1), which gives $h(q^2 - q + 2) \geq q + 1$; every Bose–Chowla set has a gap of at least 2q − 1 (Theorem 7.4). Whether the best multiplier's gap is unbounded relative to q — which would refute h(N) = √N + O(1), O'Bryant's Open Question 5 — remains open.
- **Formalization (Part XI).** Lean 4 statements of the barrier side — the L^4 identity, the renewal lower bound, the linear certificate, and an exact rational certifier for the pair-level dual — are written on the Hulak–Ramos–de Queiroz Singer library. They have not yet been compiled.

Dependence on the smoothing inequality. Every statement about h(N) in this document rests on a smoothing inequality: Theorem 2.1 of Hou and Zhao (arXiv:2607.01169v3) for symmetric certificates, which now suffices for the best verified constant, and its two-sided extension (Proposition 3.1) for the linear-kernel certificate. The two-sided version has been proved in full, following Hou and Zhao step by step with the O(1) term made explicit, and checked mechanically on explicit Sidon sets; neither proof has been refereed. The statements about the framework and the relaxations themselves — the lower bound $2\sqrt{2}/3$, the extremal certificates, Theorems 6.6 and 6.8 — concern optimization problems and do not depend on it. Several constants are ineffective, because the decay constants C_0 and κ of the renewal density (Lemma 4.4(b)) are not explicit.

Erdős Problem 30 remains open. The evidence favours a true answer, the known construction families will not refute it, and on the upper-bound side every method built on one-, two- and three-point statistics has been followed to the same wall at $(2\sqrt{2}/3)\cdot N^{(1/4)}$. That wall is a theorem about those methods, not about h(N). Moving the exponent needs constraints that hold for every Sidon set and cut off the explicit extremal points of Theorems 6.6 and 6.8; arithmetic tied to positions is the one candidate the tests have not ruled out.

Part I · The problem and its status

Statement

A Sidon set is a set of integers all of whose pairwise sums are distinct; equivalently all pairwise differences are distinct. Let h(N) denote the maximum size of a Sidon set contained in {1,…,N}. Erdős Problem 30 asks whether, for every ε > 0, h(N) = N^(1/2) + O_ε(N^ε).

Known bounds

- **Upper bound.** Erdős and Turán (1941) proved h(N) ≤ N^(1/2) + O(N^(1/4)); their argument and Lindström's give the constant 1. Since 2021 the constant has fallen in steps to 0.94349 (Hou–Zhao, 2026) and, in this work, to 0.9428096 (Version 4) and 0.9428095 (Version 6, with symmetric certificates only); Part V shows that $2\sqrt{2}/3$ is the limit of the method that produced the last three (Table 11).
- **Lower bound.** Singer's construction gives a Sidon set of size p+1 in an interval of length p^2+p+1 for every prime p, so the lower-bound error is controlled entirely by gaps between primes near √N: unconditionally N^0.2625 (Baker–Harman–Pintz), or N^0.26 on R. Li's preprint, N^(1/4) log N under RH, N^ε under Cramér-type hypotheses. For typical N the deficit is far smaller: $\sqrt{N} - h(N) \le$ N^(1/43+ε) outside a set of density zero (Part IX).
- The unconditional uniform lower-bound exponent, 0.2625 or 0.26, is therefore worse than the proved upper-bound exponent 1/4 — an asymmetry that persists for the worst N. For almost all N it reverses: the lower error is then smaller than the upper by a power of N (Part IX).

Exact values

A Sidon subset of {1,…,N} of size k is equivalent to a Golomb ruler with k marks and length at most N−1. Optimal Golomb ruler lengths are known exactly through 28 marks (OGR-28, length 585), so h(N) is exactly determined for all N ≤ 586. Exactness at the upper endpoint follows from strict monotonicity of the optimal lengths: a 29-mark ruler of length ≤ 585 would, on deleting an end mark, give a 28-mark ruler of length < 585, contradicting OGR-28.

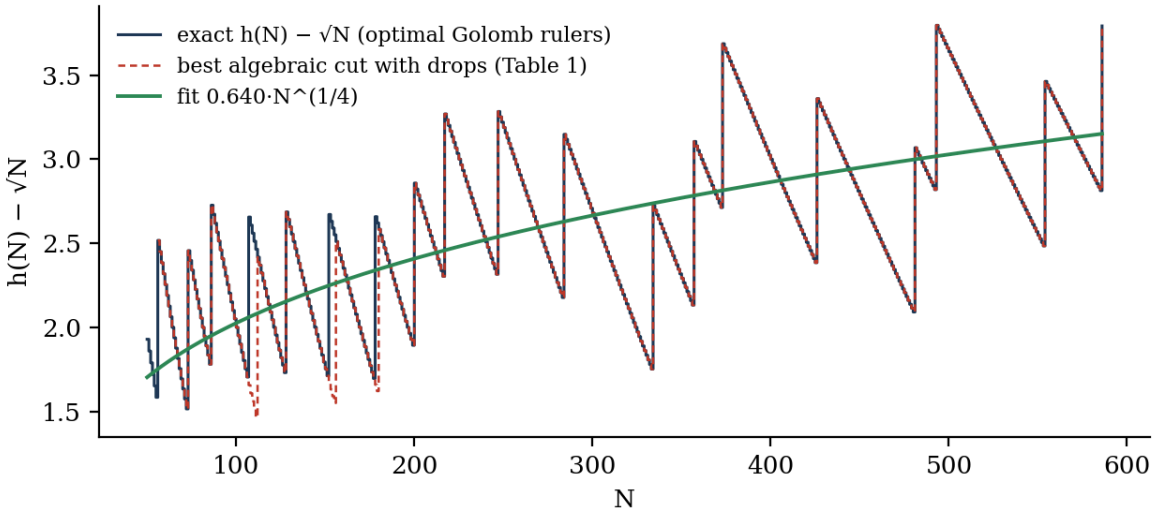


Figure 1. *Exact h(N) − √N from optimal Golomb rulers over N = 50…586, the best single-term fit 0.640·N^(1/4), and the excess obtained from the best algebraic cut with drops of Table 1 (defined for N ≥ 56). The two step functions coincide at all but 11 of the 531 values of N where both are defined and never differ by more than 1. The sawtooth is an artifact of h(N) being a step function.*

What the exact data supports

Fits over N = 50…586 (537 points): the model γ·N^(1/4) gives γ = 0.640 with RMSE 0.385; a constant gives c = 2.64 with RMSE 0.490; a free-exponent fit A·N^α returns α = 0.20 (0.201 by least squares on the values, 0.198 by least squares in log–log coordinates). The N^(1/4) model beats the constant model and the free exponent lands nearer 1/4 than 0, consistent with the gap still growing. But the residual floor is the step-function sawtooth, not model error, and over a range where N^(1/4) varies by a factor of only 1.85 the candidate models are not separable.

The cleaner statistic is in k-space. Since $h(N) \le \sqrt{N} + \gamma N^{1/4}$ is equivalent to $G(k) \ge k^2 - 2\gamma k^{3/2} + O(k)$, the deficit $D(k) = k^2 - G(k)$ gives one exact point per ruler with no sawtooth. Its log-log slope is 1.43 over k = 10…28, between the two models: the implied coefficient $D(k)/(2k^{3/2})$ drifts down (0.71, 0.64, 0.65, 0.58, 0.67 at k = 10, 15, 20, 25, 28) while $D(k)/(2k)$ drifts up (2.25, 2.47, 2.92, 2.90, 3.55).

Recent context

Recent AI-assisted work leaves Problem 30 open. In the FrontierMath Erdős benchmark (Adamczewski and Bloom, 2026), a pre-release GPT-6 Astra resolved five of 68 curated open problems across all attempts. None of the seven Sidon-set problems was among them, including Problem 30. The recent improvements to the N^(1/4) coefficient (Tao and collaborators; Hou and Zhao) did not use OpenAI systems.

Two OpenAI-linked results bear on the stages of this programme. The disproof of Erdős Problem 1 (GPT-6 Astra; Lean-verified) shows that the least N admitting a sum-distinct n-set in {1,…,N} is o(2^n), so no construction with N of order 2^n is optimal up to a constant factor. Its proof, a low-determinant lattice construction, is ineffective as formalized. Stage 1 has the same logical shape — refuting the optimality of a classical construction, where an ineffective existence proof would suffice — but whether

any part of the argument transfers to gaps in Singer sets is untested. The disproof of the unit distance conjecture (Alon et al., 2026) beat Erdős's grid by letting the degree of a number field grow, while Stage 0 surveyed only the fixed-degree Singer, Bose and Ruzsa families. Separately, Alexeev and Mixon (2025) resolved Erdős Problem 707 — not every finite Sidon set extends to a perfect difference set — formalizing Hall's polarity argument for cyclic planes in Lean; they generate Singer sets as the zeros of a third-order linear recurrence, which is the m-sequence form of Stage 1.

Part II · Stage 0: the construction survey

Method

A cyclic Sidon set S in Z_n with a gap of length g can be rotated so the gap straddles the cut and read off as a Sidon set in an interval of length n − g. For Singer sets ($n = q^2+q+1$, $|S| = q+1$) this gives $h(N) - \sqrt{N} \approx ½ + g/(2q)$ at N = n − g: the interval excess is the ratio of the largest gap to the mean gap. The question of whether that ratio grows logarithmically or as a power is therefore the whole conjecture, seen from the construction side.

A general engine was built over GF(p^m) covering three families — Singer sets from the trace form in $GF(q^3)$, Bose sets from $GF(q^2)$, and Ruzsa sets in $Z_{p(p−1)}$ — with every unit multiplier enumerated, and with the largest gap replaced by the full best-arc profile: the shortest interval retaining k − d elements, for d = 0…8, since dropping a few elements may shorten more than cutting at one gap. Every generated set was verified Sidon before use.

Validation

The engine reproduces the known optimal-ruler lengths exactly at every k where an algebraic cut is optimal. The table below is the full comparison for k = 10…28.

k	OGR length	best algebraic cut	source
10	55	55	Singer q=9, drop 0
11	72	72	Singer q=11, drop 1
12	85	85	Singer q=11, drop 0
13	106	111 (+5)	Bose q=13, drop 0
14	127	127	Singer q=13, drop 0
15	151	155 (+4)	Singer q=17, drop 3
16	177	179 (+2)	Singer q=17, drop 2
17	199	199	Bose q=17, drop 0
18	216	216	Singer q=17, drop 0
19	246	246	Singer q=19, drop 1
20	283	283	Singer q=19, drop 0
21	333	333	Singer q=23, drop 3
22	356	356	Bose q=23, drop 1
23	372	372	Singer q=23, drop 1
24	425	425	Singer q=23, drop 0
25	480	480	Singer q=25, drop 1
26	492	492	Singer q=25, drop 0
27	553	553	Singer q=27, drop 1
28	585	585	Singer q=27, drop 0

Table 1. *For every k from 17 to 28 the optimal Golomb ruler equals the best algebraic cut with drops; only k = 13, 15, 16 beat algebra, by 5, 4 and 2. The exact record above N ≈ 200 is entirely the best-arc profile of algebraic sets.*

Result: no power-law growth in any family

The extreme-value model treats each of the M inequivalent sets in a family as k uniform points on a circle of length n; the largest gap over all of them is then the maximum of M Gumbel draws with scale q, predicting max gap ≈ q·[ln k + ln M]. The table below compares observed to that parameter-free prediction.

family	n	EV ratio (mean ± sd)	slope of excess vs ln N (cut)	slope (best arc)	gain from drops	mean d
Singer, odd char	24	0.963 ± 0.108	0.843	1.019	0.70	2.4
Singer, char 2	6	1.129 ± 0.157	1.115	1.197	0.34	1.0
Bose, odd char	23	0.929 ± 0.081	0.765	0.957	0.87	2.2
Bose, char 2	6	1.115 ± 0.123	0.986	1.220	0.53	1.7
Ruzsa	108	1.029 ± 0.105	0.486	0.620	0.80	2.5

Table 2. *Per-family summary, recomputed from stage0_results.csv over the instances with $q \geq 25$ (Singer, Bose) and $p \geq 53$ (Ruzsa); the four smallest Singer and three smallest Ruzsa instances in the file are excluded, which accounts for the counts 34 → 30 and 111 → 108. The standard deviation is the population value; slopes are least-squares slopes of the excess against ln N. An EV ratio near 1 means the observed largest gap matches the random model with no fitted parameter. Ruzsa's lower slope is a search-size effect: multipliers give nothing beyond the $\varphi(p-1)/2$ primitive roots, so the family is much smaller than Singer's $\approx q^2/6$.*

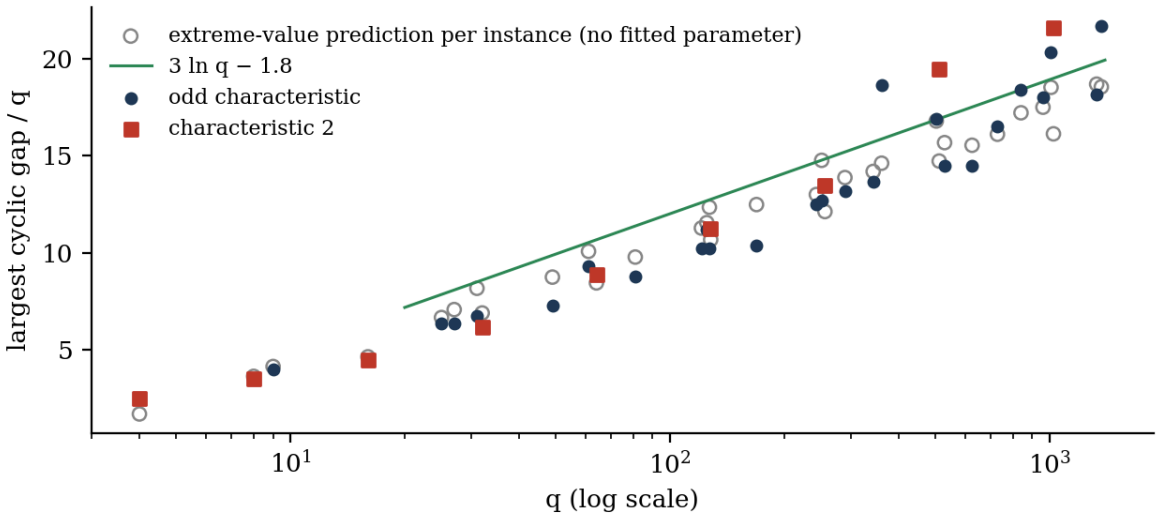


Figure 2. *Largest cyclic gap in Singer sets, normalized by q, for the 34 Singer instances of stage0_results.csv ($4 \leq q \leq 1369$, every multiplier enumerated), against the extreme-value prediction for each instance, which has no fitted parameter, and its asymptotic form 3 ln q − 1.8. Characteristic 2 in red.*

A separate gap-only run over 41 prime powers with $29 \leq q \leq 1409$, every multiplier enumerated, gives the following. Fitting max gap = A·q·ln²q + B·q·ln q + C·q returns A = −0.23: no super-logarithmic term. A naive power law returns exponent 1.26, but that is the local slope of q·ln q over this range plus a finite-size deficit at small q. The Weil bound permits gaps up to q^(3/2) log q; the actual sets use none of that room, with max gap / q^(3/2) falling monotonically from 1.2 to 0.5 across the range.

The one real structure: characteristic 2

Singer sets over GF(2^e) have EV ratios rising 1.05, 1.11, 1.32, 1.34 for $q = 2^7 \cdots 2^{10}$ and staying between 1.36 and 1.57 from $q = 2^{11}$ to 2^{14} (Table 4), and a gap distribution with a tail 3–4 times heavier than

Gumbel at the far quantiles, which odd characteristics lack. It persists at q = 2^{11}, which has no proper subfields, so it is not Baer-subplane structure. Every extreme orbit has its gap centred on the origin.

The mechanism is Frobenius: the Singer condition Tr(θ^i) = 0 satisfies Tr(x^p) = Tr(x)^p, so along the progression through the origin the zero events for j and pj coincide, cutting the number of independent constraints by a factor (1 − 1/p). This predicts that the gap containing the origin has mean 2q·p/(p−1) instead of 2q. Measured:

q	p^e	mean (gap at origin) / q	prediction	P(gap_0 > 6q)
1009	1009^1	1.99	2.00	0.017
1024	2^{10}	3.95	4.00	0.198
625	5^4	2.50	2.50	0.043
961	31^2	2.09	2.07	0.021
729	3^6	1.49	1.50 (one-sided: Tr(1)=0)	0.017

Table 3. *The Frobenius prediction verified to two decimals across characteristics. Non-origin gaps in characteristic 2 are statistically normal. The consequence is that in characteristic 2 the best cut has excess ≈ ln N rather than ≈ ¾ ln N — still logarithmic.*

Best-arc profile

Dropping d elements and keeping the shortest arc of the remaining k − d consecutive elements is measured with one convention throughout: if G is the largest sum of d + 1 cyclically consecutive gaps, the arc fits in an interval of N = n − G + 1 integers, and its excess is (k − d) − √N; d = 0 is the plain cut at the largest gap. Within the Stage 0 range (N ≤ 2.1×10^6), dropping elements gains 0.3–0.9 in excess on average, with the optimal d around 2–3; the largest single gain in stage0_results.csv is 2.6 (Ruzsa, p = 599), and the largest excess there is 12.36, from the Bose set over GF(2^{10}) with three elements dropped, at N = 1,017,354.

Characteristic 2 up to q = 2^{14}. Every multiplier class of the Singer sets over GF(2^{11}) to GF(2^{14}) was enumerated, the last at N ≈ 2.7×10^8. Version 4's figure of 14.6 for q = 2^{11} was a slip: the plain cut gives 15.13 and the best arc 16.70.

q	classes M	largest gap (EV ratio)	plain cut, d = 0	best arc, d ≤ 8 (d)	best arc, any d (d)	Rokicki–Dogon
2^{11}	54,498	59,724 (1.57)	15.13	16.70 (3)	16.70 (3)	16.70
2^{12}	139,968	112,664 (1.36)	14.27	18.73 (6)	18.73 (6)	18.73
2^{13}	728,208	260,834 (1.41)	16.43	19.80 (7)	22.13 (16)	22.13
2^{14}	1,820,448	571,140 (1.45)	17.94	23.45 (8)	24.60 (18)	24.60

Table 4. *Singer sets in characteristic 2 beyond the Stage 0 range. Excess (k − d) − √N under the convention above; d was searched up to 110, 8, 30 and 30 for q = 2^{11} to 2^{14}. The last column is the quality, marks − √length, of Rokicki and Dogon's projective record rulers (2016, Table 1), which differs from the excess by about 1/(2√N). At every q the best arc reproduces their record exactly — number of marks, length and quality — and the number of multiplier classes equals their number of inequivalent rulers. The best arcs are at N = 4,118,064, 16,583,410, 66,501,870 and 267,074,039.*

Three things change from the Stage 0 picture. First, drops matter more: the gain from dropping elements is 1.57, 4.45, 5.70 and 6.66, with optimal d = 3, 6, 16 and 18, so the Stage 0 statement that drops gain under a unit holds only in its own range. Second, the growth is steeper once d is free: the excess rises with slope 0.76 against ln N for the plain cut, 1.53 for the best arc with d ≤ 8 and 1.95 with d free. Third,

it is still logarithmic: over three doublings of q the best-arc excess grows by a factor of 1.47, where a term proportional to N^(1/4) would give 2.84.

Rokicki and Dogon observed that powers of two above 2^{10} give the rulers of highest quality and wrote that they had no explanation. The Frobenius mechanism above explains it: it enlarges the gap at the origin by the factor p/(p − 1), which is largest at p = 2. Version 5 also asked why the optimal d grows with q in characteristic 2. Part VIII answers both questions. The set is invariant under multiplication by 2, so near the origin a position is a mark exactly when its odd part is; a parameter-free chain model built on that fact reproduces the gap at the origin, the share of records held by arcs through the origin, and the bulk of the best-arc distribution at every d. Under that model the record profile across d is a plateau, whose maximizing d scatters from 0 to 21 at $q = 2^{11}$ over replicates; the sequence 3, 6, 16, 18 is consistent with draws from it, so it is not a trend.

Verdict

Outcome A, cleanly. Every family follows the random model; the single place structure appeared, it was predictable to two decimals and enlarged a constant, not an exponent. The construction side will not refute the conjecture. The corresponding Cramér-type conjecture for this problem is that the largest gap over multipliers equals q·(3 ln q + O(1)) generically and q·(4 ln q + O(1)) in characteristic 2, so that h(N) = √N + Θ(log N) along the best algebraic sequence — unbounded, hence refuting h(N) = √N + O(1), but fully consistent with the N^ε conjecture. For the weakest multiplier the same random model predicts a largest gap of q(ln q − ln ln q + O(1)); the exhaustive enumerations of Part VII match this, with median ratio 0.98, 1.00 and 1.09 for the Singer, Bose and Ruzsa families. Part VIII carries the characteristic-2 Singer family, the one with the strongest structural effect, to $q = 2^{18}$ ($N \approx 6.9\times10^{10}$) by an extreme-value analysis of every multiplier class, and bounds the tail shape formally: $\xi \le -0.011$ (one-sided 95%).

Part III · Stage 2T(a): the smoothing framework, numerically

The Hou–Zhao certificate framework generalizes the upper-bound arguments of the Erdős–Turán lineage. This part sets up the framework and its two-sided extension and reports the numerical picture — resolution scans, the effect of asymmetry, the structure of optimal certificates, and the optimal dual multiplier — which is what led to the exact results of Parts IV and V. Version 3 located the infimum at 0.942809 and proposed $2\sqrt{2}/3$ as a candidate closed form; both are now theorems.

The certificate space

At resolution (m, L, K) — m bins per kernel, L boundary blocks, K kernels mixed with weights λ_k — a certificate consists of kernels p^(k) ∈ ℝ^m, symmetric, nonnegative, with Σ_i p^(k)_i = 1; boundary weights w^(k) ∈ ℝ^(Lm), extended by w^(k)_j = 1 for j ≥ Lm; and mixing weights λ ≥ 0 with Σλ_k = 1, subject to the covering inequalities

$$\Sigma_k \lambda_k \Sigma_i p^{(k)}_i w^{(k)}_{q+i} \geq 1 \text{ for } q = 0,\dots,Lm, \quad (3.1)$$

with

$$a = m \cdot \Sigma_k \lambda_k \Sigma_i (p^{(k)}_i)^2,\ b = 1 + 2\left(\Sigma_k \lambda_k \Sigma_j (w^{(k)}_j)^2 / m - L \right),\ \gamma = \sqrt{ab}. \quad (3.2)$$

Theorem 2.1 of Hou and Zhao. If (3.1) holds and b > 0, then $h(N) \leq N^{1/2} + \sqrt{ab}\cdot N^{1/4} + O(1)$. Version 3 cited this as Lemma 2.1; the current version of arXiv:2607.01169 (v3, 4 September 2026) states it as Theorem 2.1 and certifies $\gamma_0 = 0.9434925907\dots$ with eight kernels at m = 32, L = 4. The uniform kernel with all boundary weights 1 gives a = b = 1 and recovers Erdős–Turán–Lindström. The hypothesis b > 0 is automatic: Hou and Zhao show $b \geq 1/m$, and Part IV gives $b \geq 8/(9a)$.

The optimum is non-increasing in L and K — extra boundary weights can be set to 1 and extra kernels given weight 0 — and under refinement m → 2m, since splitting each bin in two preserves feasibility exactly. Write γ*_sym for the infimum of √(ab) over all certificates at all resolutions.

Two-sided certificates

The proof of Theorem 2.1 uses the symmetry of the kernels at one point only: in the right-boundary case of the covering claim, where K_s = K_{H−1−s} reduces that case to the left one. Dropping it gives a two-sided certificate: kernels p^(k) that are nonnegative and normalized but need not be symmetric, and for each k two boundary vectors u^(k), v^(k) ∈ ℝ^(Lm), extended by 1, subject to

$$\Sigma_k \lambda_k \Sigma_i p^{(k)}_i u^{(k)}_{q+i} \geq 1 \ (q = 0,\dots,Lm),$$

$$\Sigma_k \lambda_k \Sigma_i p^{(k)}_{m-1-i} v^{(k)}_{q+i} \geq 1 \ (q = 0,\dots,Lm), \quad (3.3)$$

with a as before and $b = 1 + (1/m)\, \Sigma_k \lambda_k \Sigma_{j<Lm} [(u^{(k)}_j)^2 + (v^{(k)}_j)^2 - 2]$. A symmetric certificate with u = v = w is a two-sided certificate with the same a and b.

Proposition 3.1 (smoothing inequality, two-sided). If a two-sided certificate satisfies (3.3) and b > 0, then $h(N) \leq N^{1/2} + \sqrt{ab}\cdot N^{1/4} + O(1)$. Explicitly, with γ = √(ab) and $N_1 = N + bm$, for $N \geq N_0(a, b, m, L)$,

$$h(N) \leq \sqrt{N_1} + \gamma N_1^{1/4} + \gamma^2/2 + \gamma^3/(8N_1^{1/4}) = \sqrt{N} + \gamma N^{1/4} + ab/2 + o(1). \quad (3.4)$$

The proof follows Hou and Zhao step by step, using v^(r) and the reversed kernel on the right boundary, and replaces their final choice of scale, which calls on the Erdős–Turán bound, by the self-contained

inequality $k \le \sqrt{N_1} + \gamma\sqrt{k}$. It was implemented literally at integer positions and checked on six certificates, five explicit Sidon sets (Ruzsa sets with 30, 46 and 60 elements, an Erdős–Turán set and a greedy random set) and h = 1, 2, 3: every check passed, and swapping u and v makes the covering fail, its minimum dropping to 0.44 for the linear kernel. Write γ^* for the infimum over two-sided certificates, so that $\gamma^* \le \gamma^*_{\mathrm{sym}}$.

The continuum relaxation

Reading kernels and boundary vectors as step functions on cells of width 1/m — $\rho_k = m\cdot p^{(k)}_i$ on [i/m, (i+1)/m), and $\omega_k = u^{(k)}_j$, $\omega'_k = v^{(k)}_j$ on [j/m, (j+1)/m) — turns every discrete certificate into a continuum certificate: probability densities ρ_k on [0,1] and profiles ω_k, ω'_k on $[0,\infty)$, equal to 1 outside a compact set, such that

$$\sum_k \lambda_k \int_0^1 \rho_k(x)\,\omega_k(u+x)\,dx \ge 1 \text{ and } \sum_k \lambda_k \int_0^1 \bar{\rho}_k(x)\,\omega'_k(u+x)\,dx \ge 1 \text{ for all } u \ge 0, \quad (3.5)$$

where $\bar{\rho}(x) = \rho(1-x)$, with $a = \sum_k \lambda_k \int_0^1 \rho_k^2$ and $b = 1 + \sum_k \lambda_k \int_0^\infty (\omega_k^2 + \omega'^2_k - 2)$. At a fractional shift u = (q+s)/m the covering integral equals $(1-s)\cdot C_q + s\cdot C_{q+1}$, a convex combination of two adjacent discrete constraints, and a and b take identical values in both settings. Consequently every lower bound for the continuum problem applies verbatim to every discrete certificate at every resolution, with no limiting argument; this is how the bound of Part IV reaches the discrete framework.

Method

- **Outer:** L-BFGS over the kernels (softmax-parametrized; in the symmetric case reduced to m/2 free parameters per kernel) and the mixing weights.
- **Inner:** the exact quadratic program for the boundary weights, solved through its nonnegative dual, which is banded with bandwidth m, by a primal–dual active-set iteration warm-started from the previous outer step; for two-sided certificates it splits into one program per end. The gradient of the outer objective is read off the duals by the envelope theorem.
- **Globalization:** multi-start with 30+ random seeds at m = 32 plus basin hopping; higher resolutions are initialized by upsampling the converged lower-resolution certificate. Splitting every bin in two preserves feasibility and γ, but it also produces an exact stationary point of the finer problem, so the upsampled certificate must be perturbed before re-optimization. The m = 512 symmetric value of Version 3 (0.94281112) was under-converged for this reason; its re-optimized successor gives 0.9428108915.

The analytic gradient agrees with central finite differences to 5.5×10^{-6}, and an independent implementation of the two-sided optimizer agrees to a relative 3×10^{-8}. At Hou–Zhao's grid (m = 32, L = 4) with one symmetric kernel the optimizer returns 0.946147, the value 0.9461473014 listed in their Table 1. Every reported certificate value is verified independently of the optimizer: the certificate is rationalized, the boundary program re-solved for the rational kernels, the boundary weights rounded and repaired by a uniform increment (Hou–Zhao, Lemma 3.2), all covering inequalities checked over $\mathbb{Q}$, and a, b and γ^2 evaluated exactly.

Resolution scans

m	symmetric, K = 2, L = 4	two-sided, K = 2, L = 4	linear kernel 2x, L = 8
32	0.94324229	0.9429206679	0.9429240903

m	symmetric, K = 2, L = 4	two-sided, K = 2, L = 4	linear kernel 2x, L = 8
64	0.94292072	0.9428374016	0.9428378328
128	0.94283748	0.9428162089	0.9428162572
256	0.94281628	0.9428108628	0.9428108625
512	0.9428108915	—	0.9428095137
Richardson (last two)	0.94280910	0.94280908	0.94280906

Table 5. *Verified values for the optimized certificates — the symmetric m = 512 entry is the re-optimized certificate; the original gave 0.94281112 — and exact quadratic-program values for the linear kernel, whose m = 256 and 512 certificates were also verified in exact arithmetic (Table A1). The last row is two-point Richardson extrapolation in $1/m^2$ from the last two resolutions. For comparison, $2\sqrt{2}/3 = 0.94280904$.*

All three scans converge in $1/m^2$: successive drops fall by factors 3.86, 3.93 and 3.93 (symmetric), 3.93 and 3.96 (two-sided), and within 3×10^{-3} of 4 at every step for the linear kernel. The two L = 4 scans extrapolate to 0.9428091, about 4×10^{-8} above $2\sqrt{2}/3$. That residual is the cost of the four-width boundary box: the extremal certificate of Part V needs a boundary layer about six kernel widths deep, and at L = 4 the linear kernel loses about 10^{-4} in b, which optimized mixtures recover almost entirely. At L = 8 the linear kernel extrapolates to 0.94280906 and follows $a\cdot b = 8/9 + 2/(9m^2) - 1/(9m^4)$ to within 3×10^{-8} in b (Table 9). Version 3 proposed an m = 1024 certificate to separate $2\sqrt{2}/3$ from the Richardson value 0.9428094; that test is superseded by the proofs of Parts IV and V.

Saturation in boundary depth and number of kernels

For boundary depth, at m = 32 and K = 2, L = 3 gives 0.943245 while L = 4, 5 and 6 all give 0.943242 — the boundary layer of the symmetric optimum is under three kernel widths deep. For the number of kernels, K = 1 gives 0.946147, K = 2 gives 0.943242, and K = 3 gives 0.943242 again, identical to five decimals at both m = 32 and m = 64, with the optimizer driving one weight to about 0.04. Within the symmetric parametrization the capacity saturates at two kernels at these resolutions, where two kernels buy 2.9×10^{-3} over one. Version 6 corrects the general claim: the saturation holds only at $m \leq 64$. At fine resolution many kernels are needed to follow the extremal autocorrelation, and at m = 512 the 256-kernel family of Appendix D (0.9428108398) beats the two-kernel optimum (0.9428108915), although that two-kernel run was stopped before converging. The saturation in L is a property of these optima; it fails for the extremal two-sided certificate of Part V, whose boundary profile needs about six kernel widths.

Asymmetry

L = 4	m = 32	64	128	256	limit
symmetric, K = 1	0.946147	0.945669	0.945610	—	≈ 0.94559
two-sided, K = 1	0.942944	0.942855	0.942832	—	≈ 0.942825
symmetric, K = 2	0.943242	0.942921	0.942837	0.942816	0.9428091
two-sided, K = 2	0.942921	0.942837	0.942816	0.942811	0.9428091

Table 6. *Symmetric against two-sided certificates at L = 4. The K = 2 rows and the m = 32 entries are verified values; the remaining K = 1 entries are optimizer values. Limits are Richardson extrapolations from the last two resolutions shown; for the symmetric single kernel the convergence is irregular and the limit is rough.*

For a single kernel, asymmetry lowers the limit by almost 3×10^{-3}, to within 2×10^{-5} of the two-kernel floor: the single-kernel value 0.946147 is a property of the symmetric parametrization rather than of

one kernel as such. For two kernels, asymmetry does not move the L = 4 limit; a two-sided certificate at resolution m does as well as a symmetric one at 2m, slightly better by $5–8\times10^{-8}$. Part V explains the pattern: the extremal kernel is asymmetric, symmetric certificates can approach it only through mixtures, and with a deeper boundary layer a single asymmetric kernel reaches the limit on its own. Appendix D makes the mixture explicit: centred uniform kernels with weights $4l^3/n^4$ at resolution m = 2n have exactly the autocorrelation of the linear kernel at resolution n, which is why a symmetric certificate at 2m matches a two-sided one at m. Decoupling the kernel grid from the boundary grid cannot move the infimum at all, since such a certificate is an ordinary certificate on the common refinement of the two grids, and the continuum relaxation has no grid.

Structure of the optimal certificates

The symmetric optimum is a two-kernel mixture with $\lambda \approx (0.34, 0.66)$ — (0.3396, 0.6604) at m = 512 in the original scan, (0.3383, 0.6617) after re-optimization. The kernel carrying weight 0.34 is narrow and spiked, with peak density 2.69 and a peak-to-edge ratio of 17.3; the kernel carrying weight 0.66 is broad and nearly flat, with peak 1.87 and ratio 9.0. Both are symmetric and unimodal. Their boundary weights dip to roughly 0.08–0.10 at the outermost cell, overshoot above 1 near half a kernel width (reaching 1.37), and oscillate back toward 1 by about three kernel widths. The overshoot is what allows the covering inequality to hold while the outermost cells are heavily discounted. That the profile is confined to a few kernel widths is the dual-side statement that the method sees only a boundary layer of depth O(N^(3/4)) — the same scale as Cilleruelo's gap theorem, reached independently.

The two-sided optimum at L = 4 is organized differently: $\lambda \approx (0.15, 0.85)$ at every resolution from m = 32 to 256; the lighter kernel is strongly asymmetric, at total variation distance 0.67 from its reversal, while the heavier one is close to symmetric, at 0.19.

The optimum has flat directions. Along the curve of optimal certificates at prescribed a, the product a·b changes by less than 10^{-4} as a runs from 1.32 to 2.0, and at m = 256 the constants a and b of the two-sided optimum drift by about 10^{-3} over the last few hundred iterations while γ changes by 10^{-9}. Only γ is well determined; the individual a and b in Table A1 should not be read as converged.

The optimal dual multiplier

For fixed kernels the boundary problem is a quadratic program whose dual solution is a nonnegative measure μ on the shifts $u \geq 0$ — the multiplier of Part IV, now optimal rather than chosen. At the optimal certificates every covering constraint is active, and the optimality conditions read

$$(\mu \star R)(u) = \int_0^{\infty} R(u - v)\, d\mu(v) = 4 \text{ for all } u \geq 0, \quad (3.6)$$

where $R = \Sigma_k \lambda_k \rho_k \star \tilde{\rho}_k$ is the mixed autocorrelation of the kernels, $\tilde{\rho}(x) = \rho(-x)$, normalized so that the flat multiplier is 4·Leb. This is a Wiener–Hopf equation of the first kind; for two-sided certificates each end has its own multiplier and both satisfy (3.6) with the same R. The one-atom multiplier of Version 3 is its solution at u = 0 when the continuous part is flat: $4\cdot\tfrac{1}{2} + \beta\cdot R(0) = 4$ gives $\beta = 2/a$. Numerically, the optimal multiplier at the two-sided certificates converges, at rate 1/m, to

$$\mu_1 = 2\, \Sigma_{n\geq 0} U^{\star n}, \quad (3.7)$$

twice the renewal measure of the uniform distribution U on [0,1]: an atom of mass 2 at the origin, density $2e^u$ on [0,1), density $2(e^u - u\, e^{u-1})$ on [1,2), and so on, settling to the flat value 4 with oscillations that decay like $e^{-2.09u}$ (Figure 4a). At resolution m the atom has mass $2 - 2/m$ and the

density just after it is $2(1 + 1/m)$; the mean deviation from (3.7) on [0,1) is 0.010, 0.005 and 0.003 at m = 64, 128 and 256. The renewal identity $\mu_1 = 2\delta_0 + U \star \mu_1$ explains the flat value: the bulk density is the atom, 2, divided by the mean ½ of U. Feeding μ_1 back into (3.6) determines R on [0,1], and hence the kernel: the linear density 2x (Part V).

multiplier	bound on a·b	share of the gap
flat, 4·Leb	0.3748	0%
flat + atom at 0 (Version 3)	0.8580	94.00%
+ linear ramp on [0,1)	0.88809	99.84%
+ linear piece on [1,2)	0.88876	99.97%
exact (optimal multiplier)	0.88889	100%

Table 7. *Truncations of (3.7) evaluated at the optimal two-sided certificate (m = 256, L = 4), whose exact value is a·b = 0.888892, each family fitted by maximizing the dual value over its parameters. These are ceilings for what each family could prove on that certificate; Part IV shows that the full renewal multiplier, rescaled with the certificate, proves a·b ≥ 8/9 for every certificate.*

Before the proof was known, a three-parameter family $\mu = 4\cdot\mathrm{Leb} + \beta\delta_0 + \theta(2r(u/w) - 4)$, fitted per certificate under $\mu \geq 0$, was searched for its worst kernel over single kernels and two-kernel mixtures at m = 32, 64, 128 and 256. The worst certified a·b fell to 0.8890087, 0.8889188, 0.88889636 and 0.88889075, with drop ratios 4.006 and 4.000, extrapolating to 0.888888880. At the minimizers — mildly U-shaped kernels, not the linear one — the family selected $\beta = 2w$, $\theta = 1$ and $a\cdot w = 4/3$. That is exactly the multiplier of the proof in Part IV, which explains why the fitted family never certified less than 8/9.

Caveats

- The value of γ^* is no longer a numerical claim: the lower bound (Part IV) and the matching two-sided certificates (Part V) are proofs, and every certificate value in Table 5 is exact-rational verified or an exact quadratic-program value. What remains numerical is the value of the best symmetric certificate at each fixed resolution and kernel count; the symmetric floor itself equals $2\sqrt{2}/3$ (Appendix D).
- The optimized certificates have flat directions; only γ is converged.
- Only the implication to h(N) depends on the smoothing inequality; nothing in Parts III–V about the framework itself does.

Part IV · The lower bound $\gamma \geq 2\sqrt{2}/3$

This part proves, with no numerics in the argument, that the infimum located in Part III is at least $2\sqrt{2}/3$. Version 3 proved the weaker bound $\sqrt{((\pi^2+16)/32)} = 0.8991$ with a one-atom multiplier; that argument survives here as a stepping stone, and the gap it left is closed by replacing the atom with the full renewal measure (3.7), rescaled with the certificate.

Theorem 4.1 (lower bound). Every certificate — discrete or continuum, symmetric or two-sided, with any number of kernels — satisfies $a \cdot b \geq 8/9$, that is $\gamma = \sqrt{(ab)} \geq 2\sqrt{2}/3 = 0.94280904\ldots$. Consequently $\gamma^*_{sym} \geq \gamma^* \geq 2\sqrt{2}/3$, and no certificate of the framework proves $h(N) \leq N^{1/2} + \gamma \cdot N^{1/4} + O(1)$ with $\gamma < 2\sqrt{2}/3$.

By the continuum reduction of Part III it suffices to prove the theorem for two-sided continuum certificates.

A general dual bound

For a continuum certificate write $R = \Sigma_k \lambda_k \rho_k \star \tilde{\rho}_k$ for the mixed autocorrelation. R is even, nonnegative, supported in $[-1, 1]$, with $\int R = 1$ and $R(0) = a$. Put

$$G = \int |s| R(s)\, ds, \quad T(v) = \int_v^{\infty} R(s)\, ds \ (v \geq 0). \tag{4.1}$$

A multiplier is a nonnegative measure on $[0,\infty)$ of the form $\mu = 4 \cdot \mathrm{Leb} + \nu$, with ν a finite signed measure.

Lemma 4.2 (dual bound). For every continuum certificate and every multiplier $\mu = 4 \cdot \mathrm{Leb} + \nu$,

$$b \geq G + \int T\, d\nu - \tfrac{1}{8} \iint R(u - v)\, \nu(du)\, \nu(dv). \tag{4.2}$$

Equality holds if both covering conditions are equalities on the support of μ and $\omega_k = \tfrac{1}{4}(\mu \star \rho_k)$, $\omega'_k = \tfrac{1}{4}(\mu \star \bar{\rho}_k)$ for every k.

Proof. Write $e_k = \omega_k - 1$, so that the first end contributes $\Sigma_k \lambda_k \int (2e_k + e_k^2)$ to $b - 1$ and its covering condition reads $\Sigma_k \lambda_k \int \rho_k(x)\, e_k(u+x)\, dx \geq 0$. For fixed kernels the minimization of b is a convex quadratic program, so integrating this constraint against $\tfrac{1}{2}\mu \geq 0$ and subtracting gives a valid lower bound by weak duality. By Fubini the result is a pointwise problem in $e_k(t)$, minimized at $e_k = \tfrac{1}{4}(\mu \star \rho_k) - 1$ with value $-\Sigma_k \lambda_k \int_0^{\infty} (\tfrac{1}{4}(\mu \star \rho_k) - 1)^2$. Since $4 \cdot \mathrm{Leb} \star \rho_k = 4C_k$, with C_k the distribution function of ρ_k, the integrand is $(\tfrac{1}{4}(\nu \star \rho_k) - (1 - C_k))^2$. Expanding: $\int (\nu \star \rho_k)^2 = \iint r_k(u - v)\, \nu(du)\nu(dv)$ with $r_k = \rho_k \star \tilde{\rho}_k$; $\int (\nu \star \rho_k)(1 - C_k) = \int P(Y_k - X_k > u)\, \nu(du)$ for X_k, Y_k independent with density ρ_k; and $\int (1 - C_k)^2$ is a constant. At the second end ρ_k is replaced by $\bar{\rho}_k$, which changes neither r_k nor $P(Y_k - X_k > u)$, because X_k and Y_k are exchangeable, and turns the constant into $\int C_k^2$. The two constants sum to $1 - 2\int C_k(1 - C_k) = 1 - E|X_k - Y_k|$. Summing over the ends and the kernels gives (4.2). ∎

Two consequences are worth recording. The right-hand side of (4.2) depends on the kernels only through R, so asymmetric kernels matter only through the autocorrelations they produce. In the discrete boundary program the same is true of the optimal b only in the limit of an infinite boundary box. Version 5 stated that reversing any kernel of a two-sided certificate changes b by less than 10^{-13}; that is wrong at finite depth. Reversing every kernel only swaps the two ends and cannot change b, but reversing one kernel of the two-sided two-kernel certificates at $L = 4$ changes b by 6.1×10^{-7} at $m = 32$ and 5.1×10^{-7} at $m = 64$. Decompositions with the same R differ by $O(e^{-\kappa L})$: at $n = 32$ the linear kernel has $b = 0.667099227611$ at $L = 4$ and 0.666992220045 at $L = 8$, while the symmetric family of Appendix D, with

the same autocorrelation, has 0.666992210067 and 0.666992187500. The linear kernel's truncation loss is first order in e^(−κL) and the symmetric family's second order; this is the 10^{-4} that optimized mixtures recover at L = 4 (Part III). And the one-atom multiplier recovers Version 3's bound, as follows.

The one-atom bound and the cosine-arch inequality

With $\nu = \beta\delta_0$ one has $\int T\, d\nu = \beta T(0) = \beta/2$ — this is the kernel-independent cross term of Version 3, $\int_0^\infty (1 - C(t))\, \rho(t)\, dt = \int_0^1 (1 - C)\, dC = \tfrac{1}{2}$ — and $\tfrac{1}{8}\iint R\, \nu\, \nu = \beta^2 a/8$. So (4.2) gives $b \geq G + \beta/2 - \beta^2 a/8$, which is maximized at $\beta = 2/a$:

$$b \geq G + 1/(2a), \quad a \cdot b \geq a \cdot G + \tfrac{1}{2}. \tag{4.3}$$

The mixture collapses by Cauchy–Schwarz in the weights: with $F_k = \|\rho_k\|_2^2$ and $G_k = E|X_k - Y_k|$, $a \cdot G = (\Sigma_k \lambda_k F_k)(\Sigma_k \lambda_k G_k) \geq (\Sigma_k \lambda_k \sqrt{(F_k G_k)})^2 \geq \min_k F_k G_k$. The single-kernel product obeys a sharp inequality.

Lemma 4.3 (cosine-arch inequality). For every probability density ρ on $\mathbb{R}$ with $\rho \in L^2$, $Q(\rho) := \|\rho\|_2^2 \cdot E|X - Y| \geq \pi^2/32 = 0.3084251\ldots$, with equality precisely for the cosine arches $\rho_W(x) = (\pi/4W)\cdot\cos(\pi x/2W)$ on $[-W, W]$, $W > 0$, and their translates.

Proof. Let $x(u)$ be the quantile function of ρ and $g = x'$, so that $\rho(x(u))\cdot g(u) = 1$ for almost every u. The substitution $u = F(x)$ gives $\|\rho\|_2^2 = \int_0^1 du/g(u)$ and $E|X - Y| = 2\int_0^1 u(1-u)\, dx(u) \geq 2\int_0^1 u(1-u)\, g(u)\, du$, the singular part of dx — supported on gaps in the support of ρ — contributing nonnegatively. By Cauchy–Schwarz, $(\int_0^1 du/g)(\int_0^1 u(1-u)\, g\, du) \geq (\int_0^1 \sqrt{(u(1-u))}\, du)^2 = (\pi/8)^2$, the integral being the area of a half-disc of radius ½; hence $Q(\rho) \geq 2(\pi/8)^2 = \pi^2/32$. Equality forces the singular part to vanish and $g(u) = C/\sqrt{(u(1-u))}$, so $x(u) = c + C\cdot\arcsin(2u - 1)$: the cosine arch with $W = \pi C/2$, translated to c. Conversely $\|\rho_W\|_2^2 = \pi^2/(16W)$ and $E|X - Y| = W/2$. ∎

The cosine arch is also the stationary point of Q, but after normalizing $E|X - Y| = 1$ the constraint set $\{E|X - Y| \leq 1\}$ is a sublevel set of a concave functional and is not convex, so stationarity alone does not prove minimality; the quantile argument avoids the issue. The inequality is elementary and no claim of novelty is made for it.

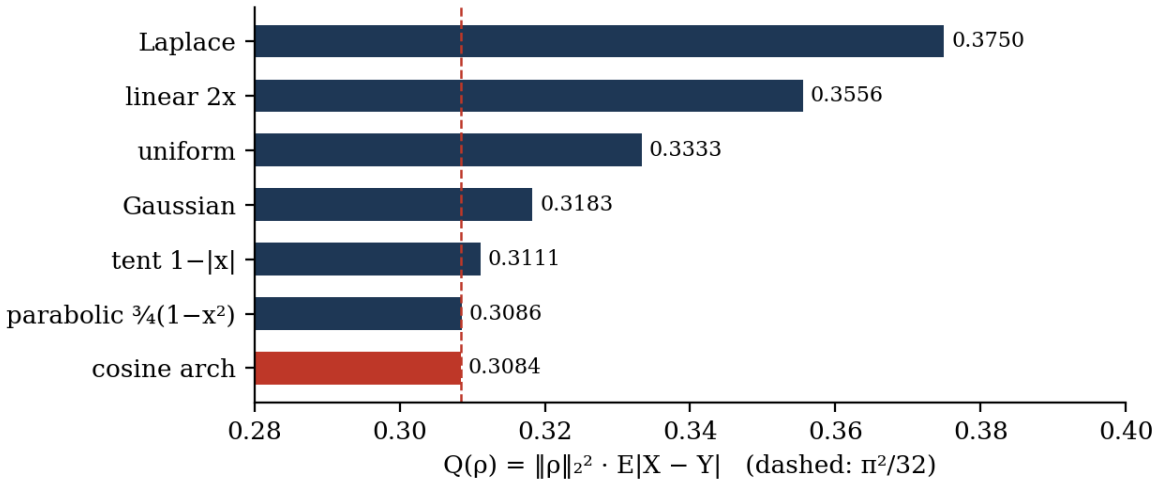


Figure 5. *The scale-invariant functional Q of Lemma 4.3 on standard densities. The cosine arch attains $\pi^2/32$; the parabolic density comes within 1.5×10^{-4} of it, and the others lie higher.*

Combining, every certificate satisfies $a \cdot b \geq \pi^2/32 + \tfrac{1}{2} = (\pi^2 + 16)/32$, that is $\gamma \geq 0.89912465\ldots$ — the theorem of Version 3, which by Lemma 4.2 now holds for two-sided certificates as well. Each inequality in the chain is sharp on its own (the cosine arch attains Lemma 4.3, and a single kernel makes the mixture step an equality), so 0.8991 is the best that the one-atom family can give, and the remaining gap of 0.0437 in γ lies entirely in the choice of multiplier. Table 8 shows how each multiplier family performs on reference kernels.

kernel	a = F	F·G (flat)	F·G + ½ (one atom)	renewal μ_w, w = 4/(3a)	F·B (exact)
uniform on [0,1]	1	$1/3 \approx 0.333$	$5/6 \approx 0.833$	$8/9 \approx 0.889$	1.000
cosine arch	$\pi^2/8 \approx 1.234$	$\pi^2/32 \approx 0.308$	$(\pi^2+16)/32 \approx 0.808$	$8/9 \approx 0.889$	0.982
linear 2x	4/3	$16/45 \approx 0.356$	$77/90 \approx 0.856$	$8/9 \approx 0.889$	$8/9 \approx 0.889$
V3 certificate, kernel 0 (m = 512)	1.708	0.369	0.869	0.894	0.922
V3 certificate, kernel 1 (m = 512)	1.130	0.333	0.833	$8/9 \approx 0.889$	0.959

Table 8. *Lower bounds on a·b for single kernels from successively richer multipliers, against the exact inner value F·B, where B is the minimum of b over boundary profiles for that kernel. By (4.3) the one-atom column is exactly the flat column plus ½. The renewal column is exactly 8/9 whenever $a \leq 4/3$ (see the remark after Theorem 4.7): the renewal multiplier cannot distinguish kernels of equal energy, and it is tight only for the linear kernel. All entries are closed forms except the cosine arch's F·B (0.9820 on grids of 64, 128 and 256 bins) and the certificate rows, which are evaluated directly from the kernels in cert_m512_kernels.txt; kernel 1 has $a \leq 4/3$, so its renewal entry is exactly 8/9. For the uniform kernel B = 1 exactly, since summing the covering condition over integer shifts gives $\int_0^\infty e \geq 0$ at each end.*

The renewal density of the uniform distribution

Let U be the uniform density on [0,1) and $M = \Sigma_{n\geq 0} U^{\star n}$ its renewal measure: an atom of mass 1 at the origin plus a density r. Put $g = r - 2$ and $G_1(t) = \int_0^t g$.

Lemma 4.4. (a) $r(x) = e^x$ on [0,1), $r(x) = e^x - x\,e^{(x-1)}$ on [1,2), and $r(x) = \int_{x-1}^x r(t)\,dt$ for $x \geq 1$. (b) g decays exponentially: $|g(x)| \leq C_0 e^{(-\kappa x)}$ for some $C_0, \kappa > 0$; in particular $g \in L^1 \cap L^2$. (c) $G_1(t) \geq -1$ for every $t \geq 0$, and $\int_0^\infty g = -1/3$.

Proof. (a) is the renewal equation $r = U + U \star r$, solved on [0,1) and [1,2) by the method of steps. For (b), the Laplace transform of g is $s\cdot\hat{q}(s)/(s - 1 + e^{(-s)})$ with $q(x) = (2x-1)\cdot\mathbb{1}_{[0,1)}(x)$. It is analytic in $\mathrm{Re}\, s > -2.0888$, because the non-zero roots of $s = 1 - e^{(-s)}$ all have real part at most -2.0888 and the double zero of the denominator at $s = 0$ is cancelled by the numerator, since $\hat{q}(0) = \int q = 0$; exponential decay then follows from standard renewal theory for distributions with a density and finite exponential moments (Stone 1965; Feller, Ch. XI). For (c), $\int_0^t r$ is the renewal function H(t), and Wald's identity gives $H(t) + 1 \geq t/E[U] = 2t$. Finally $\int_0^\infty g$ is the Laplace transform at the origin, where $\hat{q}(s) = -s/6 + O(s^2)$ and $s - 1 + e^{(-s)} = s^2/2 + O(s^3)$. ∎

Lemma 4.5 (autocorrelation of the renewal density). Let $A(t) = \int_0^\infty g(x)\,g(x+t)\,dx$. Then $A(t) = e^t - 2t - 2/3$ for $0 \leq t \leq 1$. In particular $\int_0^\infty (r(x) - 2)^2\,dx = 1/3$.

Proof. By Lemma 4.4(a), $g = U \star g + q$, so in the sense of distributions on $(0,\infty)$, $g' = g - g(\cdot - 1)\mathbb{1}_{\{x\geq 1\}} + 2\cdot\mathbb{1}_{(0,1)} - \delta_1$, where δ_1 records the jump of g by -1 at $x = 1$. For $0 < t < 1$, differentiating A under the integral and splitting at $x = 1 - t$, where $g(x+t)$ jumps, gives the functional equation

$$A'(t) = A(t) - A(1 - t) + e^{(1-t)} + 4t - 4. \quad (4.4)$$

Put $D(t) = A(t) - A(1 - t)$. Applying (4.4) at t and at $1 - t$ gives $D'(t) = e^t + e^{(1-t)} - 4$ with $D(\tfrac{1}{2}) = 0$, so $D(t) = e^t - e^{(1-t)} - 4t + 2$; then (4.4) reads $A'(t) = e^t - 2$, whence $A(t) = e^t - 2t + C$ on [0,1]. For $t > 1$

the same differentiation gives the delay equation $A'(t) = A(t) - A(t-1)$, for whose solutions $\Phi(t) = A(t) - \int_{t-1}^{t} A$ and $\Psi(t) = \int_{t-1}^{t} (s - t + 1)A(s)\,ds$ satisfy $\Phi' = 0$ and $\Psi' = \Phi$ on $(1,\infty)$. Since $g \in L^2$, $A(t) \to 0$, so $\Phi \equiv 0$ and then $\Psi \equiv 0$. Now $\Psi(1) = \int_0^1 s\,A(s)\,ds = 1/3 + C/2$, so $C = -2/3$, and $\int g^2 = A(0) = 1/3$. ∎

Lemma 4.6 (weight identity). Let $\nu_1 = 2M - 4\cdot\mathrm{Leb} = 2\delta_0 + 2g$, and for $s > 0$ put

$$h(s) = 2s + \nu_1([0, s]) - \tfrac{1}{4}\int \nu_1(du)\,\nu_1(d(u+s)). \quad (4.5)$$

Then $h(s) = 8/3$ for $0 < s < 1$, and $h(s) \geq 8/3$ for $s \geq 1$.

Proof. Here $\nu_1([0, s]) = 2 + 2G_1(s)$, and for $s > 0$ the double integral is $4g(s) + 4A(s)$, an atom–density term plus a density–density term, so

$$h(s) = 2s + 2 + 2G_1(s) - g(s) - A(s). \quad (4.6)$$

For $0 < s < 1$, Lemmas 4.4(a) and 4.5 give $g(s) = e^s - 2$, $G_1(s) = e^s - 1 - 2s$ and $A(s) = e^s - 2s - 2/3$; substituting, every term cancels except $8/3$. For $s \geq 1$, Lemma 4.4(a) gives $g(s) = G_1(s) - G_1(s-1)$, so $h(s) = 2s + 2 + G_1(s) + G_1(s-1) - A(s)$, and $A(s) \leq A(0) = 1/3$ by Cauchy–Schwarz. If $s \geq 3/2$, Lemma 4.4(c) gives $h(s) \geq 2s - 1/3 \geq 8/3$. If $1 \leq s < 3/2$, then $g \geq e - 3$ on $[1, 3/2]$, since r increases there from $r(1+) = e - 1$, so $G_1(s) \geq (3/2)(e - 3)$; and G_1 is decreasing on $[0, \ln 2] \supset [0, \tfrac{1}{2}]$, so $G_1(s-1) \geq e^{1/2} - 2$. Together, $h(s) \geq 4 + (3/2)(e - 3) + e^{1/2} - 2 - 1/3 = 2.8928\ldots > 8/3$. ∎

The weight jumps at $s = 1$ to $h(1+) = 11/3$ and increases thereafter (Figure 4c).

Proof of the lower bound

For $w > 0$ let μ_w be the renewal multiplier rescaled to width w,

$$\mu_w = 2w \cdot \Sigma_{n\geq 0} U_w^{\star n}, \quad \mu_w(du) = 2w\cdot\delta_0 + 2r(u/w)\,du, \quad (4.7)$$

with U_w the uniform distribution on $[0, w]$: an atom of mass $2w$, density $2e^{u/w}$ on $[0, w)$, settling to 4. By Lemma 4.4(b), $\nu_w = \mu_w - 4\cdot\mathrm{Leb}$ is a finite signed measure.

Theorem 4.7. Every continuum certificate satisfies $b \geq 4w/3 - (w^2/2)\cdot a$ for every $w > 0$. Taking $w = 4/(3a)$ gives $b \geq 8/(9a)$, that is $a\cdot b \geq 8/9$.

Proof. Apply Lemma 4.2 with $\mu = \mu_w$. Since R is even, $G = \int_0^\infty 2s\,R(s)\,ds$, $\int T\,d\nu_w = \int_0^\infty R(s)\,\nu_w([0, s])\,ds$ by Fubini, and $\iint R(u - v)\,\nu_w(du)\nu_w(dv) = (2w)^2 R(0) + 2\int_0^\infty R(s)\,\int \nu_w(du)\nu_w(d(u+s))\,ds$. The right-hand side of (4.2) is therefore

$$\int_0^\infty R(s)\,[\,2s + \nu_w([0, s]) - \tfrac{1}{4}\int \nu_w(du)\nu_w(d(u+s))\,]\,ds - (w^2/2)\cdot R(0).$$

By scaling, the bracket equals $w\cdot h(s/w)$, which is at least $8w/3$ for every $s > 0$ by Lemma 4.6. Since $R \geq 0$ and $\int_0^\infty R = \tfrac{1}{2}$, the bound is at least $(8w/3)\cdot\tfrac{1}{2} - (w^2/2)\cdot a$. ∎

Theorem 4.1 follows by the continuum reduction of Part III, and the hypothesis $b > 0$ of the smoothing inequality is automatic, since $b \geq 8/(9a) > 0$.

Remarks. Three features of the argument stand out. First, the multiplier depends on the certificate only through a, just as the atom $2/a$ of the one-atom bound did. This is how the proof escapes the obstruction noted in the 25 September manuscript: for a fixed multiplier the bound on $a\cdot b$ is a product of two positive affine functions of R, hence minimized at a single kernel, and no fixed multiplier certifies $8/9$ on every kernel. Second, symmetry is never used, and mixtures are handled through R with no reduction to single kernels. Third, when $a \leq 4/3$ the width $w = 4/(3a)$ is at least 1, so the kernels see only the first

window of μ_w, where the weight is constant; (4.2) with μ_w then equals $4w/3 - (w^2/2)a$ exactly, for every kernel. The renewal multiplier cannot distinguish kernels of equal energy.

Independent check. Solving the renewal equation on a grid of step 10^{-4} reproduces $\int(r-2)^2 = 1/3$ to 2×10^{-5} and $\int(r-2) = -1/3$ to 5×10^{-5}, the formula of Lemma 4.5 to 4×10^{-5} on [0,1], and $h(s) = 8/3$ on (0,1) to 5×10^{-5}; on [1, 12] the smallest value of h is 11/3, at $s = 1+$ (Appendix A.3).

Part V · Sharpness: the extremal certificate and a new constant

The lower bound of Part IV is attained. The certificate that attains it was found by solving (3.6) with the numerically observed multiplier μ_1. By the renewal identity, (3.6) for $u \geq 1$ follows from its restriction to [0,1], where it becomes a linear integral equation for R whose solution is

$$R^*(s) = (2/3)(1 - |s|)^2(2 + |s|),\ |s| \leq 1, \quad (5.1)$$

the autocorrelation of the single linear density $\rho(x) = 2x$ on [0,1]. That density is asymmetric, which is why symmetric certificates approach it only through mixtures.

The extremal continuum certificate

Theorem 5.1. Let $\rho(x) = 2x$ on [0,1], and let r be the renewal density of U[0,1]. At the end whose covering condition involves the reflected kernel $2(1 - x)$, take $\omega' \equiv 1$. At the other end take $\omega = \tfrac{1}{4}(\mu_1 \star \rho)$, that is

$$\omega(t) = e^t - 1\ (0 \leq t < 1),\ \omega(t) = \int_0^1 x \cdot r(t - x)\,dx\ (t \geq 1). \quad (5.2)$$

This is a continuum certificate with $a = 4/3$ and $b = 2/3$, so $a \cdot b = 8/9$ and $\gamma = 2\sqrt{2}/3$.

Proof. A direct computation on $0 \leq u \leq 1$, using $r(v) = e^v$ on [0,1) and $r(v) = e^v - v\,e^{(v-1)}$ on [1,2), gives $(\mu_1 \star R^*)(u) = 4$. The renewal identity $\mu_1 = 2\delta_0 + U \star \mu_1$ then gives $(\mu_1 \star R^*)(u) = \int_{u-1}^{u} (\mu_1 \star R^*)$ for $u \geq 1$, since R^* vanishes there, so $\mu_1 \star R^* \equiv 4$ on $[0,\infty)$. As $\int \rho(x)\omega(u+x)\,dx = \tfrac{1}{4}(\mu_1 \star R^*)(u)$, the covering condition holds with equality everywhere at that end. At the other end $\tfrac{1}{4}(\mu_1 \star \tilde{\rho}) \equiv 1$ — for $0 \leq t < 1$ it is $(1 - t) + \int_0^t e^v\,(1 - t + v)\,dv = 1$ — so $\omega' \equiv 1$ is both feasible and of the form required in Lemma 4.2, with equality in its covering condition. The equality conditions of Lemma 4.2 therefore hold with $\mu = \mu_1$, and since R^* lives inside the first window, the dual value is exactly $4/3 - \tfrac{1}{2} \cdot (4/3) = 2/3$, as in Theorem 4.7 with $w = 1 = 4/(3a)$. Finally $\omega - 1$ decays exponentially by Lemma 4.4(b). ∎

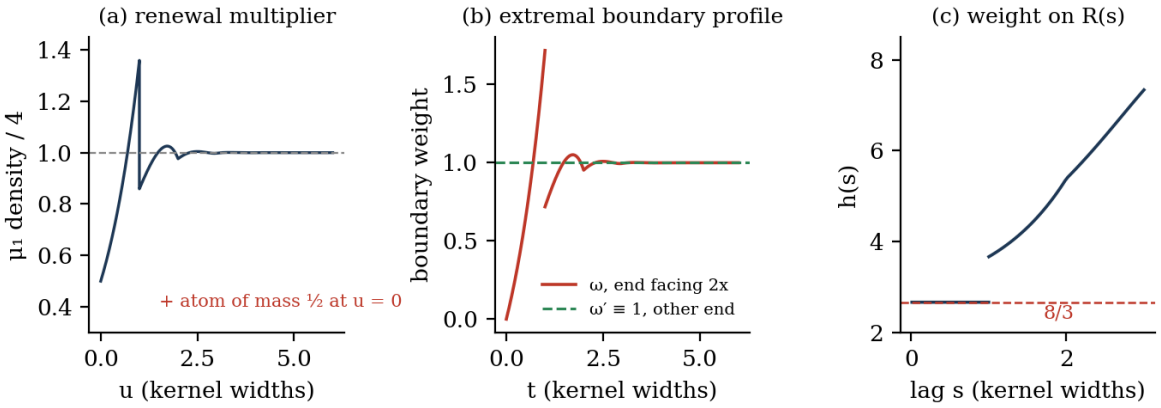


Figure 4. *(a) The renewal multiplier μ_1 of (3.7), density divided by the flat value 4, plus an atom of mass ½ at the origin in the same units; the optimal multipliers of the numerically optimal certificates converge to it at rate 1/m. (b) The boundary profiles of the extremal certificate of Theorem 5.1: $e^t - 1$ across the first kernel width at the end facing 2x, a jump to $e - 2$ at $t = 1$, and decaying oscillations; $\omega' \equiv 1$ at the other end. (c) The weight h(s) of Lemma 4.6 that the multiplier places on the autocorrelation: exactly 8/3 on the first window, 11/3 and rising beyond it.*

The boundary profile discounts the outermost cells completely, as the optimized symmetric certificates did approximately. Kernel, profile and multiplier are tied together — 2x is the density of the maximum of

two independent uniform variables — but that is an observation, not an explanation. **Independent check:** numerically $\int_0^\infty(\omega^2-1)=-0.33351$ against $-1/3$, and the covering integral equals 1 to within 10^{-4} at every shift tested (Appendix A.3).

Discretization

Proposition 5.2. For every $m \geq 2$ there is a two-sided certificate with one kernel at resolution m and boundary depth $L = \max(2, \lceil (2/\kappa)\log m \rceil)$, with $a = 4/3 - 1/(3m^2)$ and $a \cdot b \leq 8/9 + C(\log m)/m^2$, where C depends only on the constants C_0, κ of Lemma 4.4(b). Hence $\gamma^* = 2\sqrt{2}/3$.

The proof discretizes (5.2) by cell averages and repairs the covering deficits with a uniform increment $\eta = 4/m^2 + \sup_{t \geq L-1}|\omega(t) - 1|$ on the boundary vector. At interior shifts the deficit is at most $4/m^2$, since the kernel and the profile deviate from their cell averages by at most $1/m$ and $4/m$. The delicate shifts are the last ones, where only a few kernel cells fall inside the box — at the very last, only the first, of mass $1/m^2$. There the deficit carries the same small in-box mass as the effect of the increment, because every in-box cell sits at depth at least $L - 1$, where the profile is exponentially close to 1; so an increment of order m^{-2} suffices. The factor $\log m$ comes only from spreading η over the whole box; a repair localized to the interior deficits would give $O(m^{-2})$, matching the exactly optimal certificates below. Appendix D uses a repair that decays with depth, which is possible because the covering defect of its kernels decays like $e^{-\kappa u}$; it gives the rate $O(1/m^2)$ for the symmetric family. The same device should remove the factor $\log m$ here, but that has not been written out. Numerically the construction is feasible at m = 32, 64 and 128, with interior deficits at most 1.8×10^{-5} at m = 128 against the bound $4/m^2 = 2.4\times10^{-4}$.

Exactly optimal discretizations and the certified constant

With the exactly optimal boundary vectors in place of cell averages, the discrete values behave as cleanly as one could hope. The kernel $p_i = (2i + 1)/m^2$ has $a = 4/3 - 1/(3m^2)$, and with L = 8 the boundary program gives $b = 2/3 + 1/(3m^2)$ to within 3×10^{-8} for $m \leq 512$, so that $a \cdot b = 8/9 + 2/(9m^2) - 1/(9m^4)$ to that accuracy. In an infinite box $b = 2/3 + 1/(3m^2)$ exactly, according to the discrete renewal multiplier described after Theorem 5.4, so the 3×10^{-8} is truncation at L = 8, not discretization. The reversed end costs nothing: its optimal boundary vector is identically 1.

m	γ	$b - 2/3$	$1/(3m^2)$
32	0.9429240903	3.2555×10^{-4}	3.2552×10^{-4}
64	0.9428378328	8.1413×10^{-5}	8.1380×10^{-5}
128	0.9428162572	2.0377×10^{-5}	2.0345×10^{-5}
256	0.9428108625	5.1183×10^{-6}	5.0863×10^{-6}
512	0.9428095137	1.3035×10^{-6}	1.2716×10^{-6}

Table 9. *The discretized linear kernel at L = 8 with exactly optimal boundary vectors at both ends. The m = 32, 64 and 128 rows were reproduced independently for this report with a separate quadratic-program solver, to 10^{-10} in γ (Appendix A.3). The certificates at m = 256 and 512, rationalized and verified in exact arithmetic on 4098 and 8194 covering inequalities, give $\gamma < 0.942810871$ and $\gamma < 0.942809522$.*

Theorem 5.3 (computational). The verified two-sided certificate with the linear kernel at m = 512, L = 8 has a = 1.3333320618…, b = 0.6666679802… and $\gamma = 0.9428095208\ldots < 0.942809522$. Consequently, by Proposition 3.1,

$$h(N) \le N^{(1/2)} + 0.9428096 \cdot N^{(1/4)} + O(1), \quad (5.3)$$

with the O(1) term asymptotically ab/2 ≈ 0.444 by (3.4). This is 6.8×10^{-4} below Hou and Zhao's 0.9434926 and 4.8×10^{-7} above the proved limit, so no certificate in the discretized framework can improve it by more than that. It is no longer the smallest constant claimed: Chen's argument, below, works with the continuous kernel directly and reaches $2\sqrt{2}/3$ itself. Nor, since 13 October, is it the smallest verified constant: the symmetric certificates of Table 10 give $\gamma < 0.942809493$ using only the original framework (Theorem 5.4).

Two structural facts

b depends only on the autocorrelation, in the infinite box. In the limit of an infinite boundary box (see the correction after Lemma 4.2), the optimal b of a two-sided certificate depends on the kernels and mixing weights only through $R = \Sigma_k \lambda_k p_k \star \tilde{p}_k$, as does $a = R(0)$. At a single end the boundary cost also depends on the kernels' means, and these contributions cancel between the two ends. The two-sided problem is therefore to minimize $R(0)\cdot b(R)$ over the closed convex hull of autocorrelations of probability densities on [0,1], with b a convex function of R, being a supremum of affine functionals. In this language Lemma 4.3 says $R(0)\cdot\int|s|R(s)\,ds \ge \pi^2/32$, Theorem 4.7 says $R(0)\cdot b(R) \ge 8/9$, and Theorem 5.1 says the minimum 8/9 is attained at R*.

The symmetric framework. Symmetric certificates cannot contain the extremal kernel, but they can contain its autocorrelation. Version 5 left open whether they reach $2\sqrt{2}/3$, with the verified symmetric certificates confined to $[2\sqrt{2}/3, 0.942810893]$. The next section closes the question.

Symmetric certificates reach the limit

R* is convex on (0, 1] with $R^*(1) = R^{*\prime}(1) = 0$, so by Pólya's representation it is a mixture of triangles:

$$R^*(s) = \int_0^1 4h^3 \cdot T_h(s)\,dh, \quad T_h(s) = (h - |s|)_+/h^2. \quad (5.4)$$

T_h is the autocorrelation of the uniform density on an interval of length h, a symmetric kernel. In probabilistic terms, if X and Y are independent with density 2x, then $X - Y$ has the law of $H\cdot(V - V')$, with V, V′ uniform and H the maximum of four independent uniforms. Generating both boundaries from one multiplier, $\omega_k = \tfrac{1}{4}(\mu \star \rho_k)$, makes the covering sums and the cost depend on the kernels only through R (Lemma D.5), so the continuum mixture of centred uniform kernels with weight $4h^3\,dh$, with $\mu = \mu_1$, is a symmetric continuum certificate with $a = 4/3$ and $b = 2/3$.

The discrete form is explicit. At resolution $m = 2n$ take the n kernels $p^{(l)}$ uniform on the 2l central cells, $l = 1, \ldots, n$, with weights $\lambda_l = 4l^3/n^4$ for $l < n$ and $\lambda_n = (2n - 1)/n^2$. Their mixed autocorrelation equals the autocorrelation of the linear kernel $p_i = (2i + 1)/n^2$ at resolution n exactly, and $a = 4/3 - 1/(3n^2)$ exactly (Lemma D.3; checked in rational arithmetic for $n \le 80$). With exactly optimal boundary vectors at $L = 8$, $b = 2/3 + 1/(3n^2)$ to about 10^{-15} for $n \le 256$, so $\gamma^2 = 8/9 + 2/(9n^2) - 1/(9n^4)$ up to an $e^{(-2\kappa L)}$ truncation term.

Theorem D.1 and Corollary D.2 (Appendix D). *For every n there are boundary vectors making these kernels a symmetric certificate of the original Hou–Zhao framework with $a\cdot b \le 8/9 + C/n^2$, C depending only on the renewal constants C_0 and κ. Hence $\gamma^*_{sym} = \gamma^* = 2\sqrt{2}/3$, approached at rate $O(1/m^2)$.*

These certificates satisfy $b \ge 8/(9a) > 0$ by Theorem 4.7, so Hou and Zhao's Theorem 2.1 applies to them directly; no two-sided extension is needed. Members of the family were verified in exact arithmetic: kernels rounded to integers over 10^{12}, kept exactly symmetric, the boundary program re-solved for the

rounded kernels, boundary vectors rounded and repaired by a uniform increment, and every covering inequality checked over $\mathbb{Q}$ by three independent checkers, which agree to every printed digit. Corrupting one boundary entry makes the checkers fail.

m	L	K	covering inequalities	a	b	γ (exact)	certified
128	8	64	1025	1.333251953159	0.666748046890	0.942837809902	γ < 0.942837811
256	8	128	2049	1.333312988234	0.666687011774	0.942816234419	γ < 0.942816236
512	8	256	4097	1.333328247092	0.666671753640	0.942810840342	γ < 0.942810842
1024	6	512	6145	1.333332061750	0.666667938506	0.942809491335	γ < 0.942809493

Table 10. *Verified symmetric certificates from the family of Appendix D (13 October 2026). The repair increments were 1, 1, 23 and 11 ×10⁻¹². The m = 1024 certificate was checked by the project's check_lin_cert.py in four ranges of shifts, since one run exceeds the sandbox time limit. To ten places the certified bounds are 0.9428378101, 0.9428162346, 0.9428108405 and 0.9428094915. The certificate files (21 MB compressed) have not yet been deposited in the project (Appendix A.4).*

Theorem 5.4 (computational). *The verified symmetric certificate at m = 1024, L = 6 with 512 kernels has γ < 0.942809493. Consequently, by Hou and Zhao's Theorem 2.1,*

$$h(N) \le N^{1/2} + 0.9428095 \cdot N^{1/4} + O(1). \quad (5.5)$$

This is 4.5×10^{-7} above the proved limit and 2.9×10^{-8} below the linear-kernel certificate of Theorem 5.3, and, unlike (5.3), it does not depend on the two-sided smoothing inequality. Whether a bounded number of symmetric kernels reaches $2\sqrt{2}/3$ is open. A single symmetric kernel cannot reproduce R* exactly: the Fourier transform of R* is $\rho(z)\rho(-z)$, whose complex zeros (from $e^{it} = 1 + it$) are simple, so it is not the square of an entire function. Together with the rough single-kernel limit of about 0.94559 (Table 6), this suggests the one-kernel floor lies strictly above $2\sqrt{2}/3$; whether two kernels suffice is unknown.

A discrete renewal multiplier. Let U be the renewal sequence of the step law $f_j = 2/(2n-1)$ for $1 \le j \le n-1$, $f_n = 1/(2n-1)$, and $\mu_k = (2n^2/(2n-1)) \cdot U_k$. In an infinite box, the covering sum of the linear kernel at resolution n is identically 1 and the boundary at the reversed end is identically 1, which by the renewal equation follows from finitely many shifts; this was checked exactly in rationals for n = 2, …, 30, and $b = 2/3 + 1/(3n^2)$ to 25 digits at n = 2, 3, 5 and 8. It is a discrete form of Theorem 5.1, verified for small n only. It implies that the 3×10^{-8} excess in b at L = 8 in Table 9 is truncation, not discretization.

An independent proof at exactly $2\sqrt{2}/3$

On 2 October 2026 Haoyu Chen posted on the Erdős Problems forum a proof that $h(N) \le N^{1/2} + (2\sqrt{2}/3) \cdot N^{1/4} + 1$ for $N \ge 120^4$, with code and write-ups deposited on Zenodo. (The forum text reads "1204", which we take to be 120^4.) The argument is a weighted Erdős–Turán energy bound with a continuous weight: the ramp $h(t) = 2(1-t)$ on [0, 1], which is the linear kernel of Part V reflected, sampled at d/H with monotone Riemann-sum bounds. Working with the continuous kernel removes the discretization gap of 4.8×10^{-7} that separates Theorem 5.3 from the limit, and the explicit additive constant replaces the O(1) of (5.3). Chen also states an optimality result: for every even nonnegative kernel f with $\int f = 1$, the capacity $C_f(L)$ of his argument satisfies $\liminf (C_f(L) - L) \ge 8/(9f(0))$, with equality for the ramp.

The two results agree on everything they share — the constant, the extremal kernel and the value 8/9 — and reach it by different machinery. They differ in scope: Chen's barrier concerns a single fixed even kernel, whereas Theorem 4.1 covers mixtures of kernels and two-sided certificates in the Hou–Zhao framework. Chen's proof is unrefereed and we have not yet checked it line by line; the timestamps (our limit manuscript was committed to GitHub on 1 October, Chen's post is dated 2 October) indicate

independent work. A continuum version of Proposition 3.1 would give the same bound inside our own framework, with explicit constants; this is one of the open questions of Part XII.

Consequence for Erdős Problem 30

bound	γ	year
Erdős–Turán	1	1941
Lindström	1	1969
Balogh–Füredi–Roy	≈0.998	2021
O'Bryant	0.99703	2022
Carter–Hunter–O'Bryant	0.98183	2023
R. Stoltz (forum claim)	0.98173	2026
Carter, Georgiev, Gómez-Serrano, Hunter, O'Bryant, Tao, Wagner (announced)	0.97633	2026
the same group, a more efficient argument (tentative, unpublished)	≈0.947	2026
Hou–Zhao, arXiv v1	0.94601	2026
Hou–Zhao, arXiv v3 (8 kernels)	0.94349	2026
independent 11-kernel certificate	0.94301	2026
this work, Version 3 (symmetric, m = 512)	0.94282	2026
this work, Version 4 (linear kernel, m = 512, L = 8)	0.9428096	2026
this work, Version 6 (symmetric, m = 1024, 512 kernels)	0.9428095	2026
H. Chen (forum claim; + 1 for $N \geq 120^4$)	$2\sqrt{2}/3$	2026
limit of the method (Theorem 4.1, Proposition 5.2)	$2\sqrt{2}/3 = 0.9428090\ldots$	—

Table 11. *The second-order constant. Years are those of first public appearance. Hou and Zhao's version 3 certifies $\gamma_0 = 0.9434925907\ldots$; their version 1 (July 2026) gave 0.94601. The entire remaining room in the discretized framework below the Version 6 certificate is 4.5×10^{-7}; the Version 6 certificate uses only Hou and Zhao's Theorem 2.1. Forum claims are unrefereed: Stoltz's certificate was posted on 17 February 2026; the tentative ≈0.947 was reported by Tao in February 2026 and is cited by Hou and Zhao; Chen's bound is discussed above.*

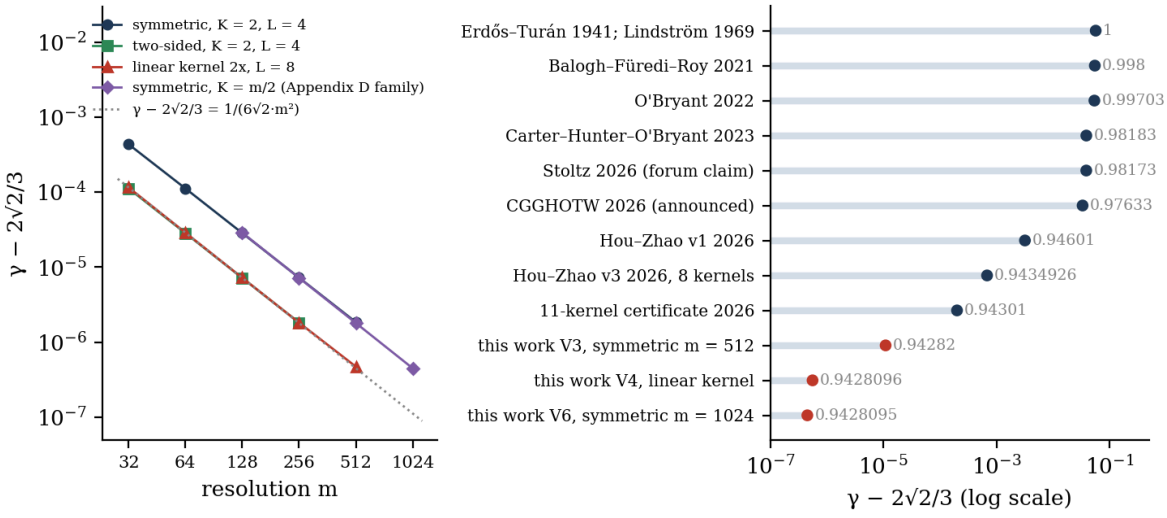


Figure 3. *Left: distance above $2\sqrt{2}/3$ of the verified values of the optimized certificates, the exact quadratic-program values of the linear kernel and the verified symmetric certificates of Appendix D (K = m/2), against the resolution m, with the reference line $1/(6\sqrt{2}\cdot m^2)$ that corresponds to $a\cdot b - 8/9 = 2/(9m^2)$. Right: the second-order constant through history, as distance above*

the proved limit on a logarithmic scale. Version 5 values for the regenerated two-sided certificates and the Version 6 symmetric certificate at m = 1024; Chen's claimed constant equals the limit and cannot be drawn on this scale.

What an improvement would have to retain. The framework reaches its limit because the argument discards two quantities: the variance term in Cilleruelo's identity for the number of representations, which is nonnegative and thrown away when the energy is bounded, and the slack in the bound $d(h) \leq 1$ on the multiplicity of small differences. Any argument that goes below $2\sqrt{2}/3$ must retain at least one of them; Part VI shows that retaining both at the level of pair and three-point statistics is not enough (Theorems 6.6 and 6.8). The qualitative conclusion, that arguments of this shape cannot reach $o(N^{1/4})$ let alone N^{ε}, already followed from any positive lower bound, including the $\pi/(4\sqrt{2})$ of the flat multiplier; what Version 4 adds is the exact location of the wall.

Part VI · Keeping the variance term: the pair-level relaxation

The natural question after Part V is whether the method has an extension that keeps what it discards, and what that extension's limit is. This part answers it at the first natural level: the extension is a semidefinite relaxation of the Sidon condition, and its second-order constant is again $2\sqrt{2}/3$. Version 6 proves the matching lower bound with an $O(1)$ error (Theorem 6.6), extends the barrier to three points (Theorem 6.8), and tests whether arithmetic constraints can break it. Throughout, $A \subset \{0,\dots,N-1\}$ is a Sidon set with k elements, a two-sided certificate at resolution (m, L, K) is fixed, and $H = mh$ with $N \geq 2LH$; K_r are the smoothing kernels, $S_r = \mathbb{1}_A \star K_r$ the smoothed set, Q_r the weight functions, C_r the autocorrelation of K_r, and $c(x)$ the covering sum at x, as in the proof of the smoothing inequality.

The identity behind the smoothing inequality

Proposition 6.1. Put $c(A) = \Sigma_{x\in A}\, c(x)$, $\Delta(A) = \Sigma_r\, \lambda_r\, \Sigma_{d\neq 0,\ d\notin A-A}\, C_r(d)$ and $V(A) = (\Sigma_r\, \lambda_r \|Q_r\|^2)(\Sigma_r\, \lambda_r \|S_r\|^2) - c(A)^2$. Then $c(A) \geq k$, $V(A) \geq 0$, $\Delta(A) \geq 0$, and

$$c(A)^2 + V(A) + (N - 1 + bH)\cdot\Delta(A) = (N - 1 + bH)\cdot(1 + (k-1)a/H). \tag{6.1}$$

The proof of the smoothing inequality discards exactly three quantities: the covering slack $c(A)^2 - k^2$, the Cauchy–Schwarz defect $V(A)$, and the difference slack $(N - 1 + bH)\Delta(A)$. For the flat kernel with $u = v \equiv 1$ ($a = b = 1$), $V(A)/(N + H - 1)$ is the variance of the window counts of A divided by H^2, and $H^2\Delta(A) = \Sigma_{0<|d|<H,\ d\notin A-A}\, (H - |d|)$ is the slack in $d(h) \leq 1$; in that case (6.1) is, up to normalization, Cilleruelo's identity.

The two discarded quantities are coupled: for fixed k and $c(A)$, (6.1) fixes $V(A) + (N - 1 + bH)\Delta(A)$, so a set that uses fewer small differences has a larger variance and conversely, and a lower bound for either alone is equivalent to an upper bound on k. Normalized as $\varphi(A) = V(A)\cdot H/((N - 1 + bH)\cdot k\cdot a)$, which is about 1 for Poisson-like fluctuations, (6.1) gives $\varphi(A) \leq 1 + t(bt - 2\gamma)/a + O(N^{-1/4})$ whenever $k \geq \sqrt{N} + \gamma N^{1/4}$ and $H = tN^{3/4}$. At the certificate's own scale $t = \sqrt{a/b}$ the right-hand side is $2(1 - \gamma/\sqrt{ab}) + O(N^{-1/4})$, so every Sidon set more than halfway to a certificate's bound is sub-Poissonian, $\varphi < 1$, at scale $\sqrt{a/b}\cdot N^{3/4}$. Measured with a discretized linear-kernel certificate, the optimal Golomb rulers with 12 to 20 marks have φ between 0.23 and 0.45, and most of them use every difference below H; the full-period sets of Bose and Ruzsa have φ between 1.1 and 1.5; a uniformly random subset of the same size and length gives 1.57.

The pair-level relaxation

Definition 6.2. For $N \geq 2$ let $\hat{h}_2(N)$ be the value of the semidefinite program that maximizes $\Sigma_x f(x)$ over $f \in \mathbb{R}^N$ and symmetric $\Gamma \in \mathbb{R}^{N\times N}$ subject to

(i) $\begin{bmatrix} 1 & f^{\mathsf T} \\ f & \Gamma \end{bmatrix} \succeq 0$; (ii) $\Gamma(x, x) = f(x)$; (iii) $0 \leq f \leq 1$, $\Gamma \geq 0$ entrywise;

(iv) $\Sigma_x \Gamma(x, x+d) \leq 1$ for $1 \leq d \leq N - 1$, with indices in $\{0,\dots,N-1\}$. (6.2)

For a Sidon set A the pair $f = \mathbb{1}_A$, $\Gamma = \mathbb{1}_A\mathbb{1}_A^{\mathsf T}$ is feasible with value k, so $h(N) \leq \hat{h}_2(N)$. Conditions (i)–(iii) are the level-one moment conditions of Lasserre and the pair conditions of Sherali and Adams for 0/1 variables; (iv) is the Sidon condition in expectation. The covariance $\Gamma - ff^{\mathsf T}$ is kept positive semidefinite rather than discarded and each difference budget is kept separately, so both quantities of Proposition 6.1 become linear functionals of (f, Γ), and the proof of the smoothing inequality becomes a dual solution.

Proposition 6.3. For every two-sided certificate and every H = mh with N ≥ 2LH, $\hat{h}_2(N)^2 \le (N - 1 + bH)\cdot(1 + (\hat{h}_2(N) - 1)a/H)$. Consequently (3.4) holds with h(N) replaced by $\hat{h}_2(N)$, and limsup_{N→∞} ($\hat{h}_2(N)$ − √N)/N^(1/4) ≤ 2√2/3.

Proof sketch. For feasible (f, Γ) with κ = Σf, summing the covering condition against f ≥ 0 and applying Cauchy–Schwarz in the weighted direct sum gives $\kappa^2 \le (\Sigma_r \lambda_r \|Q_r\|^2)(\Sigma_r \lambda_r f^T \mathscr{C}_r f)$, where $\mathscr{C}_r(x, y) = C_r(x - y)$ is a Gram matrix. By (i), $f^T \mathscr{C}_r f \le \mathrm{tr}(\mathscr{C}_r \Gamma) = C_r(0)\kappa + \Sigma_{d\neq 0} C_r(d) \Sigma_x \Gamma(x, x+d) \le C_r(0)\kappa + 1 - C_r(0)$, using (ii), $C_r \ge 0$ and (iv). The certificates of Proposition 5.2 give the limsup. ▌ Only (i), (ii), (iv) and f ≥ 0 are used.

The relaxation reaches 2√2/3

The matching lower bound comes from explicit feasible points whose shape is read off from the equality conditions of Lemma 4.2. Near each end, a set that makes the Cauchy–Schwarz step tight must have density following ¼μ₁ = ½δ₀ + ½r: an atom, then a density that starts at half the bulk value, rises to e/2 at one kernel width and settles. The key fact is that this profile is invisible to short-range pair counts.

Lemma 6.4. Let σ = ½δ₀ + ½r·Leb on [0,∞) and τ = σ − Leb, a finite signed measure with τ([0,∞)) = 1/3. For s > 0 put $\mathcal{B}(s) = 2\tau([0,\infty)) - \tau([0, s]) + \int \tau(du)\tau(d(u+s))$. Then $\mathcal{B}(s) = s/2 + (8/3 - h(s))/4$, with h as in (4.5); in particular $\mathcal{B}(s) = s/2$ for 0 < s < 1 and $\mathcal{B}(s) \le s/2$ for s ≥ 1.

Take a density on [0, $\mathscr{L}$], lengths in kernel widths, that equals σ at both ends. Its autocorrelation at a lag s ∈ (0,1) is $\mathscr{L} - s$ from the bulk plus $2\mathcal{B}(s) = s$ from the two ends, up to exponentially small terms — exactly $\mathscr{L}$. The ends supply exactly the pairs that the finite length removes; this is a second reading of the weight identity, Lemma 4.6.

The construction. Fix a bulk density δ, a scale H = t·N^(3/4), an atom fraction β ∈ (0, 1] and ℓ = ⌊1/δ⌋. The profile ψ follows δ·½r at each end, started at the centre of the atom, with bulk value δ; the atom ½δ₀ is spread uniformly over the first and last ℓ sites, with mass βδH/2 at each end; and f = ψ + atoms. The covariance is $\Gamma = ff^T + D(I - \delta\Phi_\ell)D + \mathrm{diag}(\varepsilon)$, where Φ_ℓ is the Toeplitz matrix with entries $(1 - |x - y|/\ell)_+$, $D = \mathrm{diag}(\tilde{\psi}/\sqrt{\delta})$ with $\tilde{\psi}$ a slightly reduced ψ, and ε ≥ 0 restores the diagonal. Since the Fejér symbol of Φ_ℓ takes values in [0, ℓ] ⊂ [0, 1/δ], the matrix $I - \delta\Phi_\ell$ is positive semidefinite and (i)–(iii) hold. The term $D(I - \delta\Phi_\ell)D$ is a covariance with a triangular hole: each bulk point gives up about one partner at lags below ℓ, which makes the bulk hyperuniform in the sense of Torquato and Stillinger.

Why the scale is √2·N^(3/4). Each atom contributes about $(\tfrac{1}{2}\delta H)^2$ ordered pairs at lags below ℓ, with the same triangular profile as the hole, so (iv) at a lag d < ℓ reads, to leading order, $\delta^2 N + (1 - d/\ell)\cdot[2(\tfrac{1}{2}\delta H)^2/\ell - \delta^2 N] \le 1$. The atoms refill the vacated lags exactly when $2(\tfrac{1}{2}\delta H)^2 = \ell \approx \sqrt{N}$, that is H = √2·N^(3/4) — the certificate's own optimal scale √(a/b)·N^(3/4) for a = 4/3, b = 2/3, and no other. At that scale $\delta^2 N \to 1$, and each end carries an excess b/2 = 1/3 in units of δH (½ from the atom and −1/6 from the continuous part), so Σf = δN + (2/3)δH + O(1) = √N + (2√2/3)N^(1/4) + O(1).

N	t	β	δ√N	Σ_x f(x)	(Σf − √N)/N^(1/4)	√N + (2√2/3)N^(1/4) − Σf
10^4	1.344	0.9936	1.00514	108.441	0.844	0.99
10^5	1.414	0.9698	1.00221	331.969	0.885	1.02
10^6	1.414	0.9837	1.00069	1028.794	0.911	1.02
10^7	1.414	0.9912	1.00021	3214.284	0.925	1.01

Table 12. *Explicit feasible points of (6.2). The shortfall from √N + (2√2/3)N^(1/4) in the last column is constant to within 0.04 over three decades; the normalized excess in the sixth column approaches 2√2/3 = 0.943 slowly because the shortfall is O(1). At N = 10^7, β was extrapolated from the smaller cases rather than optimized.*

Theorem 6.5 (computational). For N ∈ {10^4, 10^5, 10^6, 10^7} these points are feasible for (6.2), and $\hat{h}_2(N) \geq \sqrt{N} + (2\sqrt{2}/3)N^{1/4} - 1.03$. Conditions (i)–(iii) hold by construction, with the hypotheses checked on the computed vectors. Condition (iv) was evaluated in double precision at every d by fast Fourier transform; the largest difference sum is at most $1 - 1.3\times10^{-8}$, and lowering δ by a relative 10^{-7} leaves every sum at most $1 - 2\times10^{-7}$ at a cost of at most 3.3×10^{-4} in Σf. An explicitly formed feasible point at N = 3000 has a moment matrix with smallest eigenvalue 6.1×10^{-5}, and direct summation of (iv) agrees with the transform to 4×10^{-16}. This is a double-precision computation with an explicit safety margin, not an exact-arithmetic verification.

Theorem 6.6 (Conjecture 6.6 of Version 5). *$\hat{h}_2(N) = \sqrt{N} + (2\sqrt{2}/3)\cdot N^{1/4} + O(1)$. More precisely:*

(a) (lower half) there is N_0, depending only on the renewal constants C_0 and κ, such that for $N \geq N_0$

$$\hat{h}_2(N) \geq \sqrt{N} + (2\sqrt{2}/3)\cdot N^{1/4} - 2 - (8/3)D_+ - 2\sqrt{2}\cdot N^{-1/4} - \varepsilon(N),\ \varepsilon(N) = O(N^{1/4}\cdot e^{-\kappa N^{1/4}/2}),\quad (6.3)$$

where $D_+ = \int_0^\infty (1 - r(u)/2)_+\, du = 0.228168\ldots$, so the lower constant is $2 + (8/3)D_+ = 2.6084\ldots$;

(b) (upper half) with C_1 the constant of Proposition 5.2, $\hat{h}_2(N) \leq \sqrt{N} + (2\sqrt{2}/3)\cdot N^{1/4} + 4/9 + 3C_1/(32\sqrt{2}) + o(1)$;

(c) the lower bound (6.3) holds also for the larger relaxation $\hat{h}_2^+$ that adds the distinct-sums constraints $\Sigma_{x<y,\ x+y=s}\, \Gamma(x,y) + [s \text{ even}]\cdot f(s/2) \leq 1$ and the Sherali–Adams inequalities $f(x) + f(y) - 1 \leq \Gamma(x,y) \leq f(x)$, for $N \geq \max(N_0, 10^4)$.

The proof (Appendix B, written 14 October 2026) modifies the construction above in one way: the whole continuum measure, atoms and renewal profile together, is averaged over the same window of $\ell \approx \sqrt{N}$ sites. Averaging over a window convolves the pair function with the triangle $(\ell - |t|)_+/\ell^2$, which keeps the continuum pair function flat on (0, H) (Lemma 6.4) and turns the atoms' self-pairs into exactly the profile of the Fejér hole. Of the four errors listed in Version 5, the atom width and the short-lag atom–bulk pairs disappear, the reduction $\tilde{\psi} < \psi$ becomes the explicit constant D_+, and discretization becomes a lattice-count error of relative size $2/\ell$. The atom weight stays at β = 1 and H is lowered slightly instead, to the positive root of $H^2/(2\ell) = N - \ell - 4HD_+$. The upper half does not follow from Proposition 6.3 with one fixed certificate, which gives only the limsup; it follows by taking the certificates of Proposition 5.2 at resolution $m \approx N^{1/8}\cdot(\log N)^{1/2}$, so that the excess $C_1(\log m)/m^2$ in a·b costs $O(N^{-1/4})$ in γ and O(1) in the bound. Part (c) rests on a reflection symmetry: f is its own reflection, so every sum constraint is half a difference sum at lag |N − 1 − s| plus a diagonal term.

N	δ√N	ℓ	H/N^(3/4)	$\mathcal{L}$	max_d P(d)	Σf	shortfall	proved bound
10^4	0.990148	100	1.31881	7.51	0.970589	106.7301	2.698	2.891
10^5	0.996853	317	1.36318	13.00	0.990565	330.3431	2.650	2.768
10^6	0.999001	1000	1.38494	22.81	0.997006	1027.1704	2.644	2.698
10^7	0.999684	3163	1.39801	40.21	0.999052	3212.6723	2.623	2.659

Table 13. *The construction of Theorem 6.6 with exactly the parameters of the proof. Shortfall is $\sqrt{N} + (2\sqrt{2}/3)N^{1/4} - \Sigma f$; the proved bound is $2 + (8/3)D_+ + 2\sqrt{2}\cdot N^{-1/4}$. Condition (iv) was evaluated at every lag by FFT in double precision, and the bounds of Lemmas B.1–B.4 were checked one by one at $N = 10^4$, 10^5 and 10^6. For $N \geq 10^5$ the largest P(d) equals the flat*

value $\delta^2(N - \ell)$ to seven digits, so the lattice factor $1 + 2/\ell$ in the proof is slack. At $N = 3000$, with Γ formed explicitly, the moment matrix has smallest eigenvalue 2.0×10^{-4}.

The construction of Table 12, with t and β optimized numerically, remains about 0.1 better than this one and has no written proof; Theorem 6.5 stays as its numerical record. A sharper lattice estimate — the lattice error is a difference of fractional parts, which integrates against a function of bounded variation to O(N^(−3/4)) — would lower the proved constant to ½ + (8/3)D_+ + o(1) ≈ 1.11; enlarging δ accordingly gives 1.13 at $N = 10^7$. The constant D_+ is numerical, accurate to about 10^{-8}; only its finiteness, $D_+ \le \tfrac{1}{2}\|r - 2\|_1$, matters for the O(1) claim. Because C_0 and κ are not explicit, N_0 and the upper constant are ineffective, as in Proposition 5.2.

Structure of the optimum at moderate N

The program (6.2) was solved for $20 \le N \le 180$ with Clarabel through CVXPY, after block-diagonalizing by the reflection $x \mapsto N - 1 - x$. Table 14 compares it with h(N) and with the finite-N smoothing bound s(N) given by the linear kernel with exactly optimal weights, minimized over H.

N	h(N)	$\hat{h}_2(N)$	s(N)	$s(N) - \hat{h}_2(N)$
20	6	6.4291	6.5822	0.1531
32	7	7.9189	8.0726	0.1537
40	8	8.7405	8.8939	0.1534
50	9	9.6469	9.7993	0.1524
64	10	10.7568	10.9089	0.1521
80	11	11.8671	12.0198	0.1527
100	12	13.0938	13.2489	0.1551
120	13	14.1959	14.3522	0.1563
140	14	15.2037	15.3612	0.1575
160	15	16.1372	16.2958	0.1586
180	16	17.0104	17.1699	0.1595

Table 14. *The pair-level relaxation against the smoothing bound. The gap stays between 0.152 and 0.160 while N grows ninefold; a difference of 0.05 in the second-order constant would open it by about 0.08 over this range. For this report $\hat{h}_2$ was re-solved at N = 12, 20, 32 and 40 without the symmetry reduction, reproducing all printed values to four decimals (Appendix A.3).*

Two warnings apply. Second-order constants cannot be fitted reliably from $N \le 180$: fitting $\sqrt{N} + \gamma N^{1/4} + c_0 + c_1 N^{-1/4}$ to s(N) itself, whose limit is 0.943, returns anything from 0.85 to 0.96, so the asymptotic claim rests on Theorem 6.5, not on these fits. And Clarabel flagged some instances as possibly inaccurate, although the returned solutions satisfy every constraint of (6.2) to within 10^{-9}.

The optimal dual is the smoothing certificate. The optimal multipliers y_d of the difference constraints (iv) fit $y_d \approx c \cdot R^*(d/H_{fit})$, with R^* the autocorrelation (5.1) of 2x. The relative L^2 misfit is 6.1%, 4.1%, 3.3%, 2.9%, 2.6% and 2.5% at N = 32, 50, 64, 80, 100 and 120, and the fitted scale H_fit = 22.2, 30.7, 35.3, 39.6, 47.7, 55.6 tracks the optimal H of s(N): 22, 30, 35, 41, 49, 56. The only visible departure is at d = 1. Lindström's flat kernel fits more than twice as badly (5.8% at N = 120).

The optimal primal is the configuration above. At N = 120 the endpoints have f = 0.98, and the first six sites at each end carry total mass 2.49 against $N^{1/4}/\sqrt{2} = 2.34$ predicted, followed by a depleted stretch. In the middle half $\Gamma(x, x+d) = 0$ for $1 \le d \le 6$, against a predicted radius $\sqrt{N}/2 \approx 5.5$, and $\Gamma \approx ff^{\mathrm{T}}$

from d = 8 on; the covariance hole sums to about −1 per point; and (iv) is tight for every $d \le 56$, against $\sqrt{2}\cdot N^{(3/4)} \approx 51$. Adding the distinct-sums constraints and the Sherali–Adams inequalities $\Gamma(x, y) \le f(x)$ and $\Gamma(x, y) \ge f(x) + f(y) - 1$ lowers the optimum by only 0.019 at N = 12 and 0.006 at N = 100. Theorem 6.6(c) shows that these gains stay bounded for all N: at the explicit point the largest distinct-sum load is 0.915, 0.957, 0.978 and 0.988 at $N = 10^4$, 10^5, 10^6 and 10^7, always at the central sum s = N − 2, and the margin times $N^{(1/4)}$ falls toward $2\sqrt{2}\cdot D_+ = 0.645$, the leading term of the proved margin. In the continuum the sum law of the extremal configuration is its difference law shifted by its length, so it comes within $O(N^{(-1/4)})$ of saturating the central sums as well as the short differences.

Beyond pairs

Lemma 6.7. In any relaxation with triple variables T(x, y, z) that contains the lifted budgets $\Sigma_x T(x, x+d, z) \le f(z)$ and $\Sigma_x [\Gamma(x, x+d) - T(x, x+d, z)] \le 1 - f(z)$, the equality $\Sigma_x \Gamma(x, x+d) = 1$ forces $\Sigma_x T(x, x+d, z) = f(z)$ for every z. Wherever a difference budget is full, the number of pairs at that difference must be uncorrelated with every point indicator. In the extremal configuration every budget below H is full, and the independent atoms of the construction violate this.

Deterministic atoms. Deterministic atoms have constant indicators and satisfy every constraint involving them at every level. Take for the two atoms the two halves, m points each, of the Ruzsa set for the smallest prime $p \ge 2m + 1$, whose difference sets are disjoint, and modify the construction: replace the hole matrix by the Toeplitz matrix with entries 1 at lags in $\{0\} \cup D$, where D is the atoms' difference set; cap the profile at $c = 1/(1 + 2|D|)$; and deplete a zone of length span + max D at each end. The result is a family of feasible points whose atoms are genuine Sidon sets (Table 15), checked in exact arithmetic at every lag that carries an atom pair.

N	m	t	δ√N	Σ_x f(x)	√N + (2√2/3)N^(1/4) − Σf
10^4	7	1.2	1.0356	107.324	2.10
10^5	13	1.25	1.0358	330.962	2.03
10^6	22	1.414	1.0060	1027.411	2.40
10^7	40	1.5	1.0041	3213.196	2.10

Table 15. *Feasible points of (6.2) whose atoms are genuine Sidon sets. The shortfall is O(1), about 1.1 larger than in Table 12, and fluctuates with the integer choice of m. Any obstruction beyond the pair level must therefore live in the bulk, which must use every difference below H once in expectation, except the roughly √N/2 differences left to the atoms.*

Three points on the interval. The three-point relaxation $\hat{h}_3(N)$ adds triple variables with positive semidefiniteness of the moment matrices localized at $z \in A$ and at $z \notin A$ for every z, the lifted budgets, T(x, x+d, x+2d) = 0 (a Sidon set contains no three-term progression), the Sherali–Adams bounds, and $\Gamma \le f$.

N	12	18	20	26	32	40
h(N)	5	6	6	7	7	8
$\hat{h}_2(N)$	5.1194	6.1340	6.4291	7.2237	7.9189	8.7405
$\hat{h}_3(N)$	5.0000	6.0149	6.3271	7.1370	7.8554	8.6916
$\hat{h}_2 - \hat{h}_3$	0.119	0.119	0.102	0.087	0.064	0.049

Table 16. *Three-point information lowers the relaxation by an amount that shrinks with N, whereas a second-order improvement would require growth like N^(1/4). Solved with Clarabel, and with SCS for N ≥ 32; the two agree to four decimals at N = 26. At N = 20 the lifted budgets and the localized positive semidefiniteness do most of the work, synergistically: each alone recovers about a third of the reduction.*

A cyclic model of the bulk. To isolate the bulk, consider translation-invariant relaxations on $\mathbb{Z}_n$ with $n = q^2 - 1$, with the Sidon condition taken in $\mathbb{Z}_n$ and every difference in a symmetric forbidden set F of $q - 2$ nonzero residues excluded, so that the counting bound is exactly q and the pair-level relaxation attains it. Three forbidden sets of the same size are compared: the subgroup of nonzero multiples of $q + 1$, which are exactly the residues Bose's sets miss; short lags; and random lags. The four-point relaxation uses the moment matrix indexed by subsets of size at most 2, with exact zeros on every set of at most four elements containing a repeated or forbidden difference; it is block-diagonalized by the Fourier transform on $\mathbb{Z}_n$ and returns exactly 4 and 5 on the Singer moduli 13 and 21.

$n = q^2 - 1$	forbidden set	pair	three-point	four-point	largest genuine set
24	subgroup	5	5	5.0000	5
24	short lags	5	5	4.2983	4
24	random lags	5	5	4.2983	4
48	subgroup	7	7	7.0000	7
48	short lags	7	7	6.9720	6
48	random lags	7	7	6.9720	6
63	subgroup	8	8	8.0000	8
63	short lags	8	8	8.0000	7
63	random lags	8	8	8.0000	7

Table 17. *The cyclic model; the largest genuine sets were determined by exhaustive search. At n = 63 the four-point relaxation was solved with SCS, feasible to 3×10^{-8}; since the counting bound caps every relaxation at q, the values 8 are exact to that order.*

Only the subgroup admits a set of size q; for the other two shapes the largest genuine set has $q - 1$ elements, yet the pair and three-point relaxations return q for every shape. The four-point relaxation, which contains the Sidon condition in its native form $a + b = c + d$, closes 70% of the gap between the counting bound and the truth at $n = 24$, 2.8% at $n = 48$ and none of it at $n = 63$, and where it cuts it separates the subgroup from the other two shapes, not short lags from random ones. The obstruction is arithmetic and global: invisible at the pair and three-point levels, and what the four-point level captures of it disappears by $n = 63$. Three small sizes cannot settle whether any fixed level of the moment hierarchy captures it as n grows, but they suggest not.

Three points: the wall extends

Table 16 suggested that three-point information adds only O(1). Version 6 proves it.

Theorem 6.8 (three-point barrier). *$\hat{h}_3(N) = \sqrt{N} + (2\sqrt{2/3})\cdot N^{(1/4)} + O(1)$. More precisely $\hat{h}_3(N) \le \hat{h}_2(N)$, and there is N_0 such that for $N \ge N_0$*

$$\hat{h}_3(N) \ge \sqrt{N} + (2\sqrt{2/3})\cdot N^{(1/4)} - 10.28 - o(1). \quad (6.4)$$

The O(1) term over all N is finite but ineffective, as in Theorem 6.6.

The upper half is immediate: the formalization of $\hat{h}_3$ (triple.py) keeps the pair budgets, the main positive semidefinite block and the Sherali–Adams bounds, so every three-point feasible point is pair-feasible. The lower half needs an explicit feasible point, and the first finding was that the obvious one fails. Lifting the point of Theorem 6.6 to triple variables by the Kirkwood product $T(x, y, z) = f_x f_y f_z \rho_{xy} \rho_{xz} \rho_{yz}$, with $\rho = \Gamma/ff^T$, satisfies localized positive semidefiniteness with margin $\delta(1 - \delta\ell)$, but violates the lifted budgets of Lemma 6.7 across the whole end zone: conditioning on a bulk point within about H of an end

overloads the lags that point back at the nearer atom, by 0.51–0.78 times the atom's plateau density p_a ≈ N^(−1/4)/√2 at N = 10^4, 10^5 and 10^6. Paying for that excess with global pair slack would cost Θ(N^(1/4)), the size of the term itself. The proof therefore repairs the lift locally.

The construction. (i) The pair point of Theorem 6.6, with the hole, window and atom width narrowed to $\ell = \lfloor(1-\theta)/\delta\rfloor$, $\theta = \kappa \cdot N^{(-1/4)}/\sqrt{2}$, and with δ chosen so that every difference budget carries slack c_s·δ. (ii) The lift: the Kirkwood product times $1 - \max(r_z(|y-x|), r_x(|z-y|), r_y(|z-x|))$, the largest of the reductions seen from the triple's three conditioning points, and zero on every three-term progression. Each reduction $r_w(d) = \min(1, (\Delta_w(d) - s)_+/B_w(d))$ is defined from an explicit formula for the conditional excess, so the construction contains no numerical step.

The proof (Appendix C) rests on an exact two-scale split of the margin. The triangle kernel is a sum of box products, $\Phi_\ell = (1/\ell)\cdot BB^T$ with boxes truncated at the ends of the interval, so

$$I - \delta\Phi_\ell = \theta_\ell \cdot I + \delta \cdot Q_\ell,\ \theta_\ell = 1 - \delta\ell,\ u^T Q_\ell u = \Sigma_{\text{boxes}} [\Sigma_{\text{box}} u^2 - (1/\ell)(\Sigma_{\text{box}} u)^2] \geq 0. \quad (6.5)$$

The θ part pays for directions spread over the whole interval and the Q_ℓ part, including the truncated boxes at the ends, pays for localized ones. The reductions split the same way: the large ones, of size p_a, sit next to the atoms and are paid by the local margin, so only O(δ) per site has to be paid from θ, and θ of order N^(−1/4) costs O(1) in the count. The binding constants are a row-sum constant for far rows (proved A = 8.83, measured about 1.9), one for mirror rows (B = 2.02) and a corner constant for conditioning points within 2ℓ of an end, certified by branch and bound at most 0.925. They force κ = 20 and c_s = 2.8 (the excess bounds need c_s > e), which gives the lower constant 2.108 + c_s/2 + η/2 + κ/3 = 10.28: the narrowed hole costs κ/3 = 6.67 and the slack c_s/2 = 1.40.

check	N	parameters	result	shortfall
Kirkwood lift, budgets only	10^4–10^6	Theorem 6.6 point	lifted budgets exceeded by 0.51–0.78·p_a in the end zone	—
repaired lift, every conditioning point	800	θ_eff = 0.184, c_s = 0.6	feasible	1.72
repaired lift, every conditioning point	1200	θ_eff = 0.164, c_s = 0.9	feasible	1.89
repaired lift, sampled points	2000	threshold θ_eff ≈ 0.147	feasible above threshold	1.72
explicit Lemma C.1 point, untuned	800	κ = 2.5, c_s = 2.8	every three-point constraint, all 800 points	4.78
explicit Lemma C.1 point, untuned	1200	κ = 2.5, c_s = 2.8	every constraint for z = 0–199; min localized eigenvalue $+5.5\times10^{-4}$	4.72
Lemma C.1 at the proof constants	10^6, 10^7	κ = 20, c_s = 2.8	budgets clear $1 - c_s\cdot\delta$	11.61, 10.84 (proved 13.71, 12.07)

Table 18. *Numerical checks of the three-point construction. For comparison, the pair-level point of Theorem 6.6 at N = 800 with comparable slack has shortfall 1.26–1.30, so the three-point constraints cost about 0.45 there. The threshold hole margin tracks N^(−1/4) (ratios 0.98 and 0.96 at N = 800 and 1200), which is what an O(1) total cost requires. Checks are in double precision; the every-point check at N = 800 caught failures at z = 43–47 and 752–754 that sampling had missed, so sampled rows are lower brackets.*

The gap between the proved constant (10.3) and the numerics (about 4.7, or 1.7–1.9 with tuned parameters) comes almost entirely from the crude far-row constant; computing it in the scaling limit, as was done for the corner, should bring κ to about 3–5 and the proved constant to about 5. Three variants failed and are recorded because they locate the difficulty. With slack of only N^(−3/4), every point needs repair and positive semidefiniteness fails in the conditional-count direction, Var(k | z) < 0. Adding the

lost pair mass back on mutually distant triples breaks positive semidefiniteness. And making the end atoms genuine Sidon rulers (deterministic atoms, as in Table 15) fails differently: the lifted budgets force $\Gamma(y, z) = 0$ wherever $y + z$ is a cross sum of two marks, and restoring positive semidefiniteness in that construction family cost $\Theta(N^{1/4})$. That is a failed construction, not a lower bound, but it marks where integrality might have leverage.

Theorem 6.8 is a barrier result: it extends the wall from pair statistics to the three-point relaxation without moving the bound. Its caveats are those of Theorem 6.6 — N_0 and the $O(1)$ constant are ineffective, and the proof is new and unrefereed — plus one more: the corner certificate and the profile constants were computed in double precision, not interval arithmetic.

Arithmetic: the quotient test and its interval analogues

Theorems 6.6 and 6.8 show that pair and three-point constraints cannot push the constant below $2\sqrt{2}/3$, so an improvement has to use something else; the natural candidate is arithmetic. Arithmetic can enter an upper bound validly in two ways: through constraints every Sidon set satisfies because its counts are integers, or through a structure theorem that pushes near-extremal sets toward arithmetic structure. Version 6 tested the first in three forms.

The cyclic quotient test (positive). In the cyclic model of Table 17, a set D of size q with distinct differences avoiding a symmetric forbidden set F of $q - 2$ residues uses every other nonzero residue exactly once, since $q(q - 1) = n - 1 - |F|$. Reducing D modulo a divisor e of n gives q points in $\mathbb{Z}_e$ with prescribed difference counts $t_e(r) = n/e - [r = 0] - \#\{f \in F : f \equiv r \pmod e\}$. If no multiset of nonnegative integer counts meets these targets, D does not exist. The constraint holds for every candidate set and uses no field structure.

q	$n = q^2 - 1$	random F	genuine set exists	certified impossible	missed	undetermined
5	24	40	15	21	4	0
7	48	40	0	40	0	0
8	63	40	0	37	3	0
9	80	40	—	39	0	1
11	120	40	—	39	0	1
13	168	40	—	40	0	0

Table 19. *The cyclic quotient test on random forbidden sets. For $q \geq 9$ the search for genuine sets was not run, so undetermined cases are counted against the test. Over $n \leq 63$ a single quotient certifies 98 of the 105 non-existent cases (93%), including every case in Table 17 where the four-point relaxation fails. The failing quotients are solvable over the nonnegative reals, so integrality, not a further linear constraint, does the work.*

The cyclic test works because the whole difference budget is full. On the interval it is not: a Sidon set in $[0, N)$ uses about half of its ordered differences, and in the extremal configuration only two windows are full, the short differences $d < H \approx \sqrt{2} \cdot N^{3/4}$ and the central sums s near $N - 1$.

Short-difference coverage on the interval (negative). Every Sidon set of size $h(N)$ containing 0 was enumerated for $N = 20$–100, together with the published optimal Golomb rulers for $k = 6$–28, and the relaxation $\hat{h}_2^+$ was given one more constraint: every difference $1, \ldots, M$ is used. Genuine optimal sets often use every difference below H — OGR-28, at $N = 586$, uses all of 1–171 against $H = 168$ — and in the $k = 11$ block every genuine set misses 3 of the 35 short differences while the relaxation can use them all for free. That integrality is real but costs nothing in size. The decisive cases are those where the relaxation rounds above $h(N)$:

N	h(N)	$\hat{h}_2^+(N)$	unused short differences allowed for a set of size h(N) + 1	best genuine set of size h(N) + 1
55	9	10.05	at most 2.4 of 29	misses 0 of 29 (OGR-10, N = 56)
72	10	11.32	at most 6.3 of 35	misses 3 of 35 (N = 73)

Table 20. *Short-difference coverage cannot certify h(55) = 9 or h(72) = 10: genuine sets of the excluded size meet the relaxation's requirement in the shortest interval where they fit. What excludes them is that they do not fit in the interval — information about positions, not about which short differences appear. Clarabel reported "optimal_inaccurate"; values are consistent across runs but not confirmed by a second solver.*

The interval quotient test (negative). Three forms of integrality were imposed on $\hat{h}_2^+$ where the extremal configuration is full: integer point counts in end windows [0, w) and [N − w, N), with zero variance in the moment matrix; genuine end assignments, by branch and bound over 0/1 values taken from the outside in with $\hat{h}_2^+$ as the bound at every node; and integer residue counts mod e, the literal analogue of the cyclic test. At N = 55, where certifying h(55) = 9 needs a bound below 10, $\hat{h}_2^+ = 10.0523$; end counts lower it to 10.042 at best (w = 5), genuine ends on eight positions per side to 10.0226, and residues mod 2 and mod 3 accept 5 of 11 and 10 count vectors. End windows are weak for a structural reason: the relaxation can leave an integral window empty and place its fractional atom just inside, which shortens the interval by w at a cost of about w/√N.

At large N the test reduces to whether genuine end atoms survive the distinct-sums constraints. The points of Table 15, whose atoms are halves of a Ruzsa set, overload every one of the m^2 cross sums (one point from each atom), with load 1.46–1.48 at N = 10^4 and 10^5, while every other central sum carries about 0.5. Removing the bulk pairs at those sums restores feasibility only if the atoms shrink to about N^(1/4)/√3, with shortfalls 2.77, 4.83 and 7.00 at N = 10^4, 10^5 and 10^6 — a cost growing like N^(1/4). But the other central sums have room: raising the bulk pairs there by a smoothed compensating boost, matched in mass to what was removed, keeps the anti-diagonal change free of slowly varying components, so it no longer competes with the difference hole for positive semidefinite room (largest eigenvalue of the removed-pair pattern 312.03 against 2|D| = 312 at N = 10^5). Full-size genuine atoms are then feasible for $\hat{h}_2^+$ with shortfalls 2.11 and 2.00 at N = 10^4 and 10^5, matching Table 15; a dense end-to-end check at N = 3000 found every constraint satisfied and every cross sum at load exactly 1. The run at N = 10^6 did not finish. All of this is numerical.

So integrality at the ends, and genuine end structure checked against distinct sums, cost only O(1) at the pair level. The one form of arithmetic the interval data have not ruled out ties integrality to positions on windows of length about N^(3/4), at the scale where the N^(1/4) term lives.

Consequences

The level-one moment relaxation is the extension of the smoothing method that keeps the variance term and the difference slack. Its dual is a semidefinite program of which smoothing certificates are feasible points — the kernel autocorrelations are the multipliers of (iv), and the Cauchy–Schwarz direction is the positive semidefinite multiplier of (i) — and at moderate N its optimal dual is the extremal smoothing certificate up to local corrections. Its second-order constant is exactly 2√2/3 (Theorem 6.6), with distinct sums and Sherali–Adams added, and the three-point relaxation stops at the same constant (Theorem 6.8). Keeping the variance term therefore gains only O(1) as long as the argument sees one-, two- or three-point statistics; four-point gains shrink at the sizes that can be computed (Part X), but no theorem covers any level above three. In the cyclic model, what separates the relaxations from genuine Sidon sets is a design-type question — whether a near-perfect difference set with a prescribed set of

missing differences exists — which integrality on one quotient answers. On the interval, the tests above say that this arithmetic has to be tied to positions, or to the slack in the long differences. Part XII states the resulting requirements for an argument below $2\sqrt{2}/3$.

Part VII · Stage 1: gaps over multipliers

Stage 1 asks whether the largest gap of a Singer set, maximized over multipliers, is unbounded relative to q. A gap g in a cyclic Sidon set modulo n places its k elements in an interval of n − g + 1 integers, so if the best multiplier's largest gap is at least C·q for every C, the excess h(N) − √N is unbounded and h(N) = √N + O(1) — which Erdős thought possibly true — fails. This is Open Question 5 of O'Bryant (2024). The mean gap is about q, and the random model of Part II predicts about 3q ln q for the best multiplier and q(ln q − ln ln q) for the weakest. Version 5 records the first proved results in this direction: for every multiplier, every Singer set has a gap of about twice the mean, and so does every Bose–Chowla set.

Setup

Let q be a prime power, E = GF(q^3), n = q^2 + q + 1, and θ a primitive element of E. The Singer set is D = {i mod n : Tr(θ^i) = 0}, a planar difference set with q + 1 elements; its multipliers t ∈ (ℤ/nℤ)^× give the sets tD. For x ∈ E^× the trace factorizes as Tr(x) = x·T(x^(q−1)) with T(u) = 1 + u + u^(q+1), and ρ_0 = θ^(q−1) generates the group μ□ of n-th roots of unity. As ρ runs over the generators of μ□, the sets D_ρ = {j ∈ Z_n : T(ρ^j) = 0} run over all the multiplied sets tD. So D_ρ has L consecutive residues outside it exactly when

$$H_L(u) = \prod_{j=0}^{L-1} T(\rho^j u) \quad (7.1)$$

does not vanish identically on μ□. Expanding the product and folding exponents modulo n gives $H_L(u) = \sum_K \Phi_K u^K$ on μ□, where Φ_K collects the monomials $u^{b+(q+1)c}$, b + c ≤ L, with b + (q+1)c ≡ K. Since n ≡ 1 (mod p) is invertible in E, H_L vanishes on μ□ only if every Φ_K vanishes; one nonzero folded coefficient therefore certifies a run of length L for every multiplier at once.

The 2q bounds

Theorem 7.1 (Singer). *For every prime power q and every multiplier t ∈ (ℤ/nℤ)^×, the set tD has a gap of length at least 2q. Equivalently, every Singer difference set modulo $q^2 + q + 1$ misses some 2q − 1 consecutive residues.*

Proof sketch. Take L = 2q − 1 and K = q − 1. The only pair (b, c) with b + c ≤ L and b + (q+1)c ≡ q − 1 (mod n) is (q − 1, 0), because (q+1)(2q − 1) < 2n. Its coefficient is the elementary symmetric function $e_{q-1}(1, \rho, \ldots, \rho^{(2q-2)}) = \rho^{\binom{q-1}{2}} \cdot [2q-1, q-1]_\rho$, a Gaussian binomial at a primitive n-th root of unity. Every factor 1 − ρ^i in it has 0 < i < n, so it is nonzero. ∎ The full proof is in Stage1_gap_proofs.

Corollary 7.2. *For every prime power q, $h(q^2 - q + 2) \geq q + 1$. At $N = q^2 - q + 2$ this gives $h(N) - \sqrt{N} \geq 3/2 - 7/(8q) + O(q^{-2})$.*

Lemma 7.3. *If the largest gap of D_ρ is exactly 2q, then 2q − 1 or 2q − 2 is also a gap of D_ρ.*

The lemma is the first step of the next rung, a gap of at least 2q + 1, discussed below. Its proof uses that the gaps of a planar difference set are distinct residues, so at most one gap exceeds 2q − 3 when the largest is 2q.

Theorem 7.4 (Bose–Chowla). *For every prime power q ≥ 3 and every multiplier t, the Bose–Chowla set tB modulo $q^2 - 1$ has a gap of length at least 2q − 1. The bound is sharp at q = 3 and q = 5.*

Here B = {j mod q^2 − 1 : θ^j has trace 1 over GF(q)} for a primitive θ ∈ GF(q^2), a line of AG(2, q) not through the origin. *Proof sketch.* A window that meets every line not through 0 forms, together with 0, a

blocking set of AG(2, q), which by Jamison (1977) and Brouwer and Schrijver (1978) has at least 2q − 1 points; so every gap is at least 2q − 2, for any subset of GF(q^2)^× whatsoever. The orbit adds one. For L = 2q − 2 the sum of the analogue of (7.1) over GF(q^2) is, up to sign, a Gaussian binomial [2q − 2, q − 1]_ω with ω = θ^(1−q) of order exactly q + 1; by the q-Lucas theorem it vanishes, since 2q − 2 ≡ q − 3 (mod q + 1) and q − 3 < q − 1. The value at 0 is 1, so the sum over nonzero arguments is −1 and some line misses the window. ∎

Corollary 7.5. *For every prime power $q \geq 3$, $h((q-1)^2) \geq q$: an excess of at least 1 at $N = (q-1)^2$.*

The sources of the two bounds differ. For Bose–Chowla sets almost all of the bound comes from a classical theorem about arbitrary point sets, and the orbit adds exactly one. For Singer sets the generic blocking-set bounds stop at q + √q + 1 (Bruen 1970) or 3(q + 1)/2 for prime q (Blokhuis 1994), so the orbit structure supplies everything from there to 2q. The previous general bound for Singer sets, from the distinctness of the q + 1 gaps, is $\lceil 3q/2 + 1/(q+1) \rceil$; it is formalized in Lean by Hulak, Ramos and de Queiroz (2026) and also given by Gupta (2026). A literature check on 3 October 2026 found no earlier statement of Theorems 7.1 or 7.4. It did not cover the sequence-design literature on m-sequences, so their novelty is provisional.

family	elements, modulus	mean gap	proved: every multiplier	method	sharp at	consequence
Singer	q + 1, $q^2 + q + 1$	≈ q	gap ≥ 2q (Theorem 7.1)	one folded coefficient, a Gaussian binomial	q = 2, 3, 5	$h(q^2 - q + 2) \geq q + 1$
Bose–Chowla	q, $q^2 - 1$	≈ q	gap ≥ 2q − 1 (Theorem 7.4)	Jamison / Brouwer–Schrijver, plus q-Lucas	q = 3, 5	$h((q-1)^2) \geq q$
Ruzsa	p − 1, p(p − 1)	≈ p	none proved	—	—	—

Table 21. *Proved lower bounds on the largest gap for every multiplier. The counting steps of both proofs were checked by computer for every integer $2 \leq q \leq 300$.*

How far single coefficients can go

The method certifies a run of length L whenever one folded coefficient can be shown to be nonzero. A coefficient is usable when its class contains a single lattice point (it is *isolated*), so that it is an explicit product. This has a hard ceiling.

Lemma 7.6. *(Singer.) Isolated classes exist at every $L \leq 3q - 3$ and at no $L \geq 3q - 2$. (Bose–Chowla.) No class is isolated when $L \geq 3q - 4$.*

So arguments resting on a single isolated coefficient can prove at most a gap of 3q − 2 for Singer sets and 3q − 4 for Bose–Chowla sets. In practice even that ceiling is not reached. At L = 3q − 3 the one available Singer coefficient vanishes *spontaneously* — at a multiplier class whose window is not blocking — for about one class in q: 243 such zeros among 17,409 non-blocking classes for q ≤ 121. Meanwhile, every multiplier has a gap of at least 3q − 2 for 73 ≤ q ≤ 419.

The next rung, a gap of at least 2q + 1 for every q ≥ 7, needs more than one coefficient. If the largest gap were exactly 2q, a run of coefficients would form a geometric progression. Its first relation, δ, holds spuriously at two classes each for q = 11, 29, 59, 67 and 89 (all prime powers q ≤ 128 tested). The relation one step into the interior, through κ(2, q − 1), vanishes at none of these, and nowhere for 7 ≤ q ≤ 101. The interior coefficients have explicit formulas and an exact Pascal-type recursion. They have no closed product form: each has an irreducible factor that is not cyclotomic, of degrees 20/18, 52/52, 106/106 and 318/306 for q = 3, 4, 5 and 7. A proof for all q would need to show that two explicit

cyclotomic expressions never share a prime factor above p. A criterion based on word counts modulo p, tested at q = 41, did not separate the cases and was retired.

Computation: the weakest multiplier

Every multiplier class was enumerated for all prime powers $q \leq 419$ (Singer, Bose–Chowla) and all primes $p \leq 1447$ (Ruzsa).

family	range	minimum over multipliers of the largest gap	tight at	min/q at the top of the range	min/q ÷ (ln q − ln ln q), q ≥ 50: median (range)
Singer	prime powers $2 \leq q \leq 419$	$= 2q$ at $q = 2, 3, 5$; $\geq 2q + 2$ for $7 \leq q \leq 419$ (96 prime powers)	$2q + 2$ at $q = 7, 8, 11$	4.055 (q = 419)	0.98 (0.92–1.14)
Bose–Chowla	prime powers $3 \leq q \leq 419$	$= 2q - 1$ at $q = 3, 5$; $\geq 2q + 3$ for $8 \leq q \leq 419$	$2q + 3$ at $q = 11, 13$	4.160 (q = 419)	1.00 (0.94–1.14)
Ruzsa	primes $5 \leq p \leq 1447$	$\geq 2.2p$, and $\geq 2p + 3$ for $7 \leq p \leq 1447$	$2.2p$ at $p = 5$; $2p + 3$ at $p = 11$	—	1.09 (0.95–1.40)

Table 22. *The weakest multiplier. The maxima over multipliers agree with stage0_results.csv at every shared q. The last column is an empirical description, not a fit with error bars.*

The weakest multiplier's gap grows like $q(\ln q - \ln \ln q)$ in all three families, as the random model predicts for the smallest of about q^2 largest gaps. That is a Cramér-type statement, not a proof; turning any part of it into a theorem beyond the 2q level is the content of Stage 1. The best multiplier is what matters for h(N), and there the target is a gap of $C \cdot q$ for every C. Two routes look plausible, and neither has been tried in earnest. In m-sequence terms, each multiplier corresponds to a decimation — the m-sequence of another primitive cubic over GF(q) — and the question is about runs of nonzero terms (Hall 1938; Alexeev and Mixon 2025). Alternatively, an ineffective existence argument of the lattice type used for Problem 1 might work.

Part VIII · Extreme values of Singer constructions

Part II found logarithmic growth of the construction-side excess up to $N \approx 2.7\times10^8$, with one structural effect, in characteristic 2. This part asks the sharper question that extreme-value theory can answer: is the tail of the per-class distribution exponential, which gives logarithmic records, or heavy, which gives power laws? Characteristic-2 Singer sets are the right family to test. Their Frobenius effect is the largest of any characteristic, since the factor $p/(p-1)$ peaks at $p = 2$, and their best arcs hold the record rulers above $N \approx 200$ (Table 4). The work was done in three runs, written up on 12 October 2026, and a statistical step on 14 October.

Why extreme-value analysis, and on what

The right object is the population of multiplier classes of one family at fixed q, not the output of a randomized search. Beyond $N = 586$ every record Sidon set is a cut of a Singer, Bose or Ruzsa set, and stochastic search does not come near them, so the outcome distribution of a search measures the algorithm, not h(N).

For $q = 2^e$, $n = q^2 + q + 1$ and $D = \{i \bmod n : \mathrm{Tr}(\theta^i) = 0\}$, with θ the root of the Conway polynomial of $GF(2^{3e})$, the multiplier classes are $t\cdot\langle 2, -1\rangle$ for units t mod n. There are $M = \varphi(n)/(6e)$ of them, all of size 6e, and each class's largest gap is a maximum over $k = q + 1$ gaps. The classes therefore form a complete, deterministic population of block maxima.

The shape parameter is the growth exponent. Cutting at the largest gap G gives an excess of about $G/(2q)$. If the per-class tail had a generalized-Pareto shape $\xi > 0$, the maximum over M classes would grow like M^ξ, and M grows roughly like N; that would give $h(N) \geq \sqrt{N} + c\cdot N^\xi$ and refute the conjecture by construction. A shape $\xi = 0$ means logarithmic growth. The record sequence alone cannot decide between these: from $q = 2^{11}$ to 2^{14}, N grows 64-fold, $N^{0.065}$ by a factor of 1.31 and ln N by 1.27, so Version 5's four record points rule out an $N^{1/4}$ term but not small powers. Populations of 10^4–10^6 classes per q can.

Run 1: the largest gap is a two-component mixture

$D = 2D$ exactly, because $\mathrm{Tr}(\theta^{2i}) = \mathrm{Tr}(\theta^i)^2$. Near any point x^* of the circle $\mathbb{R}/n\mathbb{Z}$ with $2^m\cdot x^* \equiv x^* \pmod n$, position $x^* + j$ is a mark exactly when $x^* + 2^m\cdot j$ is, so only offsets not divisible by 2^m are free. Treating free positions as independent marks with density $1/q$ gives predictions with no fitted parameter: with $s = 2^m/(2^m - 1)$, the gap containing x^* is Gamma(2, sq), with mean 2sq, and when x^* is itself a mark each gap beside it is Exp(sq).

point	period m	s	prediction	measured, $q = 2^{11}$–2^{14}
origin (never a mark: Tr(1) = 1)	1	2	containing gap 4.000q	3.987q–3.999q
n/3, 2n/3, not integers (e odd)	2	4/3	containing gap 2.667q	2.658q, 2.662q
n/3, 2n/3, integers (e even): always marks	2	4/3	neighbouring gap 1.333q	1.332q, 1.329q
a·n/7, not marks	3	8/7	containing gap 2.286q	2.271q–2.288q
a·n/7, marks	3	8/7	neighbouring gap 1.143q	1.138q–1.144q
a generic point	—	1	containing gap 2.000q	1.992q–1.999q

Table 23. *Mean gaps at the short orbits of $x \mapsto 2x$. All eleven measured means fall within 0.015q of the prediction, and within 0.004q at $q = 2^{14}$.*

Proposition 8.1. *If e is even, so that 3 | n, then n/3 and 2n/3 belong to every multiplier class. Proof.* Put β = θ^(n/3). Then β^(q−1) = θ^((q³−1)/3) is a primitive cube root of unity ω, so β^q = ωβ, β^(q²) = ω²β and Tr(β) = β(1 + ω + ω²) = 0. Every unit t maps {n/3, 2n/3} to itself. ∎ When 7 | n, exactly three of the six points a·n/7 are marks in every class; this is observed, not proved.

q	classes M	mean gap at origin	KS distance to Gamma(2, 2q)	ξ, generic maximum	ξ, class maximum	ξ, mixture model
2^{11}	54,498	3.990q	0.0038	−0.005	+0.065	+0.070
2^{12}	139,968	3.987q	0.0023	−0.005	+0.052	+0.058
2^{13}	728,208	3.989q	0.0015	−0.002	+0.045	+0.046
2^{14}	1,820,448	3.999q	0.0014	−0.003	+0.033	+0.036

Table 24. *Every class at each q; ξ is the shape of a generalized extreme-value fit. The mixture model draws each class maximum as the larger of a Gamma(2, 2q) gap at the origin and a Gumbel maximum of the other gaps, with no fitted parameter.*

The apparent positive shape of the class maximum is a mixture effect: the mixture model reproduces it within 0.006 at every q, the generic maximum has $\xi \approx 0$, and the class-level value shrinks as q grows. The largest gap over all classes is the gap at the origin at every q, within about one standard deviation of the expected maximum of M Gamma(2, 2q) draws (29.16q against 28.40q, 27.51q against 30.42q, 31.84q against 33.92q and 34.86q against 35.86q). Over M classes the largest gap at a point of scale s grows like 2sq·ln q and the largest generic gap like 3q·ln q, so the coefficients are 4 at the origin, 3 for generic gaps, 8/3 at period 2 and 16/7 at period 3. Only the origin beats the generic gaps. This derives the conjecture q(4 ln q + O(1)) of Part II and a plain-cut excess of about 2 ln q.

Run 2: the record arcs straddle the origin

Every record best arc at $q = 2^{11}$–2^{14}, at every number of dropped marks d from 1 to 30, contains the gap at the origin; the record of Table 4 at $q = 2^{14}$ (excess 24.600 at d = 18, class t = 8049367) is one of them. For arcs through the origin the merged length of d + 1 gaps is G_o[d] = max over l + r = d of a_(l+1) + b_(r+1), with b_i the i-th mark to the right of 0 and a_i the distance of the i-th mark to its left, and the excess is $(k - d) - \sqrt{(n - G + 1)}$ as in Part II.

The chain model makes Version 5's half-density shell exact: near 0, x is a mark exactly when its odd part is; odd parts are marks independently with probability k/n; and the two sides are independent. Its d = 0 slice is the Gamma(2, 2q) law of Run 1. It gets four things right.

- **The share of records.** Arcs through the origin are the best arc in 92%, 89%, 82% and 76% of the top 1% of classes at $q = 2^{11}$–2^{14}. The share falls at a fixed quantile because the non-origin maximum rises by ln 2 per doubling of q.
- **The bulk.** The per-class distribution of G_o[d] matches the model at every d; the Kolmogorov–Smirnov distance at $q = 2^{13}$ is at most 0.005.
- **The optimal d is noise.** The model's record profile across d is a plateau, and its maximizing d scatters over replicates from 0 to 21 at 2^{11} and from 6 to 29 at 2^{12}. Version 5's sequence 3, 6, 16, 18 is consistent with draws from that plateau.
- **The odd-characteristic control.** At the prime q = 2003 the Frobenius factor is close to 1. Arcs through the origin are best in only 1.3% of the top 1%, the record lies elsewhere (13.07 against an independent-gaps prediction of 13.04), and the best origin arc (10.89) matches the model (10.63 ± 0.64).

Where the model fails is the far tail.

q	ratio at 10^{-3} level	ratio at 10^{-4} level	ratio at 10^{-5} level	record	model record
2^{11}	0.42	—	—	16.70	18.42 ± 0.69
2^{12}	0.69	0.71	—	18.73	20.03 ± 1.42
2^{13}	1.18	1.47	—	22.13	21.18 ± 2.00
2^{14}	1.23	1.81	2.80	24.60	22.28 ± 0.91

Table 25. *Classes whose best origin-arc excess exceeds the chain model's top-10^{-j} level, as a ratio to the expected count. Model columns use independent seeds per q and 6–8 replicates; a first analysis that reused seeds across q was corrected.*

The means agree to 0.2% at every q, so the disagreement lives in the far tail only: at small q the data are lighter than the model, and from 2^{13} on they are heavier, increasingly so further out. At 2^{14} two classes (t = 8049367 and t = 3742139) hold every record from d = 1 to 30. The record arcs are sparse, lopsided stretches: the 2^{14} record drops all 18 marks on the left, spanning −79q to +5q, and the record windows at 2^{13} and 2^{14} hold 5–16 free marks where windows that long average 31–42. Newton relations were ruled out as the cause. If $\mathrm{Tr}(\gamma) = 0$ then $\mathrm{Tr}(\gamma^3) = 3 \cdot N(\gamma) \neq 0$ for $p \neq 3$, so $D \cap 3D = \varnothing$, and in characteristic 2, given $o \in D$, $5o \in D$ exactly when $7o \in D$; both were checked exactly at 2^{11} and 2^{12}, but each changes a window's probability by only about 1/q.

Run 3: the tail width levels off

Only the marks nearest 0 matter for arcs through the origin. For each class, $t \cdot d/n \bmod 1$ was computed in floating point for every mark d, and the few within a safety margin of 0 were recomputed exactly; this skips the sorting and window passes of Run 2 (q = 2^{13} took 3.5 minutes against 23). Every class was enumerated at 2^{11}–2^{14}, and 2×10^6 random classes at each of 2^{15}–2^{18}, against three replicates of up to 2×10^6 simulated classes of the chain model per q.

q	classes analysed	10^{-3} quantile	10^{-4} quantile (model)	10^{-5} quantile (model)	ratio at 10^{-4}	ratio at 10^{-5}	GPD shape ξ
2^{13}	all 728,208	12.87	15.99 (15.65)	—	1.37	—	−0.080
2^{14}	all 1,820,448	12.97	16.49 (15.64)	19.37 (18.44)	1.79	2.25	−0.030
2^{15}	2×10^6 of 11.7×10^6	13.13	16.57 (15.59)	20.00 (18.48)	2.21	3.05	−0.052
2^{16}	2×10^6 of 23.2×10^6	13.30	16.88 (15.54)	20.35 (18.38)	2.26	4.05	−0.001
2^{17}	2×10^6 of 142.9×10^6	13.30	17.00 (15.53)	20.37 (18.12)	2.53	5.00	−0.019
2^{18}	2×10^6 of 424.2×10^6	13.31	17.03 (15.69)	20.87 (18.26)	2.33	4.60	−0.031

Table 26. *Upper quantiles of each class's best origin-arc excess, with the chain model's in brackets; the model's 10^{-3} quantile stays between 12.60 and 12.67 throughout. Ratios are real counts above the model's top-10^{-j} level over the expected count. ξ is the shape of a generalized Pareto fit to the top 0.1%.*

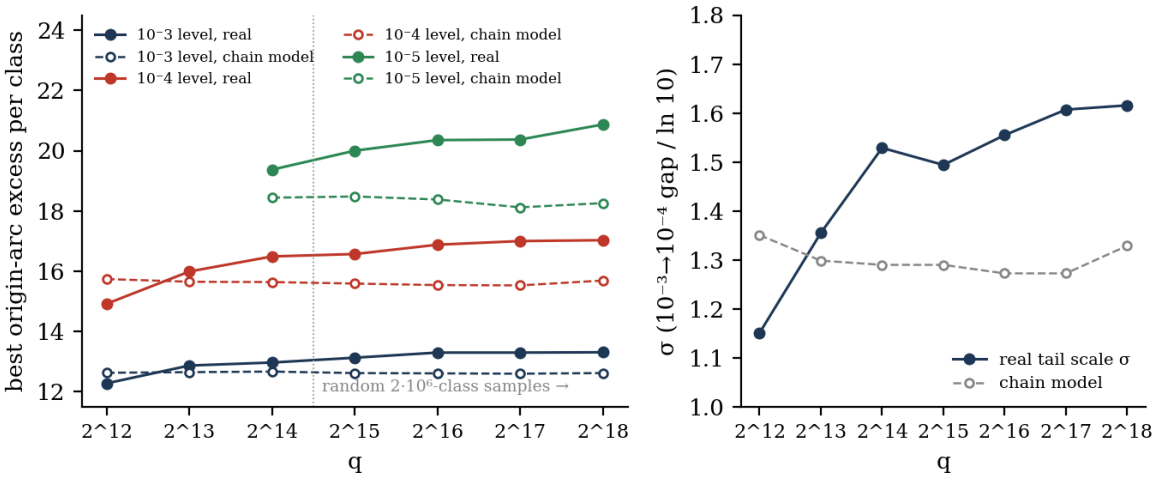


Figure 6. *Left: the 10^{-3}, 10^{-4} and 10^{-5} quantiles of the per-class best origin-arc excess (solid) against the chain model (dashed); every class to the left of the dotted line, random samples of 2×10^6 classes to its right. Right: the tail scale σ, the $10^{-3} \to 10^{-4}$ quantile gap divided by ln 10, which rose until $q = 2^{14}$ and has moved by less than 0.15 since, against 1.27–1.35 for the model.*

Three things are visible. The upper body has converged: the 10^{-2} and 10^{-3} quantiles stopped moving at 2^{16}. The deeper quantiles are slowing: the 10^{-4} quantile moved +0.08, +0.31, +0.12 and +0.03 per doubling from 2^{14}, and the 10^{-5} quantile about +0.4 per doubling on average, though each of its values rests on about 20 classes. And the shape stays exponential: the real generalized-Pareto shape lies between −0.05 and 0.00, the same family as the model's (−0.04 to −0.02), with a scale about 1.25 times larger ($\sigma \approx 1.6$ against 1.3).

Records. Extrapolating the real tail from its deepest measured quantile u, at level f, to the 1-in-M level gives

$$R(M) = u + \sigma\cdot(\ln(fM) + \gamma_E),\ \ sd = \sigma\pi/\sqrt{6}, \quad (8.1)$$

with γ_E Euler's constant. At $q = 2^{12}$–2^{14}, where every class was enumerated, this reproduces the actual records (18.62 against 18.73, 22.58 against 22.13, 24.69 against 24.60). For every class at 2^{15}–2^{18} it projects records of 28.0, 29.7, 33.0 and 35.3, each ± about 2. The records of the samples, which are actual constructions, sit below the projections, as they should, since a sample of 2×10^6 classes reaches only the 1-in-2×10^6 level. The sample record at 2^{16}, excess 28.82 at class t = 81422675 with d = 17, has not yet been checked against published ruler tables and should not be called a record until it is.

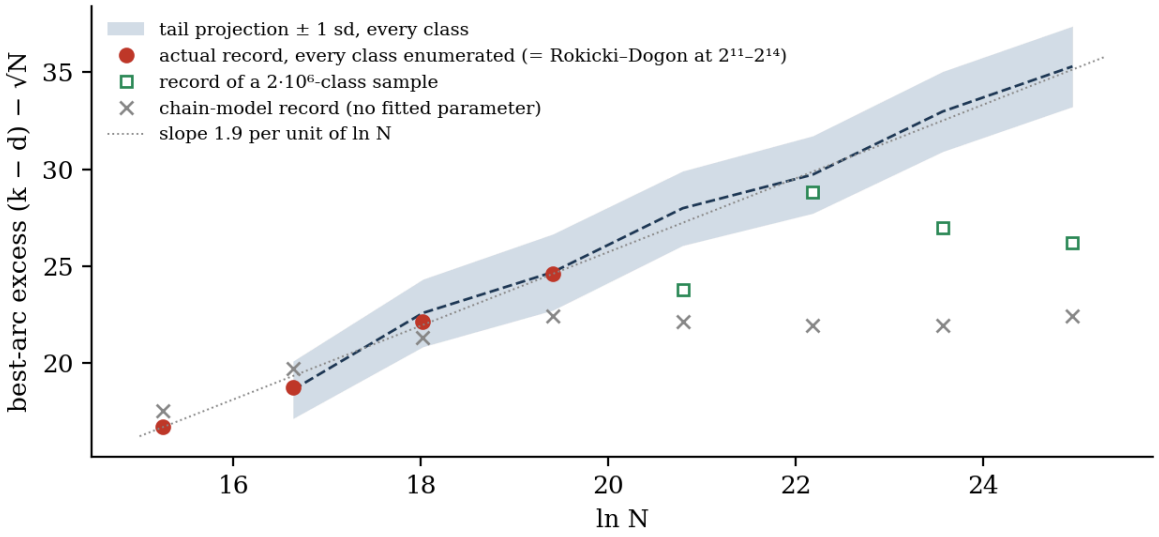


Figure 7. *Best-arc excess against ln N. Actual records with every class enumerated ($q = 2^{11}$–2^{14}, equal to Rokicki and Dogon's rulers), tail projections (8.1) for every class at 2^{12}–2^{18} with one standard deviation, records of the 2×10^6-class samples, and the chain model's record. The dotted line has slope 1.9 per unit of ln N.*

Growth. From 2^{14} to 2^{18} the record series rises 1.90 per unit of ln N. Of that, 1.53 is the tail scale σ times the growth of ln M, and the remaining 0.37 is the deepest quantiles still creeping. If the creep stops, the slope settles near 1.6; if it continues at its current pace, it stays near 2. Either way the growth is linear in ln N; a power law would need the creep to accelerate, and it is slowing from the bulk outward.

Next to the origin. The chain model assumes that free positions next to the origin are marks independently. At small q they are not.

q	dispersion, both sides	dispersion, control window	corr(left, right)	corr × q
2^{11}	0.803	0.921	−0.1094	−224
2^{13}	0.930	0.968	−0.0343	−281
2^{14}	0.961	0.976	−0.0196	−321
2^{16}	0.988	0.994	−0.0048	−315
2^{18}	0.996	1.001	−0.0016	−419

Table 27. *Counts of free marks in [−40q, 40q]: variance-to-mean ratio on both sides together and in a control window far from the short orbits of $x \mapsto 2x$, and the correlation between the two sides (chain model: dispersion 1, correlation 0).*

The anti-correlation fades roughly like 1/q. It makes sparse stretches rarer than the model allows, which explains the deficit at small q and its disappearance. By 2^{16}–2^{18} the counts are binomial to within 1%, including their low tails, yet the excess persists. The excess is therefore in how the free marks are arranged, not in how many there are; its mechanism is unidentified.

A formal bound on the tail shape

The shapes in Table 26 were turned into a confidence bound. Profile-likelihood generalized-Pareto fits were made per q at five thresholds, from the top 1% to the top 0.1%, with a joint common-ξ profile, DerSimonian–Laird pooling across q and Cochran's Q for heterogeneity. Three calibrations precede the result. On synthetic generalized-Pareto tails (200 replicates per cell, $\xi_0 = -0.05$, 0 and +0.05, with 200

and 2,000 exceedances) the intervals cover at 94.0–96.5% two-sided and 92.5–95.0% one-sided. As a positive control, the gap at the origin divided by q, which is Gamma(2, 2q) by Run 1, gives intervals covering the Gamma(2) penultimate shape, about −0.01, at every q, with scale near 2. And re-running the class enumeration at 2^{11}–2^{14} reproduces every record exactly and the shapes −0.099, −0.041, −0.080 and −0.030 to three decimals.

statistic	data	estimate	one-sided 95% upper bound
pooled shape ξ	every class, 2^{11}–2^{14}, thresholds top 1% to top 0.1%	−0.034 to −0.050	−0.029 to −0.008
pooled shape ξ	Run 3 shapes, 2^{14}–2^{18}, top 0.1%	−0.027 ± 0.010	−0.011
drift in N of the GPD scale	2^{14}–2^{18}, top 0.1%	+0.012	+0.031
drift of σ from the $10^{-3} \to 10^{-4}$ gap	2^{14}–2^{18}	+0.013	+0.023
drift of the 10^{-4} quantile	2^{14}–2^{18}	+0.007	+0.009
drift of the 10^{-5} quantile	2^{14}–2^{18}	+0.012	+0.017

Table 28. *The tail shape and the local exponent in N of the tail scale and of fixed-level quantiles (slope of the log of the statistic against ln N). At 2^{15}–2^{18} the standard errors are asymptotic, computed from exceedance counts. Cochran's Q over 2^{14}–2^{18} is 2.93 on 4 degrees of freedom (p = 0.57, I^2 = 0): the shapes scatter no more than independent-class standard errors allow. A true ξ above 0.024 would have been detected with 80% power.*

Result. For characteristic-2 Singer arcs through the origin at q = 2^{14}–2^{18}, the per-class tail shape is $\xi \leq -0.011$, and the tail scale and fixed-level quantiles drift at most like N^0.03, each at one-sided 95%. Under stationarity of the per-class law, the family's excess therefore grows no faster than about N^0.03, and the slowing drift favours logarithmic growth. Four caveats remain: only arcs through the origin were fitted (other arcs were 24% of the top 1% at 2^{14}); the standard errors at 2^{15}–2^{18} are approximate until the profile fits are re-run on the full per-class data; the maximum-likelihood estimate is biased by about −0.01 at 200 exceedances, though negligibly at 2,000; and all classes are images of one set, so bootstrap spreads and count z-scores that treat them as independent are optimistic.

What the runs establish

For q = 2^{11}–2^{18}, empirically: the plain cut is fully explained by a two-component model with no fitted parameter, the largest gap grows like 4q·ln q and the excess like 2 ln q; the best-arc law stops changing with q apart from its deepest quantiles, its tail is exponential, and records grow like σ·ln M, with c ≈ 1.6–2.0 per unit of ln N; and no component has a positive tail shape. Not established: why the tail scale is 1.25 times the chain model's; behaviour beyond 2^{18}, where the projections reach about 2.6 decades past the measured tail against 1–1.3 decades in the validated cases; and other families — Bose, Ruzsa and odd-characteristic Singer sets were not run at this depth, although the control at q = 2003 showed no origin effect, as the small Frobenius factor predicts.

These results close the characteristic-2 Singer route to a "false" answer, down to the detection floor of about N^0.03. They say nothing about the upper bound. The cleanest next test is to enumerate every class at 2^{15} (about 2.2 hours) and compare its record with the projected 28.0 ± 1.9; larger samples at 2^{16}–2^{18} would settle whether the creep of the 10^{-5} quantile stops, which decides between slopes 1.6 and 2.

Part IX · The lower side: an almost-all bound

Versions 4 and 5 recorded the lower side of the problem as $h(N) \geq \sqrt{N} - O(N^{0.2625})$, from the Baker–Harman–Pintz bound on prime gaps, and called it an asymmetry, since the lower exponent is worse than the proved upper exponent 1/4. That comparison is a worst case over N. The exponent 0.2625 is set by the longest prime gaps near $\sqrt{N}$, and those are rare. R. Li's preprint arXiv:2308.04458 (v8, 2025) proves that $[x - x^{0.52}, x]$ contains a prime for all large x, which lowers the uniform figure to $\sqrt{N} - O(N^{0.26})$; Li states the Sidon consequence himself (his Theorem 10). The asymmetry with 1/4 survives either way.

This part asks how large $\sqrt{N} - h(N)$ is for typical N. The answer comes from exceptional-set theorems for primes in short intervals, the input van Doorn and Tang use for gaps in multiplicative Sidon sets. No new construction is needed: Singer sets and a transfer lemma from the Hulak–Ramos–de Queiroz library suffice. The other half of van Doorn and Tang's method, a local-lemma repair, does not transfer: a near-extremal Singer core admits no single addable point, freeing one point needs about 0.15q deletions, and a prime gap can be bridged from below by at most one extra element (repair_feasibility.py).

The deficit is a backward prime gap

Lemma 9.1 (Singer transfer). *Let $p \geq 5$ be prime with $p^2 \leq N$. Then $h(N) \geq p + 1$.*

Proof. Singer's construction gives a Sidon set of size $p + 1$ modulo $M = p^2 + p + 1$. Deleting its largest cyclic gap leaves an interval Sidon set of full size whenever $N \geq p^2 + p + 2 - \lfloor 3p/2 \rfloor$ (Hulak, Ramos and de Queiroz, §4.5; in Lean, IsSidonMod.exists_full_intervalSidon_of_singer_parameters_clean). Since $\lfloor 3p/2 \rfloor = p + \lfloor p/2 \rfloor$, the threshold is $p^2 + 2 - \lfloor p/2 \rfloor \leq p^2$ for $p \geq 4$. ∎

Write $P(x)$ for the largest prime $\leq x$ and $G(x) = x - P(x)$ for the backward prime gap. Lemma 9.1 with $p = P(\sqrt{N})$ gives, for $\sqrt{N} \geq 5$,

$$\sqrt{N} - h(N) \leq G(\sqrt{N}) - 1 < G(\sqrt{N}). \quad (9.1)$$

For $0 < \alpha < 1$ let $B_\alpha = \{x \geq 1 : (x - x^\alpha, x] \text{ contains no prime}\}$. If $\sqrt{N} \geq 5$ and $\sqrt{N} \notin B_\alpha$, then $\sqrt{N} - h(N) < N^{(\alpha/2)}$. Everything below bounds how many N have $\sqrt{N} \in B_\alpha$.

Lemma 9.2. *On $[1, X]$, B_α is a union of at most $\pi(X) + 1$ intervals. Proof.* $\varphi(x) = x - x^\alpha$ has $\varphi' \geq 1 - \alpha > 0$. For x in a prime gap $[p_k, p_{k+1})$, $x \in B_\alpha$ exactly when $\varphi(x) \geq p_k$, a single interval. ∎

Lemma 9.3. *If $B \subseteq [1, X]$ is a union of K intervals of total length m, then $\#\{N \leq X^2 : \sqrt{N} \in B\} \leq 2Xm + K$. Proof.* An interval $[u, v]$ contains $\sqrt{N}$ for at most $v^2 - u^2 + 1 \leq 2X(v - u) + 1$ integers N. ∎

Density one: a deficit of $N^{(1/43+\varepsilon)}$

The input is R. Li's theorem on primes in almost all short intervals (arXiv:2407.05651, Theorem 1.1): with $\theta = 1/21.5 + \varepsilon$, for every $B > 0$ and all large X, all integers $n \in [X, 2X]$ outside a set of $O(X(\log X)^{-B})$ have a prime in $[n, n + n^\theta]$. It sharpens Jia's $\theta = 1/20 + \varepsilon$ (1996).

Proposition 9.4. *Fix $\varepsilon > 0$ and $B > 0$. For all $N \leq Y$ outside a set of $O_{\varepsilon,B}(Y(\log Y)^{-B})$ integers, $h(N) > \sqrt{N} - 3N^{(1/43+\varepsilon)}$.*

Proof. Apply the input with 2ε in place of ε, so $\theta/2 = 1/43 + \varepsilon$. Put $x = \sqrt{N}$ and $n = \lfloor x - 2x^\theta \rfloor$. If n is not exceptional, a prime $p \in [n, n + n^\theta]$ satisfies $x - 3x^\theta \leq p \leq x - x^\theta < x$, and Lemma 9.1 gives $h(N) \geq p +$

$1 > \sqrt{N} - 3N^{(\theta/2)}$. The map $x \mapsto x - 2x^{\theta}$ has derivative above ½, so each n arises from an x-interval of length at most 2, which by Lemma 9.3 contains $\sqrt{N}$ for at most 8n + 1 integers N. Summing over dyadic blocks [Z, 2Z] of exceptional n: blocks with $Z \geq Y^{(1/4)}$ contribute $O(Z^2(\log Y)^{(-B)})$ each and $O(Y(\log Y)^{(-B)})$ in all, and smaller blocks $O(Y^{(1/2)})$. ∎

With Jia's published exponent the same proof gives $N^{(1/40+\varepsilon)}$. Li's preprint has gone through six versions, one correcting an error, so both should be cited until it appears. A later preprint of Li claims $\theta = 1/22 + \varepsilon$, which would give $N^{(1/44+\varepsilon)}$; only its abstract has been checked.

Power-saving exceptional sets

For $0 < \alpha < 1$ let $\nu(\alpha)$ be the least $\nu \geq 0$ such that, for every $\varepsilon > 0$ and all large X, the set of $y \in [0, X]$ for which $(y, y + y^{\alpha}]$ contains no prime has measure at most $X^{(\nu+\varepsilon)}$. In van Doorn and Tang's notation $\nu(\alpha) \leq \lambda(\alpha) \leq \mu(\alpha)$.

Proposition 9.5. *For $0 < \alpha < 1$ and $\varepsilon > 0$, $h(N) > \sqrt{N} - N^{(\alpha/2)}$ for all $N \leq Y$ outside a set of $O(Y^{((1+\nu(\alpha))/2+\varepsilon)} + Y^{(1/2)})$ integers.*

Proof. If $x \in B_{\alpha} \cap [1, X]$ and $y = \varphi(x)$, then $(y, y + y^{\alpha}] \subseteq (y, x]$ has no prime, so y is exceptional; as $\varphi' \geq 1 - \alpha$, $B_{\alpha} \cap [1, X]$ has measure at most $(1 - \alpha)^{(-1)}X^{(\nu+\varepsilon)}$. Lemmas 9.2 and 9.3 with $X = \sqrt{Y}$ finish. ∎ The conversion costs a square root: measure X^{ν} in x becomes $Y^{((1+\nu)/2)}$ in N.

Corollary 9.6 (unconditional). *By Gafni and Tao's bounds, $\mu(\alpha) \leq 10/11$ for every $\alpha > 10/33$ (as read by van Doorn and Tang). Hence $h(N) > \sqrt{N} - N^{(5/33+\varepsilon)}$ for all $N \leq Y$ outside $O(Y^{(21/22+\varepsilon)})$.*

Corollary 9.7 (Lindelöf). *The Lindelöf Hypothesis gives $\mu(\alpha) \leq 1 - \alpha$ (Bazzanella). Taking $\alpha = 2\delta$: for every $\delta > 0$, $h(N) > \sqrt{N} - N^{\delta}$ for all $N \leq Y$ outside $O(Y^{(1-\delta+\varepsilon)})$.*

Corollary 9.7 is the lower half of Erdős #30 with a power-saving exceptional set, for every exponent at once.

prime input	deficit $\sqrt{N} - h(N) \ll$	exceptional $N \leq Y$	assumption
Baker–Harman–Pintz (2001): all gaps $\ll x^{0.525}$	$N^{0.2625}$	none	none
R. Li (2023): all gaps $\ll x^{0.52}$	$N^{0.26}$	none	none (preprint)
Stadlmann (2022): $\lambda(\alpha) \leq 1.23 - \alpha$	$N^{(\alpha/2)}$, any $\alpha > 0.23$	$Y^{((2.23-\alpha)/2+\varepsilon)}$	none
Gafni–Tao (2025): $\mu(\alpha) \leq 10/11$ for $\alpha > 10/33$	$N^{(5/33+\varepsilon)} \approx N^{0.152}$	$Y^{(21/22+\varepsilon)} \approx Y^{0.955}$	none
Jia (1996): $\theta = 1/20 + \varepsilon$	$N^{(1/40+\varepsilon)}$	$o(Y)$	none
R. Li (2024): $\theta = 1/21.5 + \varepsilon$	$N^{(1/43+\varepsilon)} \approx N^{0.023}$	$Y(\log Y)^{(-B)}$	none (preprint)
R. Li (2025): $\theta = 1/22 + \varepsilon$	$N^{(1/44+\varepsilon)}$	$o(Y)$	none (preprint)
Cramér (1920): all gaps $\ll \sqrt{x} \log x$	$N^{(1/4)} \log N$	none	RH
Selberg (1943): almost all $(x, x + \psi(x) \log^2 x]$	$\psi(N) \log^2 N$, any $\psi \to \infty$	$o(Y)$	RH
Bazzanella (2009): $\mu(\alpha) \leq 1 - \alpha$	N^{δ}, any $\delta > 0$	$Y^{(1-\delta+\varepsilon)}$	Lindelöf
subpolynomial gaps (Cramér model)	N^{ε}, any $\varepsilon > 0$	none	conjecture

Table 29. *The lower side of the problem under each prime input. The θ rows go through Proposition 9.4, the λ and μ rows through Proposition 9.5, and the uniform rows through Lemma 9.1 alone; the λ and μ inputs are as stated by van Doorn and Tang. Under RH the uniform deficit $N^{(1/4)} \log N$ matches the upper-bound scale, while Selberg's almost-all bound is logarithmic.*

A numerical look. Up to 10^{14} the backward gap D(N) = √N − P(√N), which bounds the deficit by (9.1), has a median near 0.6·ln √N, and on the top two decades it never exceeds 0.12·N^(1/4).

Y	median D(N)	median / ln √N	99.9th percentile	max D(N)	max D(N)/N^(1/4), N ∈ (Y/100, Y]	share of N ≤ Y with D(N) > N^(1/8)
10^8	4.94	0.568	32.3	36	0.922	27.1%
10^{10}	6.39	0.581	51.6	72	0.406	16.5%
10^{12}	7.96	0.599	70.0	114	0.218	7.3%
10^{14}	9.52	0.610	85.6	154	0.113	2.1%

Table 30. *Medians and percentiles from 2×10^6 uniform samples of N per row; maxima and shares exact over all prime gaps below √Y. The maxima are the known maximal prime gaps below 10^4, …, 10^7, a check on the code (almost_all_illustration.py). This is an illustration, not evidence for an asymptotic claim: D(N) only bounds the deficit, and h(N) is known exactly only for N ≤ 586.*

What this changes

For all N outside a set of density zero, Proposition 9.4 and Theorem 5.4 give

$$-3N^{(1/43+\varepsilon)} < h(N) - \sqrt{N} \le 0.9428095\cdot N^{(1/4)} + O(1), \quad (9.2)$$

so for typical N the lower error is the smaller of the two by a power of N, and the asymmetry of Versions 4 and 5 reverses. If the upper half holds — h(N) ≤ √N + C_ε N^ε for every ε > 0, the SidonUpperBoundHypothesis of Hulak, Ramos and de Queiroz — then |h(N) − √N| ≪ N^ε for almost all N: unconditionally for every ε > 1/43, and under Lindelöf for every ε > 0 with O(Y^(1−ε+o(1))) exceptions. The all-N version needs subpolynomial prime gaps, beyond the √x log x that RH gives; Hulak, Ramos and de Queiroz prove in Lean that subpolynomial gaps together with the upper half imply Erdős #30 (their Corollary 6.3, erdos30_of_gap_and_upper).

A negative answer therefore needs one of two things: prime gaps of size x^c infinitely often for some c > 0, against the Cramér model, with no other construction filling them; or a failure of the upper half, some c ∈ (0, 1/4] with (h(N) − √N)/N^c unbounded. Under the first route the failing N have density zero for each fixed ε > 1/43. The second route is the one Parts III–VI and VIII bear on.

The result says nothing about the exceptional N: monotonicity of h cannot reach them, since bridging from a good N′ < N costs √N − √N′, again a prime gap. It needs no new construction and suggests none: Singer, Bose–Chowla and Ruzsa sets all need prime or prime-power parameters, and covering every N needs short prime gaps everywhere or a construction not indexed by primes. The uniform Sidon consequence of prime-gap bounds is classical, and Li states it for gaps x^0.52; the almost-all version was not found in any source checked, but it is a short consequence of known theorems and is presented as a remark, crediting the exceptional-set idea to van Doorn and Tang. Three literature checks remain open: O'Bryant's annotated bibliography and its updates, the applications sections of Li's 2024 and 2025 preprints, and a zbMATH or MathSciNet search.

Part X · What data can and cannot decide

Version 6 assessed which statistical and predictive methods can bear on a true/false answer (14 October 2026). None can decide it, in either direction, and the limit comes from the form of the statement rather than from the methods. What they can do is narrower: move a calibrated credence, bound what particular construction families achieve, and help find checkable objects — constructions, certificates and mechanisms — that a proof then uses.

Why no dataset decides the problem

The statement has three alternating quantifiers,

$$\forall\ \varepsilon\ >0\ \exists C\ \forall N : |h(N) - \sqrt{N}| \le C \cdot N^{\varepsilon}, \quad (10.1)$$

so it is Π_3: every instance is computable, but neither the statement nor its negation can be settled by finitely many. A "false" answer needs a proved infinite family, like the ineffective lattice argument for Erdős #1; a "true" answer needs an upper bound below N^(1/4), plus the lower half of Part IX.

There is also an identifiability wall. $N^{\varepsilon} = e^{(\varepsilon \ln N)}$ is indistinguishable from $1 + \varepsilon \cdot \ln N$ until $\varepsilon \cdot \ln N$ is about 1, which needs $\ln N \approx 10$ for $\varepsilon = 0.1$ ($N \approx 2\times10^{4}$), 33 for $\varepsilon = 0.03$ ($N \approx 3\times10^{14}$), 100 for $\varepsilon = 0.01$ ($N \approx 10^{43}$) and 1000 for $\varepsilon = 0.001$ ($N \approx 10^{434}$). The largest N reached in Part VIII is 6.9×10^{10}, with $\ln N \approx 25$. Data can only ever say that a family gives an exponent of at most c_0, with c_0 shrinking roughly like $1/\ln N_{max}$; Part VIII's $c_0 \approx 0.03$ is such a statement. The project's own records show the wall:

model for the four full-population records, q = 2^{11}–2^{14}	fit	RMSE	prediction at N = 10^{100}
logarithmic	1.95·ln N − 13.2	0.281	435
power law	3.88·N^0.096	0.272	1.4×10^{10}

Table 31. *Two models fit the records equally well. A Box–Cox fit $a + b\cdot(N^{\alpha} - 1)/\alpha$ has a flat residual sum of squares from $\alpha \approx -0.1$ to 0.25, and with record-to-record noise of about $\sigma\pi/\sqrt{6} \approx 2$ nothing in $\alpha \in [-0.3, 0.5]$ can be excluded. Over the exact range N = 50…586, corr(N^(1/4), ln N) = 0.997 and corr(N^0.05, ln N) = 0.9999. The tail statistics of Part VIII, not the records, are what separate the models.*

Which methods help

method	can it decide #30?	best use here	value
extreme-value and tail statistics	no	confidence bound on the exponent per family (Part VIII)	high
Bayes' theorem	no	explicit evidence ledger; isolates the "new family" gap	high
linear regression	no	gain-vs-N triage of relaxations; the bridge from algebraic cuts to optimal rulers	medium
logistic regression, gradient boosting	no	finding the arrangement statistic behind the 1.25× tail scale	medium
symbolic regression, program search	no, but can find objects	conjectured closed forms; new certificates or constructions	medium
central limit theorem	no	window counts; a heuristic for why N^(1/4)	low to medium
exponential moving average	no	smoothed local exponent of tail quantiles	low
Fibonacci sequences	no	numerical utilities only	negligible

Table 32. *Tree ensembles are piecewise constant in every feature and cannot extrapolate in N, so they are useful only for the mechanism hunt, validated by splitting on q or on orbits rather than at random. Gaussian-process extrapolation is decided by the kernel's asymptote, and conformal prediction needs exchangeability across N, which fails here.*

Two readings deserve a sentence each. The central limit theorem is the wrong limit theorem for the excess, which is driven by maxima of gaps and governed by Fisher–Tippett–Gnedenko; but N^(1/4) = √(√N) is exactly the standard deviation of a count with mean √N, and the Erdős–Turán, Lindström and Cilleruelo arguments are second-moment arguments whose leftover term is the size of fluctuations they cannot rule out. On that heuristic reading a "true" proof must show that near-extremal Sidon sets are far more regular than CLT scale, which needs information beyond one- and two-point statistics — where Parts IV–VI end. And the only machine-learning method that has moved this problem's numbers is program search (the AlphaEvolve-assisted 0.97633); Theorem 4.1 closes that route inside the smoothing framework, so its remaining value is in searching new certificate frameworks or new constructions.

An explicit evidence ledger

Bayes is the right frame for credence, and its main value is to split the probability of "false" into parts that the evidence affects very differently. The ledger (Erdos30_Bayes_ledger.xlsx) has five routes and seven evidence items; priors and likelihood ratios are editable inputs, and posteriors, the ceiling and a sensitivity grid are formulas.

route	prior	product of likelihood ratios	posterior
true: h(N) = √N + O_ε(N^ε)	0.60	1	0.744
false via known families, c ≥ 0.04	0.10	0.025	0.003
false via known families, 0 < c < 0.04	0.05	1	0.062
false via new constructions	0.20	0.72	0.179
false on the lower side	0.05	0.2	0.012

Table 33. *Default (illustrative) priors. The evidence items and their ratios: exact h(N) for N ≤ 586 (ratio 1 for every route); algebraic cuts matching optimal rulers at k = 17–28 (0.8 against new constructions); the Stage 0 survey (0.5 against known families with c ≥ 0.04); Runs 1–3 with the tail-shape bound (0.05 against the same); stochastic search failing beyond N = 586 (0.9 against new constructions, since search is known to be weak there); the 85-year stall and the floor theorems (ratio 1: the stall is a property of the methods, expected under every route); and prime-gap data tracking (ln p)² (0.2 against the lower-side route).*

With these priors P(true) moves from 0.60 to 0.74. Its ceiling, if the known-family route with c ≥ 0.04 were driven to zero, is 0.746, however strong the construction-side statistics become, and 94% of the remaining probability of "false" sits on routes the data cannot reach: new constructions, and power laws in known families below the detection floor. The judgments that matter are therefore not about the data. Moving along a row of the sensitivity grid — stronger construction-side evidence — changes P(true) by at most 0.04; moving down a column — a larger prior on new constructions — changes it by 0.27.

prior on new constructions	LR 0.001	LR 0.01	LR 0.05	LR 0.2	LR 1
0.02	0.913	0.912	0.910	0.902	0.862
0.10	0.841	0.841	0.839	0.831	0.794
0.20	0.746	0.746	0.744	0.737	0.703
0.30	0.644	0.644	0.642	0.636	0.605

Table 34. *P(true | evidence) as the prior on new constructions (rows; the prior on "true" absorbs the change) and the likelihood ratio of the Part VIII evidence against known-family power laws (columns) vary. Recent disproofs of "the classical*

construction is optimal" conjectures — Erdős #1, the unit distance conjecture, #707 — argue against setting the prior on new constructions very small.

Triage of upper-bound strengthenings

A strengthening of an upper-bound relaxation lowers the N^(1/4) constant only if its gain Δ(N) = c·N^(1/4) + b + o(1) has c > 0, and removes the term only if c reaches 2√2/3. The triage fits each strengthening's measured gains against N, with two calibrations whose answers are known.

strengthening	N range	result	verdict
calibration: pair level against the smoothing bound (c = 0 by theorem)	20–180	$c \le 0.006$	drop (correct)
calibration: exact h(N) against pair level (an ideal strengthening)	20–180	$c \approx 0.41$ (≤ 0.57)	keep (correct)
three-point against pair level	12–40	gain shrinking, slope ≤ -0.97; $c \le 0.019$	drop; now Theorem 6.8 (c = 0)
window lift and bridge identity from the Erdős #152 proof	12–40	gains $\le 1.2\times10^{-6}$	drop
four-point with window polytope, or with H_val	12–26	solver noise	drop
four-point against three-point	20–28	gains 0.236, 0.210, 0.200, 0.181 at N = 20, 22, 25, 28; slope ≤ -0.42; $c \le 0.079$	drop (weakest evidence)

Table 35. *Gain-vs-N regressions (gain_vs_N.py). The four-point value at N = 28, 7.20944 against a three-point value of 7.39092, is new; the pair, three-point and four-point values at N = 20 reproduce Table 16 and results.csv. The bounds on c are one-sided 95% limits from the fits.*

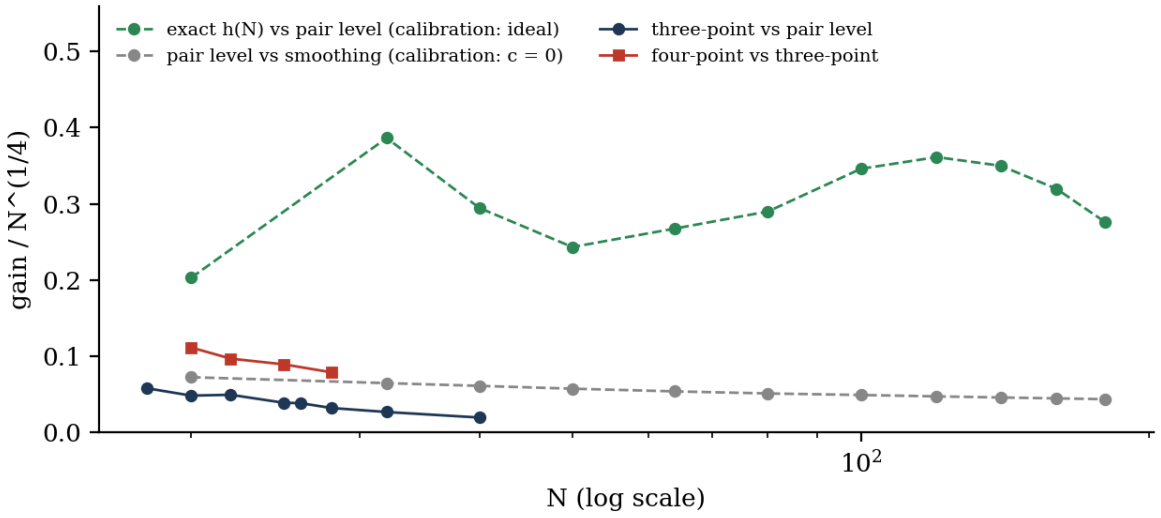


Figure 8. *Gain of each strengthening divided by N^(1/4). Moving the N^(1/4) constant requires this ratio to level off above zero; the three- and four-point gains fall, like the pair-level calibration whose limit is zero by theorem.*

The triage is the only data analysis aimed at the side from which a "true" answer must come, and its three-point verdict is now a theorem. The four-point verdict rests on N ≤ 28; four-point values at N = 30–40 would strengthen it, at about 13 minutes per point with SCS and over 3 GB of memory with Clarabel. The same reasoning points to a provable intermediate target on the other side: logarithmic

growth of the best Singer excess means that $h(N) = \sqrt{N} + O(1)$ is very likely false, which is exactly the Stage 1 question of Part VII.

Part XI · Formalization in Lean 4

The project's Lean work builds on the library of Hulak, Ramos and de Queiroz (github.com/d0d1/singer-theorem-lean, commit 0c89058), which formalizes Singer's construction and Sidon-set infrastructure and already contains several results this report uses: the full-size interval transfer of Lemma 9.1, the Lindström bound, the additive-energy identity for Sidon sets, the distinct-gap bound $\lceil 3q/2 + 1/(q+1) \rceil$ cited in Part VII, and the conditional reduction from subpolynomial prime gaps and the upper half to Erdős #30. The package SidonBarriers (Lean v4.29.0, Mathlib v4.29.0) states the barrier side on that library in three modules and one generated example.

module	statements	checked	difficulty
FourierL4	$\int_0^1 \lvert S_A(\theta)\rvert^4 \, d\theta = E(A)$ for every finite $A \subset \mathbb{Z}$, using Mathlib's addEnergy; $2k^2 - k$ for Sidon sets	numerically; the old sketch failed on $A = \{-5, 0\}$	easy to moderate
WeakDuality	renewal density in closed form; Lemma 4.4; the Wiener–Hopf identity for R*; continuum certificates with integrability fields; Theorem 4.7 and $\gamma \geq 2\sqrt{2}/3$	numerically at 7–8 points each	hard: the main target on this side
WeakDuality	the linear certificate, $a = 4/3$ and $b = 2/3$; rational two-sided certificates and Proposition 3.1 against sidonMaximum; headline $h(N) \leq \sqrt{N} + 0.942809522 \cdot N^{(1/4)} + c_0$	exact-rational Python check of the certificate	data import plus Proposition 3.1
SDPCertifier	exact rational dual certificates for $\hat{h}_2$: an explicit dual matrix and an LDL^T factorization witness; sidonMaximum $N \leq \lfloor V \rfloor$	dual optimum matches $\hat{h}_2(12) = 5.1194$ and $\hat{h}_2(40) = 8.7405$	easy to moderate
PairCert12	$h(12) \leq 5$ from a generated rational certificate	exact LDL^T in Python	depends on the speed of decide

Table 36. *Status: statements only, not yet compiled; every sorry is a proof still owed. The soundness argument of the certifier is short because the primal point of a Sidon set has rank one.*

The package replaces an earlier sketch with three defects, each found by checking statements numerically before formalizing them. Its lower-bound theorem was false as stated, because b was a free variable and $b = 0$ satisfied its hypothesis; b is now computed from the certificate data, and integrability is part of the certificate, since a non-integrable function integrates to 0 in Lean. Its L^4 identity summed over differences in [1, max A], which misses differences when A has negative elements. And its trace-form module is not carried over: the "field-theoretic multiplier" over $GF(q^3)$ that it was meant to support is identically -1, since Σ over $\gamma \in GF(q^3)^\times$ of $\chi(\mathrm{Tr}(\gamma\theta^u)) = q^2 \cdot \Sigma_{y \in GF(q)} \chi(y) - 1 = -1$, and a constraint that holds only for Singer sets bounds the maximum over that family, not h(N). One caution carries over from the literature: the Formal Conjectures statement of Erdős #707 was wrong because it allowed modulus 0, so the statement of #30 used by FrontierMath Erdős should be checked before anything is called machine-checked against it.

Part XII · Where this leaves the programme

stage	content	status
Stage 0	Construction survey across the Singer, Bose and Ruzsa families and all characteristics	Complete. Outcome A: no power-law growth (Part II). Extended in characteristic 2 to $N \approx 6.9\times10^{10}$ (Part VIII): excess $\approx c\cdot\ln N$ with $c \approx 1.6$–2.0; tail shape $\xi \le -0.011$ and drift $\le N^{0.03}$ at one-sided 95%.
Stage 1	Prove that the largest gap over multipliers is not O(q), refuting h(N) = √N + O(1) (O'Bryant's Open Question 5)	Open, with first results (Part VII). Proved for every multiplier: gap $\ge 2q$ (Singer), $\ge 2q-1$ (Bose–Chowla); single-coefficient arguments cannot pass $3q-2$ and $3q-4$. Computed: minima $\ge 2q+2$ (Singer, $7 \le q \le 419$), growing like $q(\ln q - \ln\ln q)$.
Stage 2T(a)	Determine the floor of the smoothing framework	Closed. $\gamma^* = \gamma^*_{\text{sym}} = 2\sqrt{2}/3$ (Theorem 4.1, Part V, Appendix D). Verified certificates: 0.9428096 (two-sided, linear kernel) and 0.9428095 (symmetric, original framework). Whether a bounded number of symmetric kernels suffices is open. Chen (forum) claims the explicit bound at $2\sqrt{2}/3$.
Stage 2T(b)	Extract the forced structure of a hypothetical dense Sidon set	Partial. Sets more than halfway to a certificate's bound are sub-Poissonian at scale $\sqrt{(a/b)}\cdot N^{(3/4)}$. The extremal configuration — two atoms and a hyperuniform bulk — is attained with proofs at the pair level (Theorem 6.6) and lifted to three points (Theorem 6.8); these are statements about the relaxations, not genuine Sidon sets.
Stage 2T(c)	Find the second-order constraint separating log N from N^c	Barrier proved through three points: $\hat{h}_2$, $\hat{h}_{2^+}$ and $\hat{h}_3$ all stop at $2\sqrt{2}/3$ up to O(1). Four-point gains shrink at $N \le 28$ (numerical). Arithmetic: the cyclic quotient test is positive; short-difference coverage and the interval quotient test are negative.
Lower side	$h(N) \ge \sqrt{N} - O(N^{\varepsilon})$	Not a stage of the original plan. Uniform $N^{0.2625}$, or $N^{0.26}$ on Li's preprint; $N^{(1/43+\varepsilon)}$ for almost all N (Part IX); all N needs subpolynomial prime gaps.

Table 37. *Stage status after Version 6.*

What an argument below $2\sqrt{2}/3$ must supply

No argument below $2\sqrt{2}/3$ is known, inside or outside the project; searches on 15 and 20 October found none, and erdosproblems.com lists the problem as open. The results above turn into four requirements. An argument below $2\sqrt{2}/3$ must:

- use only constraints that every Sidon subset of {1, …, N} satisfies — constraints true only of Singer, Bose or Ruzsa sets, such as trace identities, bound the maximum over those families, which is what Stage 0 measures, not h(N);
- cut off the explicit point of Theorem 6.6, its three-point lift in Theorem 6.8, and every feasible point of $\hat{h}_{2^+}$ within $o(N^{(1/4)})$ of its value;
- act on the bulk, since boundary atoms meet a full difference budget at O(1) cost (Lemma 6.7, Table 15), even when they are genuine Sidon rulers checked against distinct sums;
- on the interval, tie arithmetic to positions or to the slack in long differences, since short-difference coverage and end integrality fail (Part VI).

Lowering the constant would still leave the exponent at 1/4; reaching N^ε needs the exponent itself to move.

Open questions

Roughly in order of how much they could change the picture:

- **Arithmetic tied to positions on the interval.** The one candidate the tests have not ruled out: integrality on windows of length about N^(3/4) at the ends, where genuine cluster structure meets the distinct-sums constraints. The failed deterministic-atom lift of Theorem 6.8's proof suggests that is where integrality has leverage.
- **Higher fixed levels.** A four-point analogue of Theorem 6.8, or a proof that every fixed level of the moment hierarchy keeps an N^(1/4) term, would make the negative evidence of Tables 17 and 35 precise.
- **Constants.** Tighten Theorem 6.8 (the far-row constant computed in the scaling limit should give $\kappa \approx 3$–5 and a proved constant near 5); make C_0 and κ explicit, for instance through the delay equation and the invariant $\int$ from $t-1$ to t of $(s-t+1)\cdot g(s)\,ds = 0$ for $t \geq 1$; and replace double precision by interval arithmetic in Theorem 6.5, the corner certificate of Theorem 6.8, the root bound $\mathrm{Re}\, s \leq -2.0888$ of Lemma 4.4 and Proposition 5.2.
- **The bound at exactly $2\sqrt{2}/3$ inside this framework.** A continuum version of Proposition 3.1, with monotone Riemann-sum bounds, would give $h(N) \leq \sqrt{N} + (2\sqrt{2}/3)\cdot N^{(1/4)} + C$ for $N \geq N_0$ with explicit C and N_0; improving Chen's $N_0 = 120^4$ or his additive 1 would add to his result rather than duplicate it. Chen's optimality proof should be read in full and the correspondence recorded: his ramp $2(1-t)$ with the linear kernel $2x$, the capacity defect $C_f(L) - L$ with b, and $f(0)$ with a.
- **Symmetric certificates with bounded K.** Does a fixed number of symmetric kernels reach $2\sqrt{2}/3$? The same decaying repair as in Appendix D should remove the factor log m from Proposition 5.2.
- **Stage 1.** (i) The rung $2q+1$ for all q: the δ and $\kappa(2, q-1)$ relations must never hold together at a primitive n-th root of unity. (ii) A Ruzsa analogue of Theorems 7.1 and 7.4. (iii) The O(q) question itself, through runs in m-sequences beyond character-sum bounds or an ineffective existence argument. (iv) A Lean formalization of Theorem 7.1 on the Hulak–Ramos–de Queiroz library.
- **Extreme values.** Every class at $q = 2^{15}$; larger samples at 2^{16}–2^{18}; the mechanism of the 1.25× tail scale, through a sparse logistic model or boosted trees on arrangement features of the free marks (spacings, positions in the dyadic shells, t mod 7, left–right imbalance), validated by splitting on q or on orbits; Bose, Ruzsa and odd-characteristic Singer sets at the same depth, with hierarchical pooling of σ and ξ across families; and a check of the 2^{16} sample record against published ruler tables.
- **The lower side.** The three literature checks of Part IX.
- **Formalization.** Compile SidonBarriers; prove the easy statements first (difference counts, the energy identity, the certifier's soundness, PairCert12), then Theorem 4.7, the main target on this side.
- When the tentative ≈0.947 argument of Carter, Georgiev, Gómez-Serrano, Hunter, O'Bryant, Tao and Wagner appears, check whether it lies in the class Theorem 4.1 covers; a search on 20 October

found no paper yet. A conceptual proof of the weight identity, Lemma 4.6, is still missing; Lemma 6.4 gives one reading of it.

Four things are worth avoiding: further certificate search inside the framework, where the proved room is below 5×10^{-7}; further exact-h(N) tabulation beyond N = 586, which costs years of distributed computation for one more point on a sawtooth the Singer data already explain; higher levels of the moment hierarchy as a route below $2\sqrt{2}/3$ unless they are combined with arithmetic input; and predictive methods that cannot extrapolate in N, used as evidence on the growth law.

Status of the write-ups. "Vector-valued smoothing for Sidon sets cannot beat $2\sqrt{2}/3$" (26 September 2026; author block 5 October) records Parts III–VI as of Version 5. Before submission its Conjecture 7.7 should become Theorem 6.6 of this report, Appendix D should be added, and its abstract can then say that the pair-level relaxation has second-order constant exactly $2\sqrt{2}/3$ up to O(1); it still needs the promised deposit of code and certificates. Its predecessor of 25 September is superseded. Version 6 draws on eight notes written between 7 and 21 October: the literature note on OpenAI-linked results (7 October), the extreme-value draft (12 October), Appendix D (13 October), the proof of Conjecture 6.6 with the interval short-difference test (14 October), the predictive-analytics assessment and the three V6 steps (14 October), the almost-all section (16 October), and the revised Sections 6 and 7 of the companion manuscript "Erdős Problem 30: Theoretical Barriers, SDP Relaxations, and Spectral Mechanics" (21 October), whose requirements are adopted above and whose previous draft's trace-form multiplier and claim that the conjecture is true have been removed. The three-point theorem (20 October) and the interval quotient test (20 October) exist so far as README files with their scripts.

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Appendix A · Data, checks and file inventory

A.1 Verified certificates

certificate	m	L	K	inequalities	a	b	γ
symmetric, original scan	32	4	2	129	1.294133696	0.6874915809	0.94324229
symmetric, original scan	64	4	2	257	1.313278418	0.6770076151	0.94292072
symmetric, original scan	128	4	2	513	1.3216771522	0.6725867335	0.94283748
symmetric, original scan	256	4	2	1025	1.3262775741	0.6702236072	0.94281628
symmetric, original scan	512	4	2	2049	1.3264253589	0.6701415970	0.94281112
symmetric, re-optimized	512	4	2	2049	1.3286071522	0.6690407889	0.9428108915
two-sided, optimized	32	4	2	258	1.3131371443	0.6770803718	0.9429206679
two-sided, optimized	64	4	2	514	1.3230846885	0.6718711005	0.9428374016
two-sided, optimized	128	4	2	1026	1.3281840982	0.6692614413	0.9428162089
two-sided, optimized	256	4	2	2050	1.3306205359	0.6680284116	0.9428108628
linear kernel 2x	256	8	1	4098	1.3333282471	0.6666717951	0.9428108696
linear kernel 2x	512	8	1	8194	1.3333320618	0.6666679802	0.9428095208
symmetric family of Appendix D	128	8	64	1025	1.333251953159	0.666748046890	0.942837809902
symmetric family of Appendix D	256	8	128	2049	1.333312988234	0.666687011774	0.942816234419
symmetric family of Appendix D	512	8	256	4097	1.333328247092	0.666671753640	0.942810840342
symmetric family of Appendix D	1024	6	512	6145	1.333332061750	0.666667938506	0.942809491335

Table A1. *Every certificate was rationalized, its boundary program re-solved for the rational kernels, its boundary vectors rounded and repaired by a uniform increment, and all covering inequalities checked over $\mathbb{Q}$ (Lm + 1 for symmetric, 2(Lm + 1) for two-sided certificates); a, b and γ^2 were then evaluated exactly. Mixing weights λ_1: 0.3592, 0.3440, 0.3414, 0.3397, 0.3396 for the original symmetric scan, 0.3383 for the re-optimized certificate, and 0.6570, 0.6630, 0.6669, 0.6710 for the two-sided scan. The two-sided certificates were regenerated on 8–9 October 2026 with the original code, since the Version 4 files had not been preserved: γ agrees with Version 4 to eight digits and is lower by about 10^{-9}, while the mixing weights differ from Version 4's ($\lambda_1 \approx 0.15$), confirming that only γ is determined. The deposited symmetric certificates are the exact rational certificates rounded to double precision; read as exact binary numbers they verify after a uniform repair of at most 6.2×10^{-18}, which leaves a, b and γ unchanged to ten digits. All certificates are deposited as text (A.4) with two independent exact verifiers. Because the optimum is flat in some directions, a and b of the optimized rows are not individually converged; only γ is. For the linear kernel, $a = 4/3 - 1/(3m^2)$ exactly. The last four rows, added in Version 6, are the symmetric certificates of Table 10 (13 October 2026): kernels rounded to integers over 10^{12}, repaired by uniform increments of at most 2.3×10^{-11}, checked over $\mathbb{Q}$ by three checkers including check_lin_cert.py; for them $a = 4/3 - 1/(3n^2)$ with $n = m/2$ before rounding. Their text files are not yet deposited in the project.*

A.2 Stage 0 survey (stage0_results.csv)

174 rows: 34 Singer, 29 Bose and 111 Ruzsa instances. Columns are family, q, p, e, n, k, maxgap, maxgap/q, the extreme-value prediction and ratio, the excess at zero drops and at the best drop count, the best drop count d, the resulting interval length N, and the full excess profile for d = 0…8. The excess at zero drops is $k - \sqrt{(n - \text{maxgap} + 1)}$. Representative rows at the top of the range:

family	q	n	k	max gap	EV ratio	excess (cut)	excess (best arc)	d	N
singer	1009	1019091	1010	20503	1.097	10.71	11.50	5	987047
singer	1024	1049601	1025	22134	1.340	11.36	12.14	2	1021844

family	q	n	k	max gap	EV ratio	excess (cut)	excess (best arc)	d	N
singer	1369	1875531	1370	29683	1.169	11.38	12.10	4	1833045
bose	1024	1048575	1024	20886	1.249	10.25	12.36	3	1017354
bose	1331	1771560	1331	24888	1.066	9.38	11.58	2	1735605
ruzsa	1447	2092362	1446	16574	0.898	5.24	6.90	1	2068136

Table A2. *Selected rows from the top of the Stage 0 range, taken directly from stage0_results.csv; N is the interval length at the best drop count d. The Singer and Ruzsa rows printed in Version 3 did not match the file and are corrected here.*

A.3 Independent checks for Versions 4 to 6

what was checked	how	result
Renewal identities, Lemmas 4.4–4.6	renewal equation solved on a grid of step 10^{-4} over [0, 40]	$\int(r-2)^2 = 0.333312$, $\int(r-2) = -0.333383$; A(t) formula to 4×10^{-5} on [0,1]; h = 8/3 on (0,1) to 5×10^{-5}; min h on [1, 12] = 11/3 at s = 1+
Extremal continuum certificate, Theorem 5.1	same grid	$\int(\omega^2 - 1) = -0.33351$; covering integral 1 to 10^{-4} for $u \in [0, 6)$; $\omega(1-) = e - 1$, $\omega(1+) = e - 2$
Linear-kernel boundary program at L = 8, Table 9	CVXPY with Clarabel, both ends solved separately	γ = 0.942924090, 0.942837833, 0.942816257 at m = 32, 64, 128; b − 2/3 as printed; reversed end optimal at $v \equiv 1$
Pair-level relaxation, Tables 14 and 16	CVXPY with Clarabel, no symmetry reduction	$\hat{h}_2$ = 5.1194, 6.4291, 7.9189, 8.7405 at N = 12, 20, 32, 40
Hou–Zhao constants	arXiv 2607.01169, current version and v1 abstract	$\gamma_0 = 0.9434925907\ldots$, eight kernels, m = 32, L = 4; one symmetric kernel 0.9461473014; v1: 0.94601
Part I fits	optimal Golomb ruler lengths, $k \le 28$	γ = 0.6404 (RMSE 0.3847); c = 2.639 (RMSE 0.4899); α = 0.2013 on values, 0.1977 in log–log; D(k) slope 1.426
Stage 0 summaries, Table 2 and Table A2	stage0_results.csv	Table 2 reproduced to ≤ 0.002 after excluding $q \le 16$ (Singer) and $p \le 43$ (Ruzsa), except the Bose odd-characteristic cut slope (0.765, printed 0.756 in V3)
Figure 5	quadrature with 2×10^5 points	cosine arch 0.308425 = $\pi^2/32$; parabolic 0.308571; all other densities higher
Deposited certificates (Version 5)	exact integer arithmetic from the text files alone: certify2.py, the independent reader check_lin_cert.py, and verify_sym_certs.py	linear kernel m = 256, 512: $\gamma <$ 0.942810871, 0.942809522; symmetric scan m = 32–512: Table A1 to every digit; regenerated two-sided m = 32–256 and re-optimized symmetric m = 512: Table A1
Explicit feasible points, Table 12	recovered construction code re-run	Σf = 108.4407, 331.9689, 1028.7938, 3214.2844; every difference sum $\le 1 - 1.3\times10^{-8}$; N = 3000 end-to-end check, smallest eigenvalue 6.1×10^{-5}
Part VI relaxations, spot checks	recovered SDP code	$\hat{h}_2(12)$ = 5.1194, $\hat{h}_2(20)$ = 6.4291, $\hat{h}_3(12)$ = 5.0000; cyclic model at q = 5: 5.0000 / 4.2983 / 4.2983 against genuine 5 / 4 / 4; Table 15 at $N = 10^4$: Σf = 107.324

what was checked	how	result
Interior coefficients, Part VII	recovered scripts re-run	δ, E, F agree for every prime power $q \leq 128$; false positives q = 11, 29, 59, 67, 89; the $\kappa(2, q-1)$ relation excludes all of them, $q \leq 101$; generating-function formula and recursion exact for q = 3, 4, 5, 7
Stage 1 enumerations, Tables 21 and 22	singer_enum.py, bose_ruzsa_enum.py, isolated_class_checks.py, check_proofs.py	as stated; maxima agree with stage0_results.csv at every shared q
Characteristic 2, Table 4	singer_best_arc.py, run independently on Linux and Windows	identical output at $q = 2^{11}$; all four rows equal Rokicki–Dogon's records
Theorem 6.6 construction (Version 6)	c66_verify.py, numpy only; FFT at every lag, then direct summation	Table 13 at $N = 10^4$–10^7; Lemmas B.1–B.4 one by one at 10^4–10^6; renewal tables against $\int g = -1/3$, $\int g^2 = 1/3$ and A(s) = e^s − 2s − 2/3; N = 3000 explicit Γ: smallest eigenvalue 2.0×10^{-4}; direct summation at 10^4 reproduces 0.970588621
Theorem 6.8 (Version 6)	lift_psd.py, stress3.py, chunk.py, lemma1.py, lemma7.py, corner_cert.py, assembly.py	Table 18; identities (P1)–(P3) to 10^{-16}; corner constant ≤ 0.925 by branch and bound in double precision; with $\theta = 0$ the code reproduces the Theorem 6.6 shortfalls 2.70–2.62
Appendix D and symmetric certificates (Version 6)	appD_construct.py, appD_checks.py; certify_sym.py, check_sym_cert.py and check_lin_cert.py	every covering sum ≥ 1 for n = 4–32; $n^2 \cdot \sup\lvert R_n - R^*\rvert$ = 0.33–0.50; certificates of Table 10 agree across three checkers, and a corrupted entry is rejected
Runs 1–3 (Version 6)	singer_origin_decomp.py, singer_origin_arcs.py, singer_run3.py	every class at 2^{11}–2^{14} agrees with Table 4 and across runs; sampled classes at 2^{15}–2^{16} by brute force, at 2^{16} in exact integers
Tail-shape bound (Version 6)	xi_bound.py with --simulate and the origin-gap control; drift.py	coverage 94.0–96.5% two-sided; shapes −0.099, −0.041, −0.080, −0.030 reproduced to three decimals
Arithmetic tests (Version 6)	cyclic_arith_test.py; interval_short_diff.py with isd_enum.c; interval quotient scripts	published rulers k = 6–28 verified Sidon and optimal; $\hat{h}_2^+(46) = 9.2885$ and $\hat{h}_2^+(55) = 10.0523$ reproduced; dense check of the boost construction at N = 3000
Almost-all section (Version 6)	almost_all_illustration.py; repair_feasibility.py	maxima of D(N) equal the known maximal prime gaps 36, 72, 114, 154
Gain-vs-N triage (Version 6)	gain_vs_N.py, run_gain.py	calibrations recovered ($c \leq 0.006$ where c = 0, $c \approx 0.41$ kept); pair, three- and four-point values at N = 20 reproduce Table 16

Table A3. *Checks carried out for Versions 4 to 6, independently of the code that produced the original results where possible. The Version 4 checks are in v4_checks.py; the Version 5 and 6 checks are in the packages listed in A.4. The last rows were added in Version 6.*

A.4 Files

Every certificate is deposited as plain text; binary .npz and .pkl files are not. Each package has its own README. Three Version 6 bundles exist so far only as outputs of working sessions and are marked "pending".

topic	files
manuscripts	The_limit_of_vector-valued_smoothing_for_Sidon_sets (26 September 2026; author block 5 October): $\gamma^* = 2\sqrt{2}/3$, the two-sided smoothing inequality, the pair-level relaxation. Sidon_smoothing_barrier (25 September 2026): superseded
this report	Erdos_Problem_30_Report_V3 and V3_changes.md; Version 4 and V4_changes.md; A_Data_Science_Analysis_of_Erdos_Problem_30_V5 and V5_changes.md; this Version 6 and V6_changes.md; v4_checks.py, v4_figures.py, v5_figures.py, v6_figures.py (Figures 1–8)
Version 6 notes	literature note on OpenAI-linked results (7 October); extreme-value draft, Runs 1–3 (12 October); Appendix D (13 October); Conjecture 6.6 proved, with the interval short-difference test (14 October); predictive analytics (14 October); README_V6_steps.md (14 October); almost-all section (16 October); revised Sections 6 and 7 of the companion manuscript (21 October)
Stage 0	engine.py, run_all.py, run_rest.py, tail.py, ogr_table.py; stage0_results.csv, ogr_vs_algebraic_cuts.txt
characteristic 2 (Table 4)	singer_best_arc.py; singer_q{2048, 4096, 8192, 16384} best-arc summaries and classes; Singer_powers_of_two_best_arc.xlsx; Singer_powers_of_two_profile_by_d.csv
extreme values (Part VIII)	singer_origin_decomp.py, singer_origin_arcs.py, singer_run3.py, evt_singer.py; singer_origin_summary.csv; Erdos30_V6_Run1_results.xlsx, Run2, Run3; xi_bound.py, drift.py, xi_bound_out_11_14.txt, xi_bound_reported_out.txt, drift_out.txt, xi_bound_summary.csv
symmetric scan	floor2.py, certify.py, verify_lemma.py, search.py, step.py, one.py, upsample.py, lscan.py; floor_scan.csv; cert_m512 kernels, boundary and λ files; verified_m{32, …, 512}_L4_R2 P, W and λ files; verify_sym_certs.py
linear-kernel certificates	triangle8.py, certify2.py, export_cert.py, check_lin_cert.py; lin_m256 and lin_m512 L8 files (README_linear_kernel_certificates.md)
two-sided and re-optimized certificates	hz2.py, run32.py, refine.py, chunked.py, sym512c.py, run_chain.sh; asym_m{32, …, 256}_L4_K2 and sym_m512_L4_K2 files (README_K2_certificates.md)
symmetric family (Part V, Appendix D)	appD_construct.py, appD_checks.py. Pending: symcert.py, polya.py, build_polya.py, certify_sym.py, check_sym_cert.py, identities.py, discrete_renewal.py, symmetric_certificates_txt.zip (21 MB, twelve certificate files), README_symmetric_certificates.md
pair-level relaxation (Part VI, Appendix B)	38 scripts recovered from the 30 September session, smoke_tests.sh, run_table71.sh, constr JSON files (README_SDP_and_construction.md); Section7_variance_term_pair_relaxation.md; c66_verify.py, out_c66_verify.txt
three-point relaxation (Theorem 6.8, Appendix C)	triple.py; lift_budgets.py, profile.py, lift_psd.py, repair.py, ablate.py, stress.py, stress2.py, stress3.py, chunk.py, bisect_theta.py, compensate.py, det3.py, band_*.py, piece2_*.py, margin2.py, rigidity.py, excess_bounds.py, rowsum_*.py, corner*.py, pi_limit.py, remainder*.py, r2b_bound.py, r34_*.py, lemma1.py, lemma7*.py, assembly.py, results_raw.jsonl. Pending: the step1_three_point README with the full lemma statements

topic	files
arithmetic tests (Part VI)	cyclic_arith_test.py, out_cyclic_arith_test.txt; interval_short_diff.py, isd_enum.c, out_interval_short_diff.txt. Pending: the interval_quotient_test bundle (relax.py, q55.py, bb.py, sums73.py, det_sums.py, det_boost.py, lam_diag.py, wscan.py, fullcheck_sums.py, fullcheck_boost.py, logs and README)
lower side (Part IX)	almost_all_illustration.py and its output; repair_feasibility.py and its output
evidence (Part X)	Erdos30_Bayes_ledger.xlsx; gain_vs_N.py, run_gain.py, gain_vs_N.csv, gain_vs_N_out.txt, new_relaxation_values.csv; fourlag.py, pairlag.py, diag_cuts.py, run_one.py, check_bridge.py, validity.py, results.csv; Erdos30_status_toward_N_eps.xlsx, Erdos30_beyond_pair_statistics_status.xlsx, Erdos30_related_problems_assessment.xlsx
Stage 1 (Part VII)	Stage1_gap_proofs; notes Singer_gaps_Stage1, Singer_gaps_2q_plus_1_rung, Singer_gaps_interior_coefficients, Singer_gaps_interior_coefficients_closed_form, Singer_gaps_modp_word_counts_q41, Bose_gaps_2q_bound, Singer_gaps_literature_check; singer_enum.py, bose_ruzsa_enum.py, isolated_class_checks.py, modp_q41.py, check_proofs.py and their CSV outputs (Stage1_code_README.md); kcore.py, kappa1.py, kappa2.py, omega.py, fp.py, dep3.py, dep4.py, factor_k.py, factor_k2.py, pal.py, gf_recursion_check.py, run_all.sh
Lean (Part XI)	SidonBarriers.lean, FourierL4.lean, WeakDuality.lean, SDPCertifier.lean, PairCert12.lean, lakefile.toml, make_pair_cert.py; README.md. The project's README.md is now the SidonBarriers README; the Version 5 README of the Stage 1 interior-coefficient scripts should be re-deposited under another name
archived	FLOOR_RESULT.md and POSITIVE_INFIMUM.md; Erdos_30_Lean.md (the original Lean sketch, superseded by SidonBarriers); "Upper-Bound Barriers and Global Arithmetic Constraints" (superseded by the revised Sections 6 and 7)

About this document. The computations, proofs and drafting were carried out by John Akwei with Anthropic's Claude models, as stated on the title page. Every computational claim can be reproduced from the files listed above, except where a bundle is marked as not yet deposited. The two-sided smoothing inequality and the results of Parts IV–VII and IX and of Appendices B–D have not been refereed.

Appendix B · Proof of Theorem 6.6

This appendix condenses the note of 14 October 2026. Throughout, r is the renewal density of the uniform law on [0, 1), g = r − 2, $G_1(t) = \int_0^t g$ and $A(s) = \int_0^\infty g(x)g(x+s)\,dx$. Four facts are used, all proved earlier: **(R1)** $1 \le r \le e$, so $-1 \le g \le e-2$ (from Lemma 4.4(a) on [0, 2) and r(x) = ∫ from x − 1 to x of r beyond; the upper bound is Lemma B.1(b) of the limit manuscript); **(R2)** $|g(x)| \le C_0 e^{(-\kappa x)}$ (Lemma 4.4(b)); **(R3)** $\int_0^\infty g = -1/3$ and $\int_0^\infty g^2 = 1/3$ (Lemmas 4.4(c) and 4.5); **(R4)** $\mathcal{B}(s) = 1/6 - \tfrac{1}{2}G_1(s) + \tfrac{1}{4}(g(s) + A(s)) \le s/2$ for every s > 0, with equality on (0, 1) (Lemma 6.4). One new constant enters, $D_+ = \int_0^\infty (1 - r(u)/2)_+\,du = 0.228168\ldots$, with $D_+ \le \tfrac{1}{2}\|g\|_1$.

The Fejér hole of Part VI is used in this form (Lemma 7.5 of the limit manuscript): *if $0 \le f \le 1$, $0 \le \tilde{\psi} \le f$, $(1-\delta)\tilde{\psi}^2 \le \delta(f - f^2)$ and $1 \le \ell \le 1/\delta$, then $\Gamma = ff^T + D(I - \delta\Phi_\ell)D + \mathrm{diag}(\varepsilon)$, with $D = \mathrm{diag}(\tilde{\psi}/\sqrt{\delta})$ and $\varepsilon = f - f^2 - (1-\delta)\tilde{\psi}^2/\delta$, satisfies (i)–(iii) of (6.2), and its difference sums are*

$$P(d) = \Sigma_x f(x)f(x+d) - (1 - d/\ell)_+ \cdot \Sigma_x \tilde{\psi}(x)\tilde{\psi}(x+d). \quad \text{(B.1)}$$

The construction

Given N, put $\delta = (N + 2\sqrt{N})^{(-1/2)}$, $\ell = \lfloor 1/\delta \rfloor$, $c = (\ell - 1)/2$, $b = N - 1 - c$, H the positive root of $H^2/(2\ell) = N - \ell - 4HD_+$, $\mathcal{L} = (N - \ell)/H$ and $m_a = \delta H/2$. On [c, b] let

$$\rho(dy) = \delta \cdot \varphi((y - c)/H)\,dy + m_a(\delta_c + \delta_b), \quad \varphi(u) = 1 + \tfrac{1}{2}g(u) + \tfrac{1}{2}g(\mathcal{L} - u) \quad (0 \le u \le \mathcal{L}), \quad \text{(B.2)}$$

which in units of H is the measure of Part VI with both ends in place: bulk density 1 and, at each end, an atom ½ followed by the density ½r, that is ¼μ_1. By (R1), $\varphi \ge 0$. Then average over windows and form the hole:

$$f(x) = \ell^{-1} \cdot \rho([x - \ell/2, x + \ell/2)), \quad \tilde{\psi}(x) = \min\{f(x), \sqrt{(\delta f(x)(1 - f(x))/(1 - \delta))}\}. \quad \text{(B.3)}$$

For $N \ge N_0$: f vanishes outside $\{0, \ldots, N-1\}$; it carries the plateau m_a/ℓ on the first and last ℓ sites; $\Sigma f = \rho(\mathbb{R}) = \delta(N - \ell) + \delta H(1 + G_1(\mathcal{L}))$, since every point lies in exactly ℓ windows; $\sqrt{N} - 1/\sqrt{N} < \ell \le \sqrt{N} + 1$, $1 - \delta < \delta\ell \le 1$ and $\delta N \ge \sqrt{N} - 1$; $\sqrt{(2\ell(N - \ell))} - 4\ell D_+ \le H \le \sqrt{(2\ell N)}$ and $\mathcal{L} \ge N^{(1/4)}/2$; and $m_a/\ell \ge \delta$ and $f \le N^{(-1/4)} + 2N^{(-1/2)} < 1 - \delta$. So the hypotheses of the Fejér hole hold.

Four lemmas

Lemma B.1 (window identity). *Let ρ be a finite positive measure on $\mathbb{R}$, $\ell \ge 1$ an integer, $f(x) = \ell^{-1}\rho([x - \ell/2, x + \ell/2))$ for $x \in \mathbb{Z}$, and μ the law of $z - y$ under $\rho \otimes \rho$. Then $\Sigma_x f(x) = \rho(\mathbb{R})$, and for every integer $d \ge 1$*

$$\Sigma_x f(x)f(x+d) = \int \Lambda_\ell(d - t)\,\mu(dt) + E(d), \quad \Lambda_\ell(t) = (\ell - |t|)_+/\ell^2, \quad |E(d)| \le \ell^{-2} \cdot \mu(\{t : |t - d| < \ell, t \notin \mathbb{Z}\}). \quad \text{(B.4)}$$

Proof. A point y lies in the window of x exactly when $x \in (y - \ell/2, y + \ell/2]$, an interval holding ℓ integers. For the pair sum, with $w = z - d$, the number of common integers in the windows is the number of integers in a half-open interval of length $(\ell - |y - w|)_+$, which differs from that length by less than 1 and equals it when $z - y \in \mathbb{Z}$. ∎

For the measure (B.2), μ has atoms only at 0 (mass $2m_a^2$) and at $\pm(N - \ell)$, all integers, so E(d) involves only the absolutely continuous part, and the atoms' self-pairs contribute exactly $(2m_a^2/\ell)(1 - d/\ell)_+$ — the profile of the Fejér hole.

Lemma B.2 (continuum pair density). *Let $\varrho = \varphi(u)du + \tfrac{1}{2}\delta_0 + \tfrac{1}{2}\delta_{\mathcal{L}}$ on $[0, \mathcal{L}]$. The law of $v - u$ under $\varrho\otimes\varrho$ is $\tfrac{1}{2}\delta_0 + \tfrac{1}{4}(\delta_{\mathcal{L}} + \delta_{(-\mathcal{L})}) + p(s)ds$ with p even, zero for $|s| > \mathcal{L}$, and, for $0 < s < \mathcal{L}$,*

$$p(s) = \mathcal{L} - s + 2\mathcal{B}(s) + W(\mathcal{L} - s) + \Theta(s), \quad W(v) = \tfrac{1}{3} + G_1(v) + \tfrac{1}{2}g(v) + \tfrac{1}{4}(g*g)(v), \tag{B.5}$$

with $|\Theta(s)| \le (C_0/\kappa + C_0^2/(4\kappa) + C_0^2\mathcal{L}/4)e^{(-\kappa\mathcal{L})}$. If $\mathcal{L}$ is large enough, sup over $0 < |s| < \mathcal{L}$ of $p(s) \le \mathcal{L} + \varepsilon_{\mathcal{L}}$ with $\varepsilon_{\mathcal{L}} \le C_2(1 + \mathcal{L})e^{(-\kappa\mathcal{L}/2)}$.

Proof. Write $\varrho = \lambda + \tau_1 + \tau_2$, with λ Lebesgue measure on $[0, \mathcal{L}]$, $\tau_1 = \tfrac{1}{2}\delta_0 + \tfrac{1}{2}g\cdot\mathbb{1}_{[0,\mathcal{L}]}$ and τ_2 its reflection. Expanding and folding τ_2 onto τ_1 by the reflection, and using $\int g = -1/3$, the untruncated terms combine to $\mathcal{L} - s + 2\mathcal{B}(s) + W(\mathcal{L} - s)$; three exponentially small remainders form Θ. For $s \le \mathcal{L}/2$, (R4) gives $2\mathcal{B}(s) - s \le 0$ and $W(\mathcal{L} - s) \le \varepsilon_{\mathcal{L}}$; for $s > \mathcal{L}/2$, $p(s) - \mathcal{L} \le 2\sup|\mathcal{B}| + \sup|W| + 1 - \mathcal{L}/2 \le 0$ once $\mathcal{L} \ge 6.8$ (numerically $\sup|\mathcal{B}| = 0.50$ and $\sup|W| = 0.372$). ∎ The formula agrees with a brute-force discretization to 3×10^{-4} at $\mathcal{L} = 4$, 6 and 9.

Lemma B.3 (hole). *If $f \ge \delta$ on the first and last ℓ sites and $f \le 1 - \delta$ everywhere, then for $1 \le d < \ell$, $\Sigma_x \tilde{\psi}(x)\tilde{\psi}(x + d) \ge \delta^2(N - d - 4HD_+)$.*

Proof. With $m = \min(f, \delta)$, one has $\tilde{\psi} \ge m$ in both cases $f \le \delta$ and $\delta \le f \le 1 - \delta$. Off the plateaus, by convexity and since each y lies in at most ℓ windows, $\Sigma_x (\delta - m(x)) \le \delta H\int_0^{\mathcal{L}} (1 - \varphi)_+ \le 2\delta HD_+$. For values in $[0, \delta]$, $m(x)m(y) \ge \delta^2 - \delta(\delta - m(x)) - \delta(\delta - m(y))$; sum over the $N - d$ pairs at lag d. ∎

Lemma B.4 (large lags). *For $N - 2\ell < d \le N - 1$, $\Sigma_x f(x)f(x + d) \le m_a^2/\ell + 4\delta m_a + 8\delta$. Proof.* Fewer than 2ℓ values of x contribute; split f into the plateaus and the averaged density, which is below 2δ by (R1). ∎

Proof of the lower bound

By the Fejér hole, (f, Γ) satisfies (i)–(iii), and the difference sums are given by (B.1). For $1 \le d \le N - 2\ell$, ρ is δH times the image of ϱ under $u \mapsto c + Hu$, so the continuous part of μ has density $\delta^2 H\cdot p(t/H)$, and Lemmas B.1 and B.2 give

$$\Sigma_x f(x)f(x + d) \le (2m_a^2/\ell)(1 - d/\ell)_+ + \delta^2 H(\mathcal{L} + \varepsilon_{\mathcal{L}})(1 + 2/\ell). \tag{B.6}$$

Here $H\mathcal{L} = N - \ell$, and by the choice of H, $2m_a^2/\ell = \delta^2(N - \ell - 4HD_+)$. For $\ell \le d \le N - 2\ell$ there is no hole and no triangle; for $d < \ell$, Lemma B.3 gives a hole of at least $(1 - d/\ell)\delta^2(N - \ell - 4HD_+)$, which cancels the triangle. In both cases $P(d) \le \delta^2(N - \ell + H\varepsilon_{\mathcal{L}})(1 + 2/\ell)$, and this is at most 1 because $(N - \ell)(1 + 2/\ell) \le N + \sqrt{N} - 1$ for $N \ge 16$ and $H\varepsilon_{\mathcal{L}}(1 + 2/\ell) < \sqrt{N} + 1$ for $N \ge N_0$, while $\delta^{-2} = N + 2\sqrt{N}$. For $N - 2\ell < d \le N - 1$ there is no hole, and Lemma B.4 with $m_a^2/\ell \le \tfrac{1}{2}$ and $4\delta m_a = 2\delta^2 H \le 3N^{(-1/4)}$ gives $P(d) < 1$ for $N \ge 5000$. Finally $\Sigma f = \delta(N - \ell) + \delta H(1 + G_1(\mathcal{L}))$ with $1 + G_1(\mathcal{L}) \ge 2/3 - (C_0/\kappa)e^{(-\kappa\mathcal{L})}$, $\delta(N - \ell) \ge \sqrt{N} - 2$ and $\delta H \ge \sqrt{2}\cdot N^{(1/4)} - 3\sqrt{2}\cdot N^{(-1/4)} - 4D_+$, which gives (6.3). ∎

Of the constant 2, about 1.5 pays for the lattice factor $1 + 2/\ell$ and the slack in δ, and about $\tfrac{1}{2}$ for the window itself, since the bulk spans $N - \ell$ sites; the term $(8/3)D_+ \approx 0.61$ pays for lowering H so that the hole, at most $\min(f, \delta)$ per site, still covers the atoms.

Proof of the upper bound

Let (f, Γ) be feasible with $\Sigma f = \hat{\kappa}$. For $m \ge 2$ take the certificate of Proposition 5.2: one kernel, depth $L_m = \max(2, \lceil(2/\kappa)\log m\rceil)$, $a \le 4/3$, $a\cdot b \le 8/9 + C_1(\log m)/m^2$, and $2/3 \le b \le 1$ for $m \ge m_0$. Proposition 6.3 gives $\hat{\kappa}^2 \le (N - 1 + bH)(1 + (\hat{\kappa} - 1)a/H)$ for every $H = mh$ with $N \ge 2L_mH$, and the argument behind (3.4) then yields $\hat{\kappa} \le \sqrt{N_1} + \gamma N_1^{(1/4)} + \gamma^2/2 + \gamma^3/(8N_1^{(1/4)})$, with $\gamma = \sqrt{(ab)}$ and $N_1 = N + bm$, provided $N \ge 2L_mH$ for its choice $H < 2N^{(3/4)} + m$. Choose $m = \lceil N^{(1/8)}\cdot(\log N)^{(1/2)}\rceil$. Then $L_m = O(\log N)$, the

proviso holds for $N \geq N_0'$, and since $\sqrt{(8/9 + y)} \leq 2\sqrt{2}/3 + 3y/(4\sqrt{2})$ and $(\log m)/m^2 \leq (1/8 + o(1))N^{(-1/4)}$, $\gamma N_1^{(1/4)} \leq (2\sqrt{2}/3)N^{(1/4)} + 3C_1/(32\sqrt{2}) + o(1)$. With $\sqrt{N_1} = \sqrt{N} + o(1)$ and $\gamma^2/2 \leq 4/9 + o(1)$, this is part (b). ∎

Distinct sums and Sherali–Adams

Lemma B.5 (reflection). *$f(N - 1 - x) = f(x)$ for every x; consequently $(f*f)(s) = \Sigma_x f(x)f(x + d)$ with $d = |N - 1 - s|$. Proof.* ρ is invariant under $y \mapsto N - 1 - y$, and an atom sits at a window endpoint only for half-integer x. ∎

Lemma B.6. *For $N \geq \max(N_0, 10^4)$ the point (f, Γ) satisfies the Sherali–Adams inequalities and, for every s, $\Sigma_{x<y,\, x+y=s}\, \Gamma(x, y) + [s \text{ even}]\cdot f(s/2) \leq 1 - 2\delta^2 H D_+ + 2\delta < 1$.*

Proof. Off the diagonal $\Gamma(x, y) = f(x)f(y) - (1 - |x - y|/\ell)_+ \phi\tilde{}(x)\, \phi\tilde{}(y) \in [0, f(x)f(y)]$, which gives $\Gamma \leq f$ and $\Gamma \geq 0 > f(x) + f(y) - 1$, as $f < \tfrac{1}{2}$. For the sums, the left side is at most $\tfrac{1}{2}(f*f)(s) + [s \text{ even}]f(s/2)$. If $d = |N - 1 - s| \leq N - 1 - 2\ell$, Lemma B.5 and (B.6) at lag d (valid also at $d = 0$) give $(f*f)(s) \leq \delta^2(N - \ell - 4HD_+) + 1$, and $f(s/2) < 2\delta$ off the plateaus. Otherwise $s < 2\ell$ or $s > 2N - 2 - 2\ell$, few x contribute, and the left side is at most $\tfrac{1}{4} + 4N^{(-1/4)} + 8N^{(-1/2)} < 1$. The margin $2\delta(\delta H D_+ - 1)$ is positive once $\delta H > 1/D_+ \approx 4.4$, which holds for $N \geq 10^4$. ∎

Appendix C · Theorem 6.8: construction and proof outline

This appendix records the architecture of the proof assembled on 20 October 2026; the full lemma statements, proofs and scripts are in the step1_three_point bundle (Appendix A.4). The relaxation $\hat{h}_3$ is the one of Part VI: triple variables T(x, y, z), positive semidefiniteness of the moment matrices localized at $z \in A$ (M_z) and at $z \notin A$ ($M - M_z$) for every z, the lifted budgets $\Sigma_x T(x, x + d, z) \le f(z)$ and $\Sigma_x [\Gamma(x, x + d) - T(x, x + d, z)] \le 1 - f(z)$, $T = 0$ on three-term progressions, the Sherali–Adams bounds and $\Gamma \le f$, together with every pair-level constraint.

Lemma C.1 (modified pair point). *Take the construction of Appendix B with hole, window and atom width $\ell = \lfloor (1-\theta)/\delta \rfloor$, $\theta = \kappa \cdot N^{-1/4}/\sqrt{2}$, and δ the root of $\delta^2 N + K\delta - 1 = 0$ with $K = 2/(1-\theta) - (1-\theta) + c_s + \eta$. Then $\max_d P(d) \le 1 - c_s \cdot \delta$, the plateaus satisfy $f \ge \delta$, $f \le 1 - \delta$ and $\ell \le 1/\delta$, and $\Sigma f \ge \sqrt{N} + (2\sqrt{2}/3)N^{1/4} - [K/2 + (1-\theta)] - (\kappa/3)(1+\theta) - (8/3)D_+ - o(1).$* The narrowed hole costs $\kappa/3$ and the slack $c_s/2$ at second order.

The lift. T(x, y, z) is the Kirkwood product f_x f_y f_z ρ_xy ρ_xz ρ_yz on distinct triples, with $\rho = \Gamma/ff^T$ and T(x, x, z) = Γ(x, z), multiplied by $1 - \max(r_z(|y - x|), r_x(|z - y|), r_y(|z - x|))$ and set to zero on three-term progressions. The reduction $r_w(d) = \min(1, (\Delta_w(d) - s)_+/B_w(d))$ is computed from an exact formula for the conditional excess Δ_w(d) of the Kirkwood lift (Lemmas A and A′), so the construction contains no numerical step.

Exact identities (Lemma C.2). (P1) $I - \delta\Phi_\ell = \theta_\ell \cdot I + \delta Q_\ell$, as in (6.5), with Q_ℓ positive definite because of the truncated boxes. (P2) On coordinates $x \ne z$, the Kirkwood conditional covariance given $z \in A$ is $C_z^K = D_q S D_q + \mathrm{diag}(f\,q\,(1-q))$, with $S = \Gamma - ff^T$ and $q = \Gamma(\cdot, z)/(f\,f_z)$; so C_z^K splits into a θ part $W_\theta = \theta_\ell \cdot \mathrm{diag}(q^2\tilde{\psi}^2/\delta)$, a $\delta \cdot Q_\ell$ part and a nonnegative diagonal. (P3) For any lift, $f_z C_z + (1 - f_z)\bar{C}_z = S/z$, the Schur complement of S at z. All three were checked to 10^{-16}.

Margins and rigidity (Lemma C.3). $S \ge \theta \cdot \mathrm{diag}(f(1-f))$, and the same for S/z by Schur-complement monotonicity. On the atom zone the local margin is larger: the atom Schur complement has smallest eigenvalue about $0.21 \cdot p_a$ (atom rigidity), and an explicit diagonal lower bound for the local margin holds on the atom zone and the band $\ell \le x < 2\ell$.

Excess bounds (Lemma C.4). Pairs not through the conditioning point contribute nonpositively to the excess; off the atom lags the excess is at most $2\sigma_{\max} \cdot \delta$, where $\sigma_{\max} = e/2$ up to exponentially small terms, so slack $c_s \cdot \delta$ with $c_s > e$ absorbs it with no reduction; with two partners in the same zone the reduction is at most $(5/4 + o(1)) \cdot p_a$ and supported on at most ℓ lags pointing at the nearer atom zone; and for opposite zones $P(d) \le \tfrac{1}{2} + o(1)$. The lifted budgets then hold by construction, and the complement budgets need $c_s \ge 1.63$.

The slack function (Lemma C.5). The pair-budget deficit of the construction at lag tℓ, in units of p_a, is $\pi(t) = 2(1-\theta)(1-p_a)(1-t)^2/(1-\delta) + \lambda_1(1-t) + (1-\theta)t(1-t)\sqrt{(\delta/p_a)} + R$, with the remainder R split into five pieces, each bounded explicitly; the lattice part of one of them is bounded as a midpoint-rule error.

Localized positive semidefiniteness (Lemma C.6). $E_z = C_z^K - C_z$, the mass removed by the reductions and progression zeros, is dominated by a diagonal margin through a Schur test with symmetric row sums. Three row classes bind: far rows, with proved constant A = 8.83 (about 1.9 measured); mirror rows, B = 2.02; and the corner, conditioning points within 2ℓ of an end, whose scaling-limit constant was certified at most 0.925 by branch and bound with interval upper bounds and

upper Riemann sums. The test closes when $A/\kappa + B/\sqrt{\kappa} < 1$, that is $\kappa > 17.2$, and stays under the corner constant when $\kappa > 19.1$.

Complement matrices (Lemma C.7). By (P3), $(1 - f_z)\bar{C}_z = S/z - f_z\, C_z^K + f_z\, E_z$, with $C_z^K \le \mathrm{diag}(f)$ and $E_z \ge -\mathrm{diag}(\text{row sums})$, so $M - M_z \ge 0$ once $\theta(1 - f_x) \ge f_z(1 + \text{row sum}_x/f_x)$; this needs only $\kappa > 1$. Nonnegativity and the Sherali–Adams bounds follow from $0 \le \Gamma \le ff^{T}$, which gives $T \le \min(\Gamma_{xy}, \Gamma_{xz}, \Gamma_{yz})$.

Assembly. $\kappa = 20$ and $c_s = 2.8$ satisfy every condition, and the lower constant is $2.108 + c_s/2 + \eta/2 + \kappa/3 = 10.28 + o(1)$. N_0 is the largest of the per-lemma thresholds — the crude explicit error terms of Lemma C.6 need about 10^{10} — and is ineffective through C_0 and κ, as in Theorem 6.6. For $N < N_0$ any Sidon set is a feasible integral point, since it has no three-term progression, so the shortfall there is at most $\sqrt{N_0} + (2\sqrt{2}/3)N_0^{1/4}$; for $N \ge N_0$ the bound of Lemma C.1 decreases in N. ∎

Appendix D · Symmetric certificates reach $2\sqrt{2}/3$

This appendix condenses the note of 13 October 2026. It uses Lemma 4.2 and its equality case, the constant weight of the renewal multiplier (Lemma 4.6 and Theorem 4.7), the Wiener–Hopf identity $(\mu_1 \star R^*)(u) = 4$ of Theorem 5.1, the decay $|g(x)| \le C_0 e^{-\kappa x}$ of Lemma 4.4(b), and $|g| \le 2$. Constants $C, C', \dots$ depend only on C_0 and κ.

Fix $n \ge 1$ and $m = 2n$. For $1 \le l \le n$ let $p^{(l)} \in \mathbb{R}^m$ be uniform on the $2l$ central cells, $p^{(l)}_i = 1/(2l)$ for $n - l \le i \le n + l - 1$, with weights $\lambda_l = 4l^3/n^4$ for $l < n$ and $\lambda_n = (2n-1)/n^2$. Each $p^{(l)}$ reads the same reversed, so these are kernels of the original Hou–Zhao framework.

Theorem D.1. *There are constants C, C' and $L_0 \ge 3$ such that for every $n \ge 1$ and $L \ge L_0$ there are boundary vectors $w^{(l)} \in \mathbb{R}^{(Lm)}$ making (p, w, λ) a symmetric certificate at resolution (m, L, n) with $a = 4/3 - 1/(3n^2)$ and $b \le 2/3 + C/n^2 + C'Le^{-\kappa L}$. With $L = \max(L_0, \lceil (3/\kappa)\log(n+1)\rceil)$, $a \cdot b \le 8/9 + C''/n^2$.*

Corollary D.2. *$\gamma^*_{sym} = 2\sqrt{2}/3$, and the certificates of Theorem D.1 approach it at rate $O(1/m^2)$. Proof.* Theorem D.1 gives $\gamma^*_{sym} \le 2\sqrt{2}/3$ and Theorem 4.1 the reverse. ∎ Since $b \ge 8/(9a) > 0$, Hou and Zhao's Theorem 2.1 applies to these certificates directly.

Lemma D.3 (the mixture is the linear kernel). *The mixed autocorrelation $R_n = \sum_l \lambda_l T_{(l/n)}$ of these kernels is the autocorrelation of the cell averages $\rho^{(n)} = (2i+1)/n$ on $[i/n, (i+1)/n)$ of $2x$, and $a = R_n(0) = 4/3 - 1/(3n^2)$. Proof.* The discrete autocorrelation $r(k)$ of $p_i = (2i+1)/n^2$ satisfies $n^4 r(k) = -\tfrac{2}{3}t^3 + 2nt^2 - t/3$ with $t = n - k$, whose second differences are $4l/n^4$ for $1 \le l \le n-1$ and $(2n-1)/n^4$ at $l = n$. Summation by parts gives $r(k) = \sum_l \lambda_l (l-k)_+/l^2$ with $\lambda_l = l^2 \cdot \Delta^2 r(l)$, the weights of the statement. Both R_n and the autocorrelation of $\rho^{(n)}$ are even and linear between the nodes k/n, where they agree, so they are equal. Integrating gives $\Sigma\lambda_l = 1$, and $R_n(0) = n \cdot r(0) = (4n^2-1)/(3n^2)$. ∎

Lemma D.4. *$E = R_n - R^*$ is even, continuous, supported in $[-1, 1]$, has $\int E = 0$ and $|E| \le 3/n^2$.* (Numerically $\sup|E| \approx 1/(2n^2)$.)

Lemma D.5 (boundaries generated by a multiplier). *For symmetric kernels with mixed autocorrelation R, a multiplier $\mu = 4 \cdot \mathrm{Leb} + \nu$, $\omega_k = \tfrac{1}{4}(\mu \star \rho_k)$ at both ends and $\Delta = \tfrac{1}{4}(\mu \star R) - 1$, every covering sum equals $1 + \Delta(u)$, and the cost is $1 + 2\sum_k \lambda_k \int(\omega_k^2 - 1) = B(R, \mu) + \int \Delta \, d\mu$, where $B(R, \mu)$ is the right-hand side of (4.2).* With $\mu = \mu_1$ and R supported in $[-1, 1]$, $B(R, \mu_1) = 4/3 - R(0)/2$, so $B(R_n, \mu_1) = 2/3 + 1/(6n^2)$.

Lemma D.6 (the covering defect decays). *With $\varepsilon = 3/n^2$ and $C_2 = \max(7/2, C_0)$, $|\Delta(u)| \le C_2 \varepsilon \cdot e^{-\kappa (u-1)_+}$ and $\int |\Delta| \, d\mu_1 \le 2C_2 \varepsilon (1 + e + e/\kappa)$. Proof.* By Theorem 5.1, $\Delta = \tfrac{1}{4}(\mu_1 \star E)$, which is $\tfrac{1}{2}E(u) + \int_0^u E + \tfrac{1}{2}\int g(v)E(u-v)\,dv$; for $u \ge 1$ the first two terms vanish and the third is at most $C_0 \varepsilon \, e^{-\kappa(u-1)}$. ∎

Proof of Theorem D.1. Three steps keep every covering sum at least 1 and add $O(1/n^2)$ or $O(Le^{-\kappa L})$ to b. *Step 1, a decaying repair:* $\Omega_l = \omega_l + \eta$ with $\eta(t) = C_2 \varepsilon \cdot e^{-\kappa (t-2)_+}$; since η is nonincreasing, the repair raises each covering sum by at least $C_2 \varepsilon \cdot e^{-\kappa (u-1)_+}$, which covers the defect of Lemma D.6, and by Lemma D.5 the continuum cost is at most $2/3 + C/n^2$. *Step 2, cell averages:* the kernels are constant on cells, so each discrete covering sum of the cell averages $\tilde{w}^{(l)}_j = m\int$ over cell j of Ω_l equals the continuum covering sum at $u = q/m$, with no averaging error, and Cauchy–Schwarz on each cell only lowers the cost. *Step 3, truncation:* beyond depth $L - 1$, $|\tilde{w} - 1| \le \varepsilon_L = C_6 e^{-\kappa (L-1)}$; add ε_L to every boundary entry below Lm and set the rest to 1. The far-end shifts are then covered, because every in-box cell at such a shift lies at depth at least $L - 1$, and the added cost is at most $C'Le^{-\kappa L}$. ∎

The rate is that of the exactly optimal certificates for these kernels, $a \cdot b = 8/9 + 2/(9n^2) - 1/(9n^4)$ up to truncation; Proposition 5.2 has an extra log m because Appendix B of the limit manuscript spreads a uniform increment over the whole box. A literal construction with $\kappa = 2$ and the exact renewal function (appD_construct.py) has every covering sum at least 1 in double precision and $n^2(a \cdot b - 8/9) = 1.27$, 1.31, 1.31 and 1.32 at n = 4, 8, 16 and 32, bounded as the theorem asserts and about six times the optimal 2/9, as expected from an unoptimized repair; the proof's defect bound $10.5/n^2$ compares with about $0.2/n^2$ in practice.